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(19) **United States**(12) **Patent Application Publication**
CHEN et al.(10) **Pub. No.: US 2025/0277029 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **CANNABINOID TYPE 1 RECEPTOR
BINDING PROTEINS AND USES THEREOF**(71) Applicant: **AMGEN INC.**, Thousand Oaks, CA
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(US)(21) Appl. No.: **19/037,790**(22) PCT Filed: **Jul. 26, 2023**(86) PCT No.: **PCT/US2023/071024**

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28, 2022.**Publication Classification**(51) **Int. Cl.****C07K 16/28** (2006.01)**A61K 39/00** (2006.01)**A61P 3/04** (2006.01)(52) **U.S. Cl.**CPC **C07K 16/28** (2013.01); **A61P 3/04**
(2018.01); **A61K 2039/505** (2013.01); **C07K**
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C07K 2317/92 (2013.01)

(57)

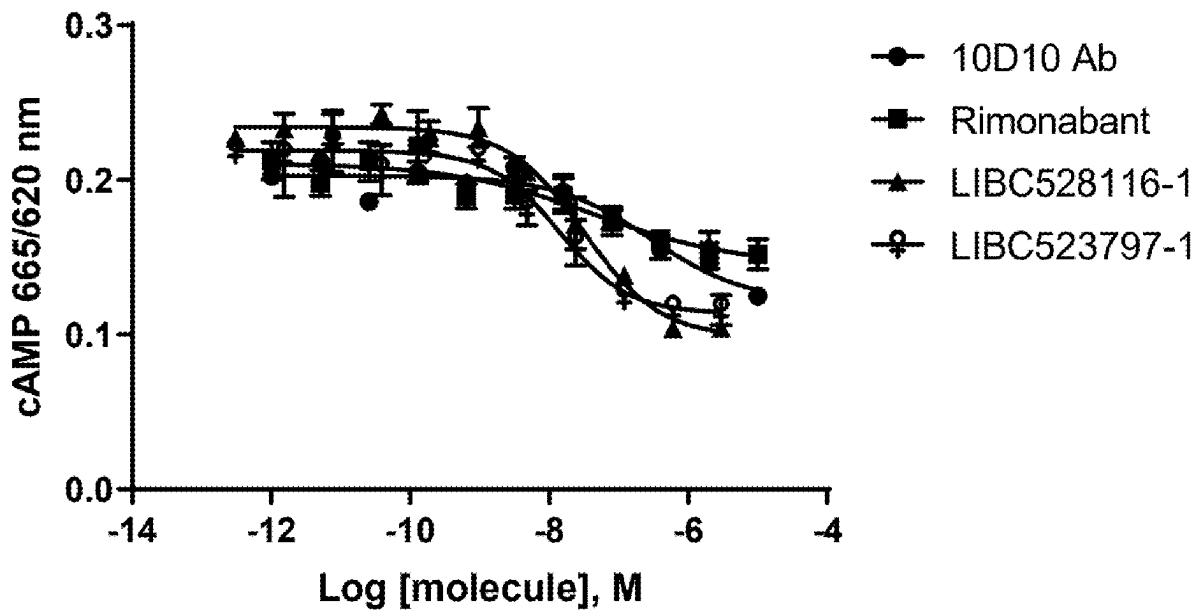
ABSTRACTDescribed herein are anti-CB1 (e.g., huCB1) antibodies and
antigen binding fragments, compositions comprising the
same and their use for the treatment of various disorders or
diseases such as obesity and its comorbidities, chronic
kidney disease, liver disease (e.g., NAFLD), and NASH.**Specification includes a Sequence Listing.**

Figure 1A.

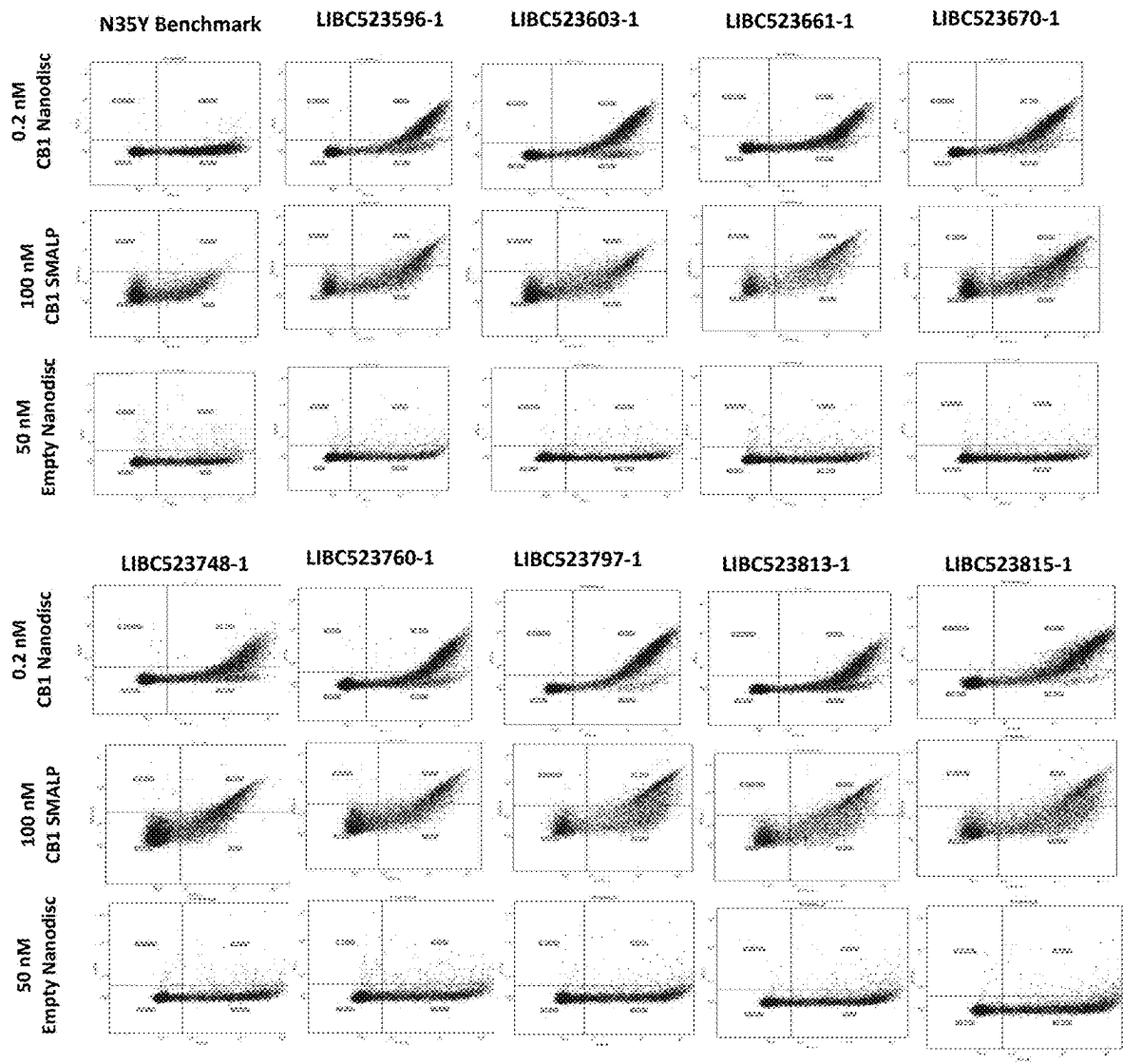


Figure 1B

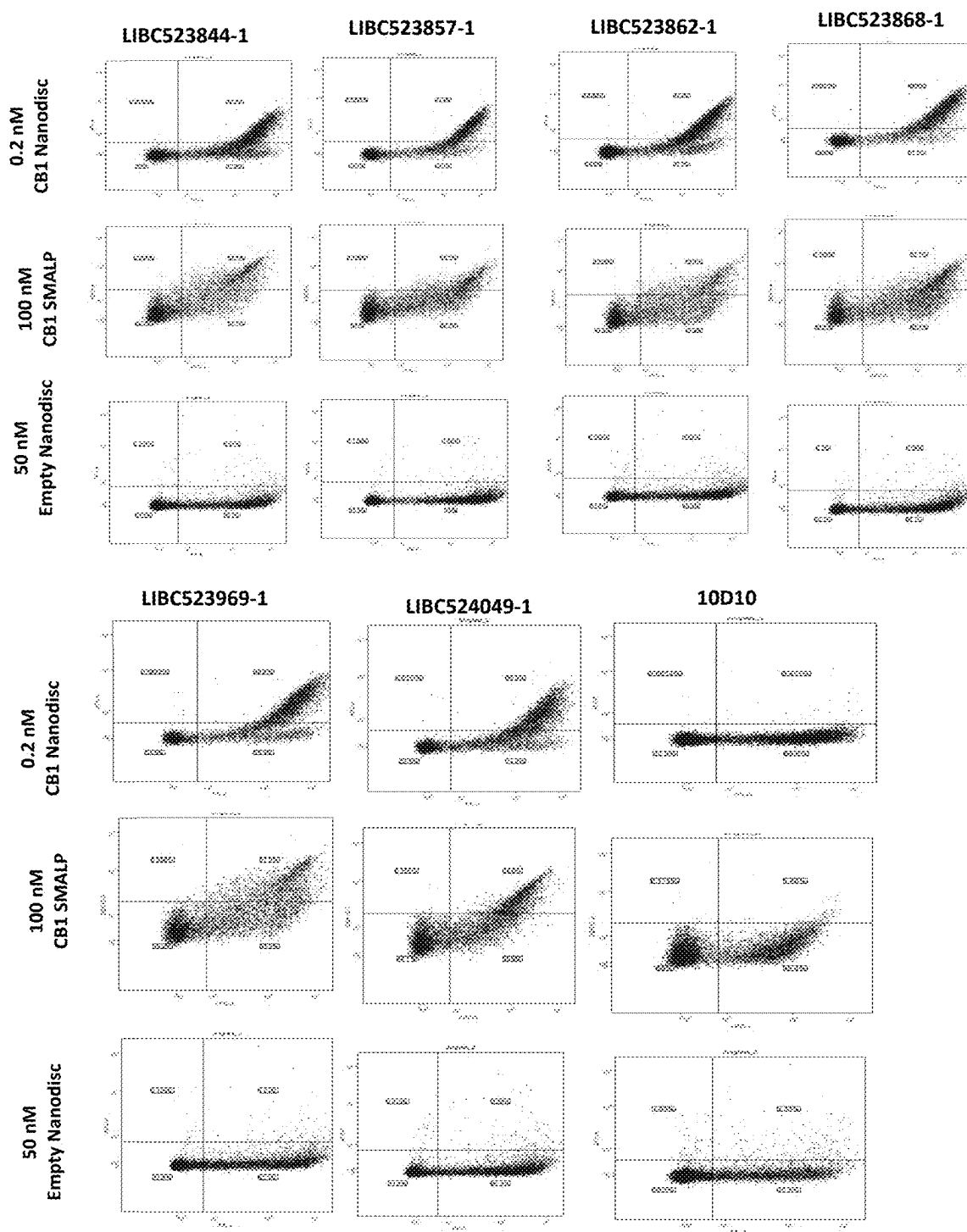


Figure 2A

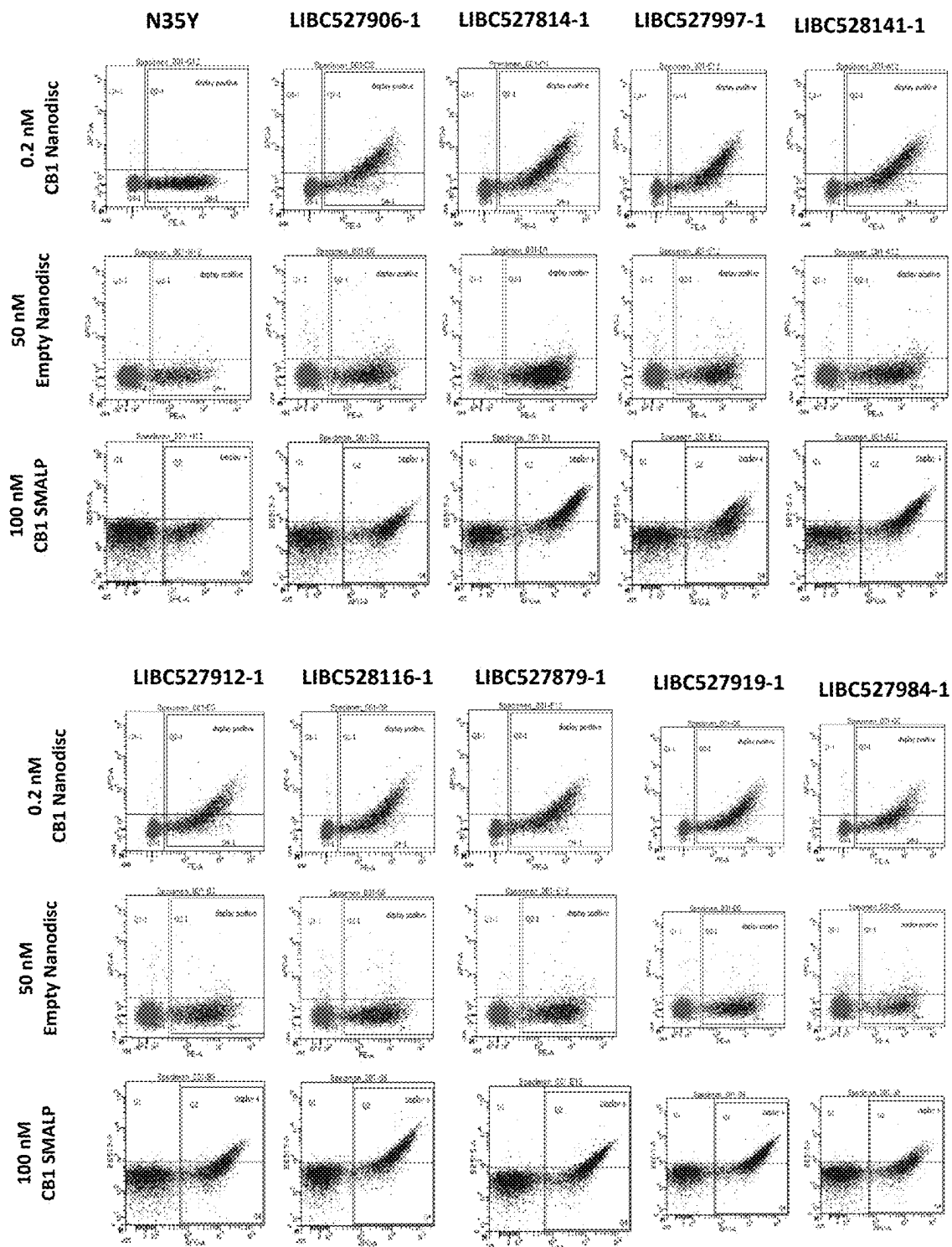


Figure 2B

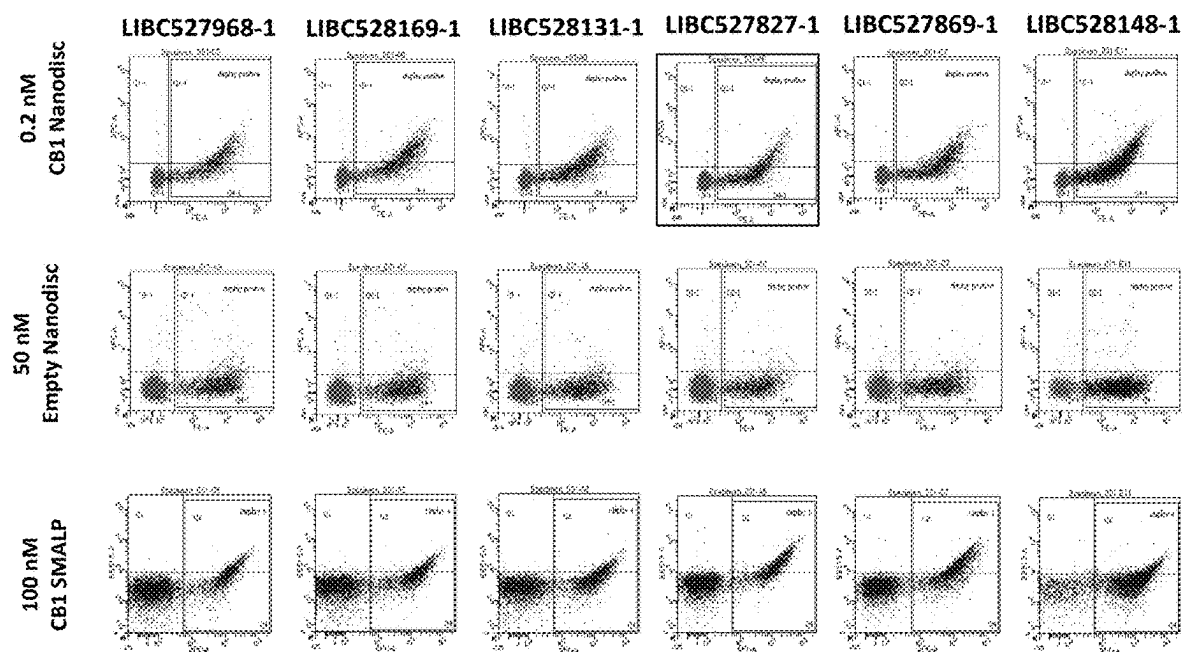


Figure 3A

Name	H_CDR1	H_CDR2	H_CDR3	H_FR3
10D10	RGGDYWS	YIYSGSTNYPNPSLKS	DYDILTGYSYYYYGMDV	RATISVDTSKNQFSLKLSVTAADTAVYYCAR
N35Y benchmark	RGGDYWS	YIYSGSTNYPNPSLKS	DYDILTGYSYYYYGMDV	RATISVDTSKNQFSLKLSVTAADTAVYYCAR
LIBC523661-1	RGGDYWS	HIYEGSTKYNPSFKG	DYDPVTGHSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523797-1	RGGDYWS	HVYTGSTKYNPNFKG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523969-1	RGGDYWS	HIYQTGQTKYNPNLKG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523748-1	RGGDYWS	HVYTGSTKYNPSFEG	DYDILTGYSYYYYGMDV	RATISVKTYKNQFSLKLSVTAADTAVYYCAR
LIBC523857-1	RGGDYWS	HIYQTGQTAYNPRLEG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523868-1	RGGDYWS	HVYEGSTAYNPSFED	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523844-1	RGGDYWS	HVYTGSTKYNPSFKD	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523760-1	RGGDYWS	HVYQTGSTKYNPKFKG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523862-1	RGGDYWS	HIYQTGSTAYNPSLKG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC524049-1	RGGDYWS	HIYTGSTAYNPSLKS	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523813-1	RGGDYWS	HVYTGQTAYNPKFKD	DYDILTGYSYYYYGMDV	RATISVKTYKNQFSLKLSVTAADTAVYYCAR
LIBC523815-1	RGGDYWS	HIYQSGQTKYNPSLKD	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523603-1	RGGDYWA	HVYQSGQTAYNPNFEN	DYDAVSGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523596-1	RGGDYWS	HVYQSGQTAYNPNFED	EYDVVSGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC523670-1	RGGDYWA	HVYQTGQTAYNPNLES	NYDVLGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
Consensus	RGGDYWX1	HX2YX3X4GX5TX6NFX7X8X9X10X11YDX12X13X14GX15SYYYYGMDV		

Figure 3B

Name	L_CDR1	L_CDR2	L_CDR3	L_FR3
10D10	RSSQSLHSHNGYNYLD	LGSNRAS	MQALQTPRT	GVPDRFSGSGSGTDFTLKIRRVEAEDVGYYIC
N35Y benchmark	RSSQSLHSHYGYNYLD	LGSNRAS	MQALQTPRT	GVPDRFSGSGSGTDFTLKIRRVEAEDVGYYIC
LIBC523661-1	RSSQSLTSSGYNYVD	QGSNRAS	RQARAIPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523797-1	RSSQSLTSSGYNYVD	QGSNRAS	RQAVGLPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523969-1	RSSQSLTSSGTANYLD	QGSNRAS	RQAVGLPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523748-1	RSSQSLHSHYGYNYLD	LGSNRAS	RQARALPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523857-1	RSSQSLTSSGTANYLD	LGSNRAS	RQAVELPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523868-1	RSSQSLTSSGYNYVD	LGSNRAS	MQARQIPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523844-1	RSSQSLSSSGANYVD	QGSNRAS	RQARALPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523760-1	RSSQSLTSSGANYVD	LGSNRAS	RQAVGTPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523862-1	RSSQSLHSTGANYVD	LGSNRAS	RQAIALPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC524049-1	RSSQSLHSHYGYNYLD	LGSNRAS	RQARTLPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523813-1	RSSQSLSSSGANYVD	LGSNRAS	MQARGIPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523815-1	RSSQSLTSSGANYVD	LGSNRAS	RQAREQIPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523603-1	RSSQSLSSYGNNYVD	QGSNRAS	RQAVGLPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523596-1	RSSQSLTSSGYNYVD	QGSNRAS	MQAVKIPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
LIBC523670-1	RSSQSLTSSGYNYLD	QGSNRAS	MQALQIPRT	GVPDRFSGSGSGTDFTLKISRVEAEDVGYYIC
Consensus	RSSQSLX16SX17GX18NYX19D	X20GSNRA	X21QAX22X23X24PRT	

Figure 4A

Name	H_CDR1	H_CDR2	H_CDR3	H_FR3
10D10	RGGDYWS	YIYSGSTNYPNPSLKS	DYDILTGYSYYYYGMDV	RATISVDTSKNQFSLKLSSVTAADTAVYYCAR
10D10 N35Y	RGGDYWS	YIYSGSTNYPNPSLKS	DYDILTGYSYYYYGMDV	RATISVDTSKNQFSLKLSSVTAADTAVYYCAR
LIBC527906-1	RGGDYWG	HIYQTGSTNYPNPFKD	DYDILTGYSYYYYGMDV	RATISVDTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527814-1	RGGDYWG	HVYQTGSTKYNPSEFKD	NYDTLTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527997-1	RGGDYWA	HIYQTGSTNYPNPFKG	NYDTLTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC528141-1	RGGDYWS	HVYQTGSTKYNPREFKG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527912-1	RGGDYWS	YIYQTGTQTKYNPNFEG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC528116-1	RGGDYWA	HVYTGSTNYPNPFKD	NYDTVTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527879-1	RGGDYWG	HIYQSGSTNYPNPKLKS	DYDIFTGYSYYYYGMDV	RATISVDTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527919-1	RGGDYWA	HVYQTGTSTAYNPNPFKD	NYDALSGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527984-1	RGGDYWS	HIYQTGKTNYPNPFKS	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527968-1	RGGDYWS	HIYQSGSTKYNPSEFKD	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC528169-1	RGGDYWS	HIYQSGSTKYNPSEFED	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC528131-1	RGGDYWS	HIYQTGTATAYNPNLEG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527827-1	RGGDYWG	HVYQSGQTAYNPKFEN	NYDTLTGHSYYYYGMDV	RATISVKTYKNQFSLKLSSVTAADTAVYYCAR
LIBC527869-1	RGGDYWA	HIYHTGKTNYPNPFKG	NYDTLSGNSYYYYGMDV	RATISVKTYKNQFSLKLSSVTAADTAVYYCAR
LIBC528148-1	RGGDYWS	HIYQSGQTCKYNPKLKG	DYDILTGYSYYYYGMDV	RATISVDTYKNQFSLKLSSVTAADTAVYYCAR
Consensus	RGGDYWX1	HX2YX3X4GX5TX6YNPX7X8X9X10	X11YDX12X13X14GX15SYYYYGMDV	

Figure 4B

Name	L_CDR1	L_CDR2	L_CDR3
10D10	RSSQSLHSHNGYNYLD	LGSNRAS	MQALQTPRT
10D10 N35Y	RSSQSLHSHGYNYLD	LGSNRAS	MQALQTPRT
LIBC527906-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC527814-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC527997-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC528141-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC527912-1	RASQSIISNYLN	AASSLRS	QQYSNYPLT
LIBC528116-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC527879-1	RASQSIISNYLN	AASSLRS	QQYSNYPLT
LIBC527919-1	RASQSIISNYLN	AASSLRS	QQYSNYPLT
LIBC527984-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC527968-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC528169-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC528131-1	RASQSVSSYLG	GASSRAT	QQYGSSPRT
LIBC527827-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC527869-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT
LIBC528148-1	RASQSIISNYLN	AASSLHS	QQYQSYPLT

Figure 5A

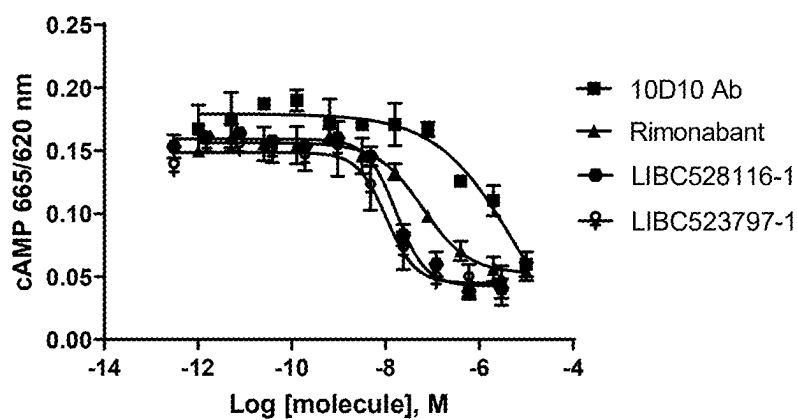
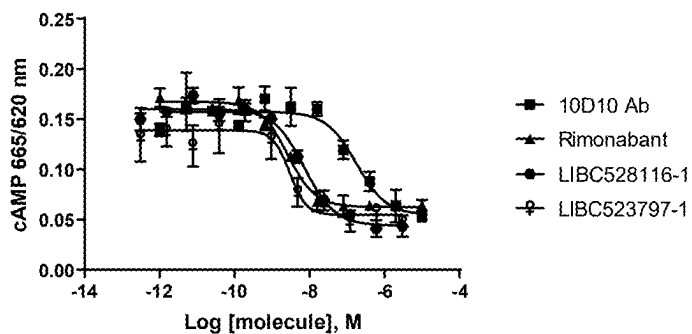


Figure 5B

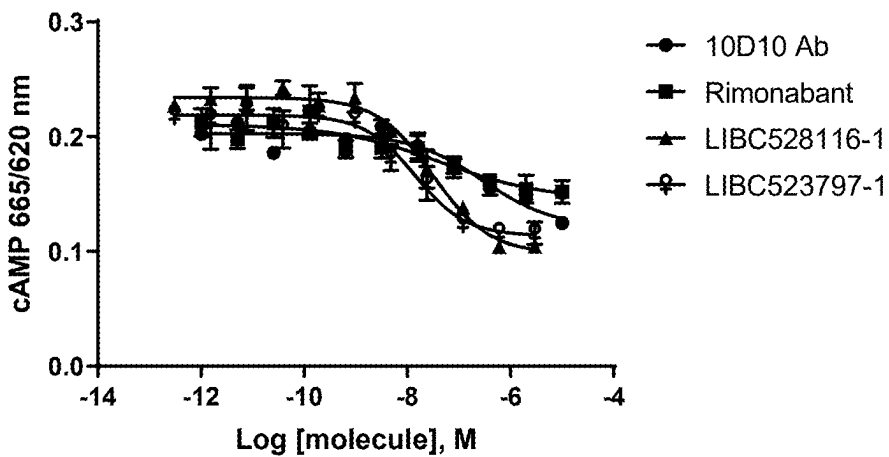


Figure 6A

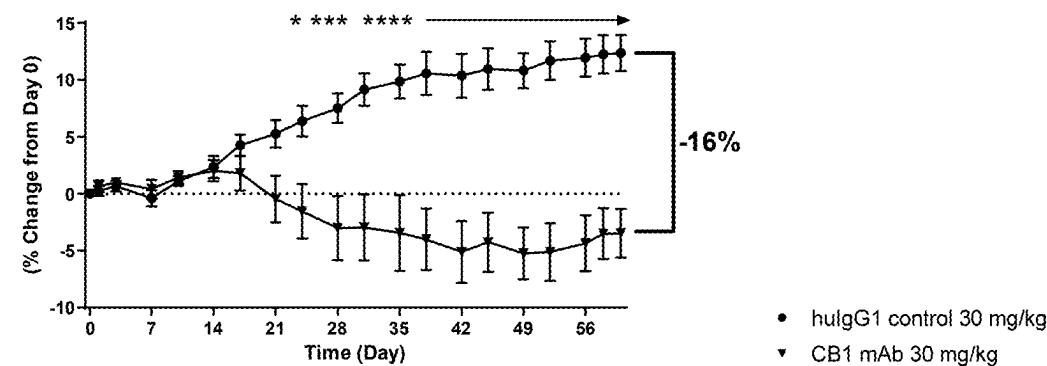


Figure 6B

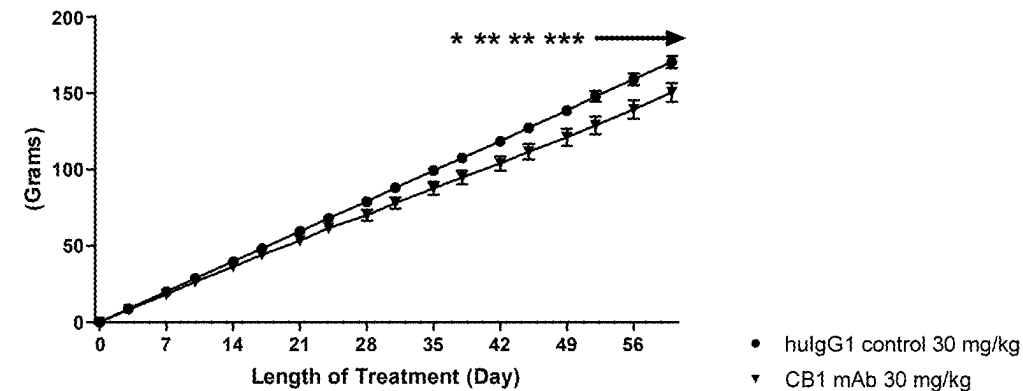


Figure 6C

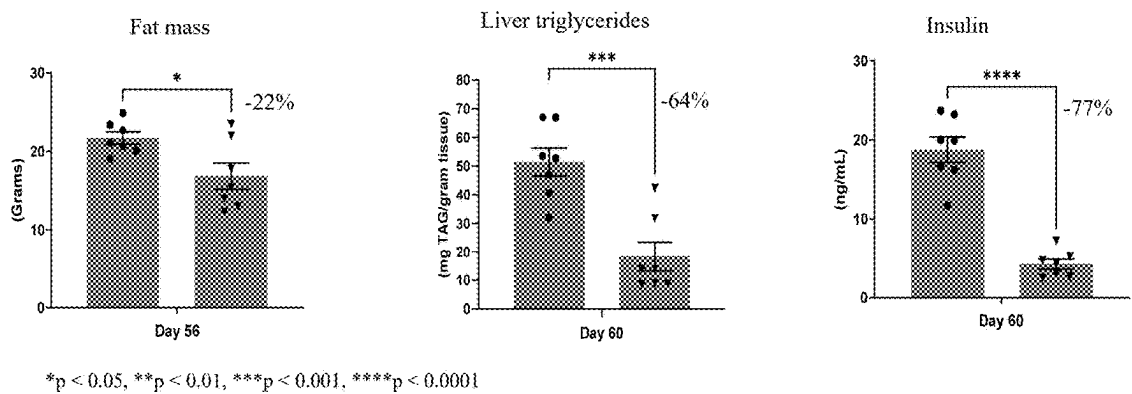


Figure 6D

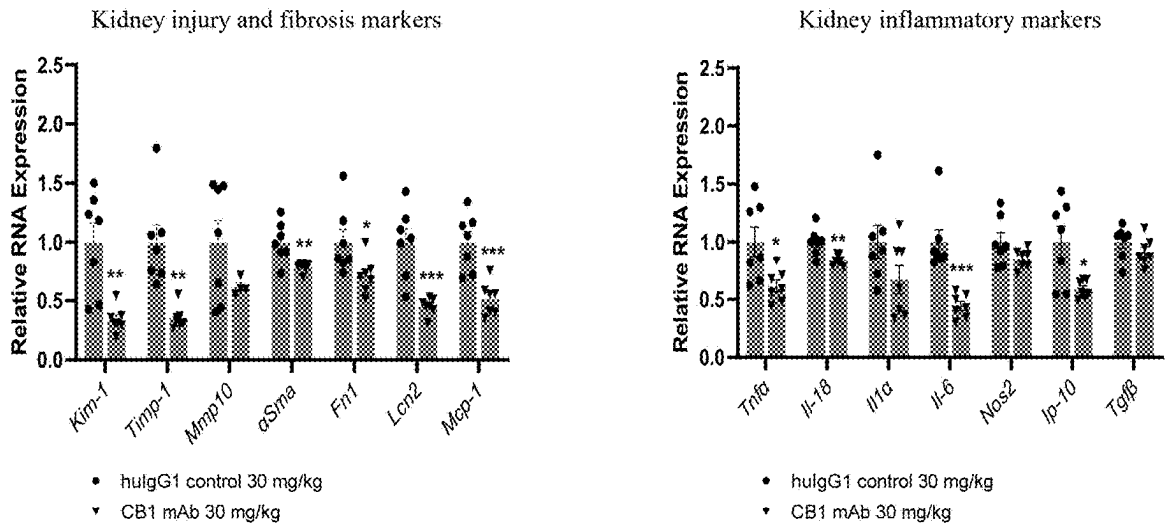


Figure 7

Name	L_CDR1	L_CDR2	L_CDR3	H_CDR1	H_CDR2	H_CDR3
LIBC528116 (Fab)	RASQSI SNYLN AASSLHS		QQQSYPLT	RGGDYWA	HVYTGSTNYPREFK	NYDTVTGYSYYYGMDV
LIBC680562- 1	RASQSI SSYLN SARRLSS		QQYAKSPIT	RGGDYWA	HIYQTGSTNYPREFK	NYDTLTGYSYYYGMDV
LIBC680574- 1	RASQSI SSYLN GARRLGS		QQYSSLEVT	RGGDYWA	HVYTGSTKYNPSFKD	NYDTLTGYSYYYGMDV
LIBC680773- 1	RASQSI SSYLN KARLLSS		QQYSRLPLT	RGGDYWA	HVYTGSTNYPREFK	NYDTVTGYSYYYGMDV
LIBC680800- 1	RASQSI SSYLN SARALSS		QQYRKFFLT	RGGDYWA	HIYQTGSTNYPREFK	NYDTLTGYSYYYGMDV
LIBC680812- 1	RASQSI SSYLN SARLLSS		QQYRSYPIT	RGGDYWG	HVYQTGSTKYNPSFKD	NYDTLTGYSYYYGMDV
LIBC681593- 1	RASQSI SSYLN NARRLGS		QQYRSSFVT	RGGDYWA	HIYQTGSTNYPREFK	NYDTLTGYSYYYGMDV
LIBC681594- 1	RASQSI SSYLN KARKLSS		QQFGSMPLT	RGGDYWA	HIYQTGSTNYPREFK	NYDTLTGYSYYYGMDV
LIBC681737- 1	RASQSI SSYLN SARRLAS		QQYVHPFIT	RGGDYWA	HVYTGSTNYPREFK	NYDTVTGYSYYYGMDV
Consensus	RASQSI SSYLN X39ARX40LX41S	QQX42X43X44X45PX46TRGGDYWX1	QQX42X43X44X45PX46TRGGDYWX1	HX2YX3X4GX5TX6YNPX7X8X9X10	X11YDX12X13X14GX15SYYYGMDV	

Figure 8A

Name	H_CDR1	H_CDR2	H_CDR3	H_FR3
LIBC529593-1 (Fab)	RGGDYWS	HIYSGSTNYPNPSLR	DYDILTGYSYYYYGLDV	RVRISVDTSKNQFSLKLSVTAADTAVYYCAR
LIBC23797-1 (Fab)	RGGDYWS	HVYTGSTKYNPNFKG	DYDILTGYSYYYYGMDV	RATISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC673948-1	RGGDYWS	HIYTGSTKYNPSLKN	GYDVSGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC673952-1	RGGDYWS	HVYTGQTYNPSFRG	DYDILTGYSYYYYGLDV	RVTISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC673965-1	RGGDYWS	HVYTGQTAYNPSLRG	GYDTSGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC673982-1	RGGDYWS	HIYTGSTTYPNPSFRG	DYDISGYSYYYYGMDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC673972-1	RGGDYWS	HIYTGQTAYNPNLRN	GYDASGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674024-1	RGGDYWS	HVYTGSTKYNPSFKN	GYDVSGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674035-1	RGGDYWS	HIYAGQTKYNPSFKN	DYDILTGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674043-1	RGGDYWS	HIYTGQTYPNPNFKG	GYDASGYSYYYYGMDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674090-1	RGGDYWS	HIYAGSTAYNPNFRG	GYDTSGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674002-1	RGGDYWS	HIYKGSSTEYNPSLRG	GYDVSGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674153-1	RGGDYWS	HVYTGSTAYNPSFRG	GYDTSGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674200-1	RGGDYWS	HVYTGQTAYNPNFRN	GYDVSGYSYYYYGMDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674214-1	RGGDYWS	HVYEGSTKYNPSFRG	GYDVSGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674216-1	RGGDYWS	HIYTGQTAYNPSLRN	GYDISGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674229-1	RGGDYWS	HIYQTGQTKYNPSFKG	GYDVSGYSYYYYGMDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674235-1	RGGDYWS	HVYAGSTTYPNPSLKS	GYDVSGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674257-1	RGGDYWS	HIYTGQTKYNPSLRG	GYDISGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
LIBC674276-1	RGGDYWS	HVYTGSTTYPNPNFKG	GYDASGYSYYYYGLDV	RVRISVNTYKNQFSLKLSVTAADTAVYYCAR
Consensus	RGGDYWS	HX25YX26X27GX28TX29YNP X30X31X32X33	X34YDX35X36X37GYYYYYYGX3 8DV	

Figure 8B

Name	L_CDR1	L_CDR2	L_CDR3
LIBC529593-1 (Fab)	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC523797-1 (Fab)	RSSQSLLTSYGINYVD	QGSNRAS	RQAVGLPRT
LIBC673948-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC673952-1	RSSQSLLHRSGYNYLD	QGSNRAS	RQARALPRT
LIBC673965-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC673982-1	RSSQSLLHRSGYNYLD	LGSNRAS	RQSVALPRT
LIBC673972-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674024-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674035-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674043-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674090-1	RSSQSLLHRSGYNYLD	QGSNRAS	MQSLQTPRT
LIBC674002-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674153-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674200-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674214-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674216-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674229-1	RSSQSLLHRSGYNYLD	QGSNRAS	MQSLQTPRT
LIBC674235-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674257-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
LIBC674276-1	RSSQSLLHRSGYNYLD	LGSNRAS	MQSLQTPRT
Consensus	RSSQSLLHRSGYNYLD	X20GSNRA	X47QX48X49X50X51PRT

Figure 9

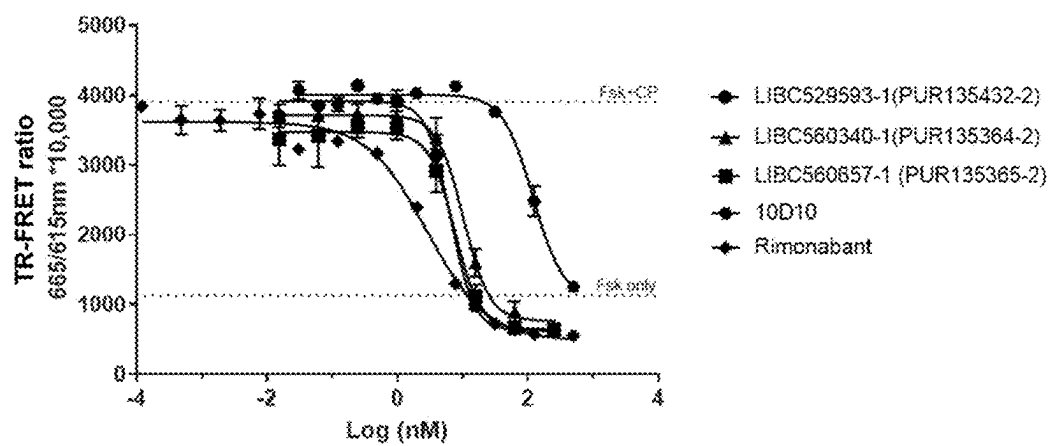


Figure 10A

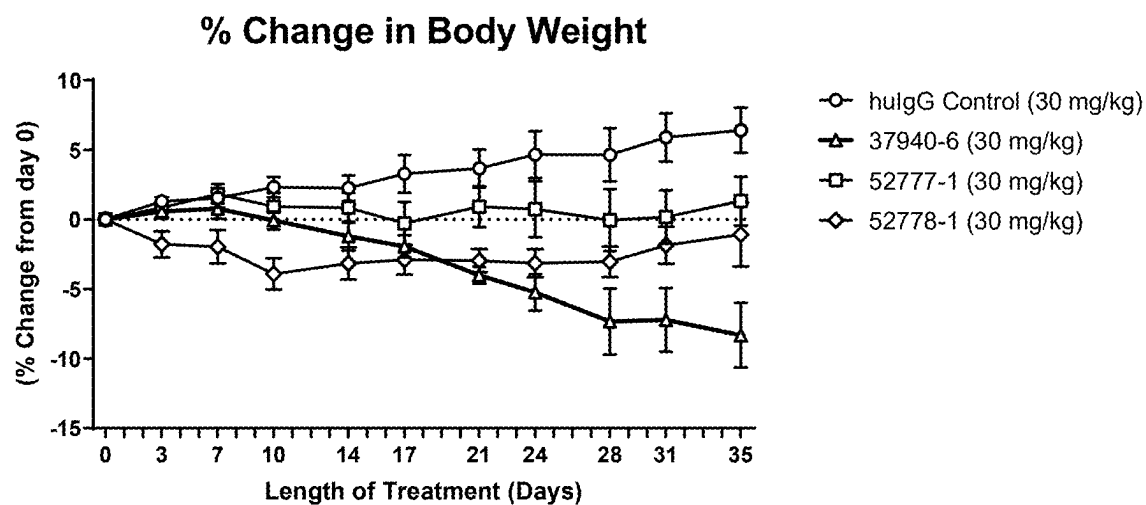


Figure 10B

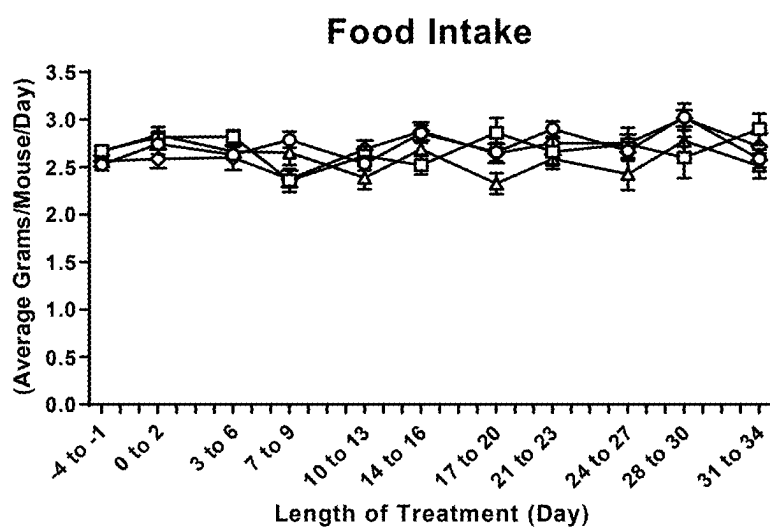


Figure 10C

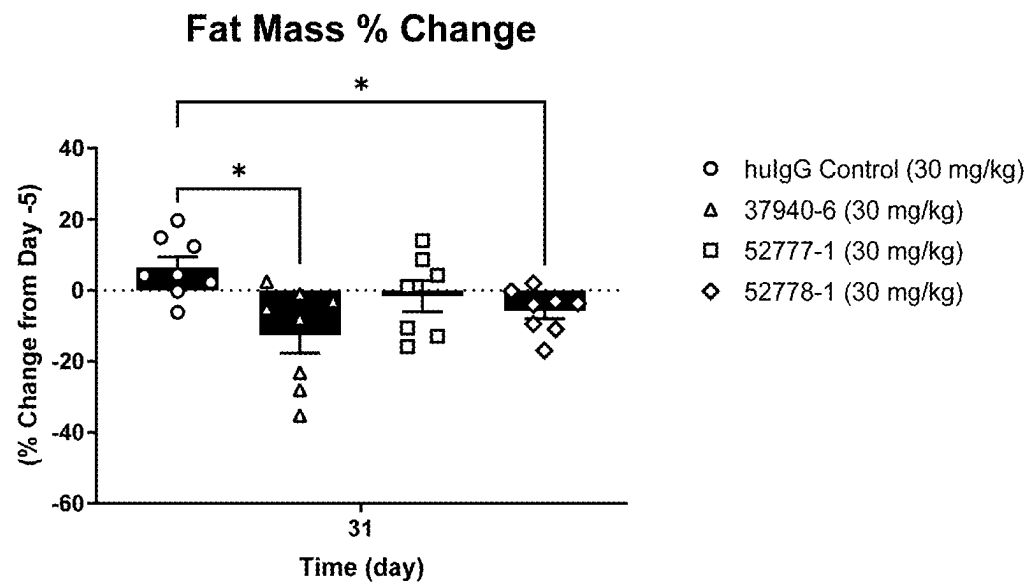


Figure 10D

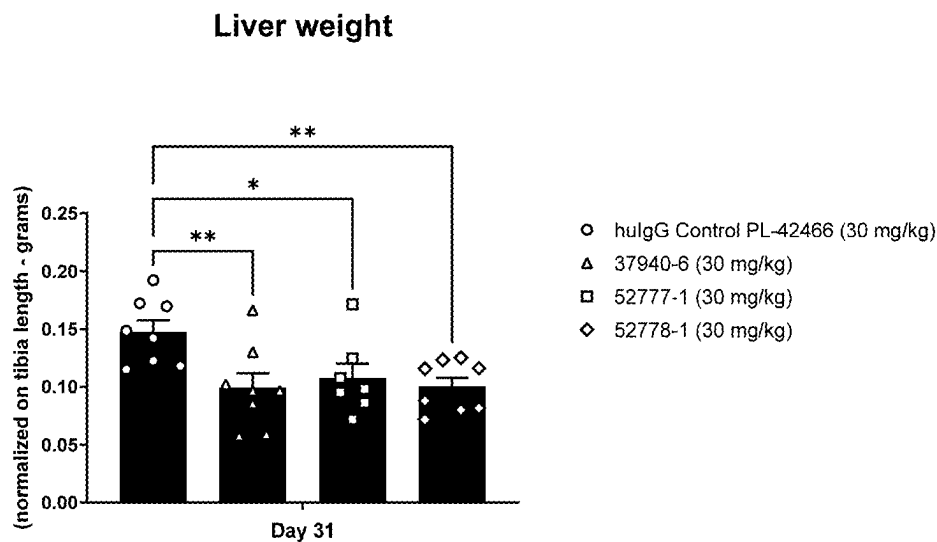


Figure 11A

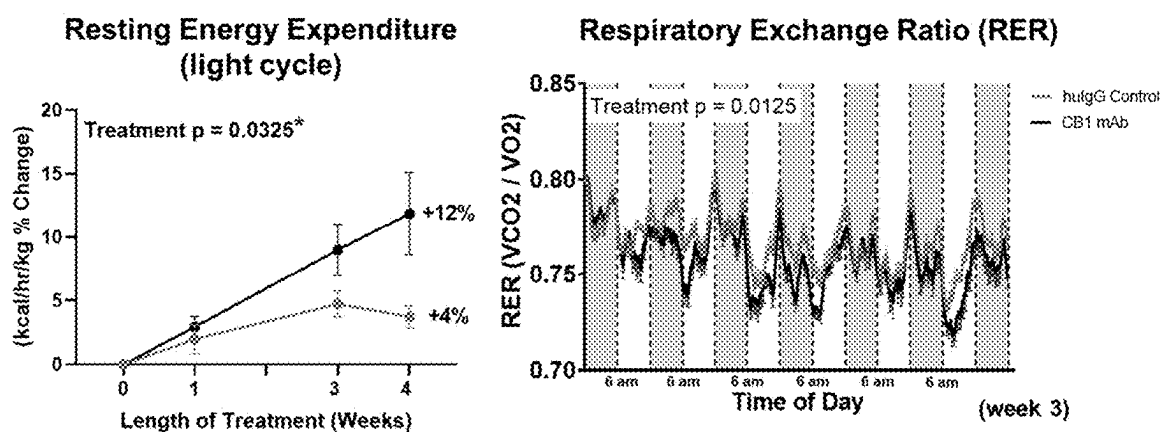
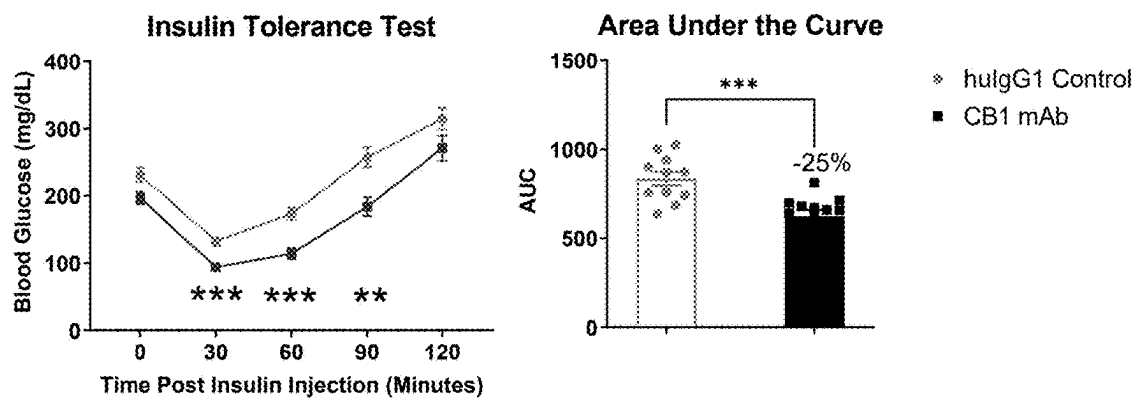


Figure 11B



CANNABINOID TYPE 1 RECEPTOR BINDING PROTEINS AND USES THEREOF

FIELD OF THE INVENTION

[0001] The present disclosure relates to cannabinoid type 1 receptor (CB1) binding proteins and use thereof for the treatment of diseases or disorders.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ELECTRONICALLY

[0002] Incorporated by reference in its entirety herein is an amino acid sequence listing submitted concurrently herewith and identified as follows: 7027 kilobyte XML document named “10129-WO01-SEC Sequence Listing.xml,” created on Jul. 20, 2023.

CROSS-REFERENCE TO RELATED APPLICATIONS

[0003] This application claims the benefit of U.S. Provisional Application No. 63/392,891, filed on Jul. 28, 2022.

BACKGROUND

[0004] The cannabinoid receptor 1 (CB1) is a G-protein coupled receptor (GPCR) that is expressed in the central nervous system (CNS) and peripheral organs such as liver, skeletal muscle, adipose tissue and endocrine pancreas. (Cinar R. et al., *Pharmacology & Therapeutics* 208 (2020) 107477.) CB1 receptors are expressed at a lower level in peripheral organs compared to the CNS and the receptors are involved in regulating metabolic homeostasis in both systems. Due to its role in energy homeostasis, there have been longstanding efforts in developing CB1 receptor antagonists and/or inverse agonists for the treatment of obesity and its comorbidities. (See e.g., Cinar R. et al., *Pharmacology & Therapeutics* 208 (2020) 107477; Thomas Murphy and Bernard Le Foll, *Biomolecules* 2020, 10, 855.)

[0005] In clinical trials, the small molecule CB1 receptor antagonist/inverse agonist rimonabant (also known as SR141716) reduced bodyweight in overweight people and significantly improved multiple cardiometabolic parameters such as waist circumference, hemoglobin A1c, HDL, plasma cholesterol and triglycerides. Rimonabant was approved as an anti-obesity drug in Europe in 2006. However, the CNS-penetrance of rimonabant also induces severe psychiatric side effects such as anxiety, depression, and suicidality, which led to its withdrawal in 2008. Additional efforts have been made in the development of peripherally restricted small molecule CB1 receptor antagonists (e.g., Tam et al, *J. Clin. Invest.* (2010) 120:2953-66; US2011/0144157), but success in this area remains to be seen.

[0006] Because antibodies do not appreciably cross the blood brain barrier and penetrate the CNS, scientists have hypothesized that CB1 antagonist antibodies with rimonabant-like potencies may confer the beneficial effects on metabolism (e.g., treating obesity and its comorbidities) without the CNS-related side effects. However, the discovery and engineering of potent antibody antagonists of CB1 have shown to be difficult in the field. For example, potencies of antagonist CB1 antibodies described in the field are in the double or triple digit nanomolar range (See e.g., WO2014/210025, WO2015/148948 and WO2019/211665); in comparison, the potency of rimonabant is in the single digit nanomolar range. (See e.g., Congy C. et al., *FEBS*

Letters 350 (1994) 240-244; Bauer M. et al., *The Journal of Biological Chemistry*, 287(44) (2016) 36944-36967). In addition, binding affinities of those antibodies to CB1 (e.g., huCB1) are inferior to that of rimonabant.

[0007] Several factors are considered to contribute to this difficulty. It has generally been difficult to raise antibodies against GPCRs with small extracellular loops such as CB1 via immunization (see e.g., Jo Migyeong and Jung Sang Take, *Experimental & Molecular Medicine* (2016) 48, e207). In addition, many CB1 antibodies raised in animal campaigns tend to be non-functional. Further, potencies and/or affinity of antagonist antibodies isolated from animal campaigns are typically not sufficiently high to meet the potency and/or affinity requirement (e.g., comparable to or better than rimonabant). Although affinity maturation can be used to improve the potency of antagonist antibodies, it has been challenging in employing this strategy for CB1 antibodies due to the lack of stable and high-quality preparations of conformationally-relevant CB1 protein.

[0008] There remains a need in the field for identifying CB1 antagonist antibodies with rimonabant-like potencies and affinities that can achieve therapeutic effects while avoid CNS-related adverse effects.

SUMMARY

[0009] The present application relates to antagonist or inverse agonist anti-CB1 (e.g., huCB1) antibodies and antigen binding fragments having high potencies (e.g., potencies comparable to or better than that of rimonabant). The antibodies disclosed herein have much higher potencies compared to previously known anti-CB1 antibodies and are suitable for safe and effective treatment of various diseases such as obesity and its comorbidities, chronic kidney disease, non-alcoholic steatohepatitis (NASH), and nonalcoholic fatty liver disease (NAFLD). Based on the disclosure provided herein, those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. Such equivalents are intended to be encompassed by the following embodiments (E).

[0010] E1: An isolated antibody that binds to huCB1, the antibody comprises a HV and a LV, wherein a) the HV comprises HCDR1, HCDR2 and HCDR3, wherein HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A, S, or G; HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is H, Y, or Q; X4 is E, T, or S; X5 is K, S, or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprises the amino acid sequence of X11YDX12X13X14GX15SYYYYGMDV (SEQ ID NO: 532), wherein X11 is D, E, or N; X12 is A, I, P, T, or V; X13 is F, L, or V; X14 is S or T; and X15 is H, N, or Y; and b) the LV is of a VK1/O2/JK4, a VK2/A19/JK1, or a VK3/A27/JK1 germline.

[0011] E2 The antibody of E1, wherein a) HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A or S; HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is Y or Q; X4 is E, T, or S; X5 is S or Q; X6 is A or K; X7 is N, S, K, or R;

X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprises the amino acid sequence of X11YDX12X13X14GX15SYYYYGMDV (SEQ ID NO: 532), wherein X11 is D, E, or N; X12 is A, I, P, or V; X13 is L or V; X14 is S or T; and X15 is H or Y; and b) the LV is of a VK2/A19/JK1 germline.

[0012] E3 The antibody of E2, wherein X4 is T, X15 is Y.

[0013] E4 The antibody of any one of E1-E3, wherein b) the LV comprises LCDR1, LCDR2 and LCDR3, and wherein LCDR1 comprises the amino acid sequence RSSQSLLX16SX17GX18NYX19D (SEQ ID NO: 533), wherein X16 is H, S, or T; X17 is S, T, or Y; X18 is A, N, I or Y; and X19 is L or V; LCDR2 comprises the amino acid sequence of X20GSNRA (SEQ ID NO: 534), wherein X20 is L or Q; and LCDR3 comprises the amino acid sequence of X21QAX22X23X24PRT (SEQ ID NO: 535), wherein X21 is M or R; X22 is L, I, R, or V; X23 is Q, E, T, A, or G; and X24 is T, I, L, or Q.

[0014] E5 The antibody E4, wherein X24 is L.

[0015] E6. The antibody of E1, wherein a) HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A, G or S; HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is H, Y, or Q; X4 is T or S; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprises the amino acid sequence of X11YDX12X13X14GX15SYYYYGMDV (SEQ ID NO: 532), wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is S or T; and X15 is H, N or Y; and b) the LV is of a VK1/O2/JK4 or a VK3/A27/JK1 germline.

[0016] E7. The antibody of claim E6, wherein a) X1 is A or G; X3 is Q or Y, X4 is T, X8 is F, X14 is T; and X15 is Y.

[0017] E8. The antibody of E6 or E7, wherein b) the LV is of a VK1/O2/JK4 germline.

[0018] E9. The antibody of E6 or E7, wherein b) the LV is of a VK1/O2/JK4 germline and comprises light chain LCDR1, LCDR2 and LCDR3, and wherein LCDR1 comprises the amino acid sequence of RASQSI-SNYLN (SEQ ID NO: 132), RASQSIISYLN (SEQ ID NO: 150), or RASQSISSYLN (SEQ ID NO: 186); LCDR2 comprises the amino acid sequence of AASSLHS (SEQ ID NO: 133) or AASSLRS (SEQ ID NO: 151); and LCDR3 comprises the amino acid sequence of QQYQSYPLT (SEQ ID NO: 134) or QQYSNYPLT (SEQ ID NO: 152); or

[0019] b) the LV is of a VK3/A27/JK1 germline and comprises light chain LCDR1, LCDR2 and LCDR3, and wherein LCDR1 comprises the amino acid sequence of RASQSVSSYL (SEQ ID NO: 168); LCDR2 comprises the amino acid sequence of GASSRAT (SEQ ID NO: 169); and LCDR3 comprises the amino acid sequence of QQYGSSPRT (SEQ ID NO: 170).

[0020] E10. The antibody of E9, wherein LCDR1 comprises the amino acid sequence of RASQSI-SNYLN (SEQ ID NO: 132), LCDR2 comprises the amino acid sequence of AASSLHS (SEQ ID NO: 133), and LCDR3 comprises the amino acid sequence of QQYQSYPLT (SEQ ID NO: 134).

[0021] E11. The antibody of E6, wherein a) HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A or G, HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is Y or Q; X4 is T; X5 is S; X6 is K or N; X7 is S or R; X8 is F; X9 is K; and X10 is G or D; and HCDR3 comprises the amino acid sequence of

X11YDX12X13X14GX15SYYYYGMDV (SEQ ID NO: 532), wherein X11 is N; X12 is T; X13 is L or V; X14 is T; and X15 is Y, and b) the LV is of a VK1/O2/JK4 germline.

[0022] E12. The antibody of E11, wherein a) HCDR1 comprises the amino acid sequence of RGGDYWA (SEQ ID NO: 414) or RGGDYWG (SEQ ID NO: 408); HCDR2 comprises the amino acid sequence of HVYYTGSTKYNPSFKD (SEQ ID NO: 427), HVYYTGSTNYNPRFKD (SEQ ID NO: 445), HIYQTGSTNYNPRFKG (SEQ ID NO: 415), or HVYQTGSTKYNPSFKD (SEQ ID NO: 409); and HCDR3 comprises the amino acid sequence of NYDTLTGYSSYYYGMDV (SEQ ID NO: 410) or NYDTVTGYSSYYYGMDV (SEQ ID NO: 446).

[0023] E13. The antibody of E11 or E12, wherein b) the LV comprises LCDR1, LCDR2 and LCDR3, and wherein LCDR1 comprises the sequence of RASQSI-SYLN (SEQ ID NO: 405); LCDR2 comprises the sequence of X39ARX40LX41S (SEQ ID NO: 536), wherein X39 is N, S, K or G; X40 is R, K, L or A; and X41 is A, G or S; and LCDR3 comprises the sequence of QQX42X43X44X45PX46T (SEQ ID NO: 537), wherein X42 is Y or F, X43 is R, A, S, G, or Y; X44 is S, K, R or H; X45 is S, L, F, Y, P, or M; and X46 is L, I or V.

[0024] E14. The antibody of E13, wherein X42 is Y.

[0025] E15. An isolated antibody that binds to huCB1, the antibody comprises a HV and a LV, wherein a) the HV comprises CDRH1, CDRH2 and CDRH3, wherein CDRH1 comprises the amino acid sequence of RGGDYWS (SEQ ID NO: 1); CDRH2 comprises the amino acid sequence of HX25YX26X27GX28TX29YNPX30X31X32X33 (SEQ ID NO: 538), wherein X25 is I or V; X26 is Y or Q; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L, X32 is K or R; and X33 is G, S, or N; and CDRH3 comprises the amino acid sequence of X34YDX35X36X37GYSSYYYGX38DV (SEQ ID NO: 539), wherein X34 is D or G; X35 is A, I, T, or V; X36 is L or absent; X37 is S or T; and X38 is M or L; and b) the LV is of a VK2/A19/JK1 germline.

[0026] E16. The antibody of E15, wherein a) X26 is Y, X34 is G, X36 is absent, and X37 is S.

[0027] E17. The antibody of E15 or E16, wherein b) the LV comprises LCDR1, LCDR2 and LCDR3, wherein LCDR1 comprises the amino acid sequence of RSSQSLLHRSGYNYLD (SEQ ID NO: 257); LCDR2 comprises the amino acid sequence of LGSNRAS (SEQ ID NO: 258) or QGSNRAS (SEQ ID NO: 264); and LCDR3 comprises the amino acid sequence of MQSLQTPRT (SEQ ID NO: 259), RQSVALPRT (SEQ ID NO: 271), or RQARALPRT (SEQ ID NO: 265).

- [0028]** E18. The antibody of E1-E14, wherein the antibody comprises a HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, or LCDR3, or a set of HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any one of LIBC523596-1, LIBC523603-1, LIBC523661-1, LIBC523670-1, LIBC523748-1, LIBC523760-1, LIBC523797-1, LIBC523813-1, LIBC523815-1, LIBC523844-1, LIBC523857-1, LIBC523862-1, LIBC523868-1, LIBC523969-1, LIBC524049-1, LIBC527906-1, LIBC527814-1, LIBC527997-1, LIBC528141-1, LIBC527912-1, LIBC528116-1, LIBC527879-1, LIBC527919-1, LIBC527984-1, LIBC527968-1, LIBC528169-1, LIBC528131-1, LIBC527827-1, LIBC527869-1, and LIBC528148-1, LIBC680562-1, LIBC680574-1, LIBC680773-1, LIBC680800-1, LIBC680812-1, LIBC681593-1, LIBC681594-1, and LIBC681737-1.
- [0029]** E19. The antibody of E15-E17, wherein the antibody comprises a HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, or LCDR3 or a set of HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any one of LIBC673948-1, LIBC673952-1, LIBC673965-1, LIBC673982-1, LIBC673972-1, LIBC674024-1, LIBC674035-1, LIBC674043-1, LIBC674090-1, LIBC674002-1, LIBC674153-1, LIBC674200-1, LIBC674214-1, LIBC674216-1, LIBC674229-1, LIBC674235-1, LIBC674257-1, and LIBC674276-1.
- [0030]** E20. The antibody of E18 or E19, wherein the antibody comprises a HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, or LCDR3 or a set of HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any one of LIBC680574, LIBC528116, LIBC673952, LIBC523797, LIBC527814, LIBC527997, LIBC527919, LIBC523661 and LIBC524049.
- [0031]** E21. The antibody of any one of E1-E20, wherein the amino acid at position 83 of the HV is N and the amino acid at position 85 of the HV is Y.
- [0032]** E22. The antibody of E21, wherein the amino acid at position 94 of the LV is S.
- [0033]** E23. The antibody of E19, wherein the amino acid at position 83 of the HV is N, the amino acid at position 85 of the HV is Y, and the amino acid at position 79 of the HV is R.
- [0034]** E24. The antibody of E23, the amino acid at position 94 of the LV is S, and the amino acid at position 76 is D.
- [0035]** E25. The antibody of any one of E1-E24, wherein HCDR2 of the antibody comprises 5 or fewer mutations, or 3 or fewer mutations, compared to HVYYTGSTNYPNPFKD (SEQ ID NO: 136).
- [0036]** E26. The antibody of any one of E1-E24, wherein HCDR2 of the antibody comprises 5 or fewer mutations compared to HVYYTGSTKYNPNFKG (SEQ ID NO: 8), and optionally, HCDR3 of the antibody comprises 4 or fewer mutations compared to DYDILTGYSYYYYGMDV (SEQ ID NO: 9).
- [0037]** E27. The antibody of any one of E1-E26, wherein the antibody comprises a HV and/or LV of any one of LIBC523596-1, LIBC523603-1, LIBC523661-1, LIBC523670-1, LIBC523748-1, LIBC523760-1, LIBC523797-1, LIBC523813-1, LIBC523815-1, LIBC523844-1, LIBC523857-1, LIBC523862-1, LIBC523868-1, LIBC523969-1, LIBC524049-1, LIBC527906-1, LIBC527814-1, LIBC527997-1, LIBC528141-1, LIBC527912-1, LIBC528116-1, LIBC527879-1, LIBC527919-1, LIBC527984-1, LIBC527968-1, LIBC528169-1, LIBC528131-1, LIBC527827-1, LIBC527869-1, and LIBC528148-1, LIBC680562-1, LIBC680574-1, LIBC680773-1, LIBC680800-1, LIBC680812-1, LIBC681593-1, LIBC681594-1, and LIBC681737-1.
- [0038]** E28. The antibody of E27, wherein the antibody comprises a HV and/or LV of any one of LIBC680574, LIBC528116, LIBC673952, LIBC523797, LIBC527814, LIBC527997, LIBC527919, LIBC523661 and LIBC524049.
- [0039]** E29. An isolated antibody that binds to huCB1 and comprises a HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, or LCDR3 or a set of HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of LIBC529593, LIBC560340, or LIBC560657.
- [0040]** E30. The antibody of E29, wherein the antibody comprises a HV and/or LV of LIBC529593, LIBC560340, or LIBC560657.
- [0041]** E31. The antibody of any one of E1-E30, wherein the antibody is a monoclonal antibody, optionally, the monoclonal antibody is a chimeric antibody, a humanized antibody, or a human antibody.
- [0042]** E32. The monoclonal antibody of E31, wherein the antibody is an antagonist and/or an inverse agonist antibody of huCB1.
- [0043]** E33. The antibody of E31 or E32, wherein the antibody has a binding affinity for huCB1 that is at least 3 times higher compared to 10D10N35Y.
- [0044]** E34. The antibody of E32 or E33, wherein the antibody has an IC₅₀ of 10 nM or less, such as 5 nM or less, 3 nM or less or 1 nM or less, as determined using a cell-based cAMP assay.
- [0045]** E35. An isolated monoclonal antibody that binds to huCB1, wherein the antibody is an antagonist and/or inverse agonist antibody of huCB1 and has an IC₅₀ of 10 nM or less, or 8 nM or less, as determined using a cell-based cAMP assay.
- [0046]** E36. The antibody of E35, wherein the antibody binds to the extracellular loop 2 of huCB1 or to a region comprised within the extracellular loop 2 of huCB1.
- [0047]** E37. The antibody of any one of E34-E36, wherein the cell-based cAMP assay is a cell-based cAMP assay with CP55,940.
- [0048]** E38. The antibody of any one of E31-E37, wherein the monoclonal antibody is a human IgG1, IgG2, IgG3, or IgG4 antibody, preferably, a human IgG1 antibody.
- [0049]** E39. The antibody of E38, wherein the antibody comprises a mutation at amino acid position N297, such as N297G, according to EU numbering in its heavy chain.
- [0050]** E40. The antibody of E39, wherein the antibody further comprises R292C and V302C mutations according to EU numbering in its heavy chain.

- [0051] E41. The antibody of any one of E37-E40, wherein the antibody comprises mutations at amino acid positions M252, S254, and T256, preferably M252Y, S254T, and T256E, according to EU numbering in its heavy chain.
- [0052] E42. A method for treating a subject in need of antagonizing or inverse agonizing the CB1 receptor, the method comprises administering the antibody of any one of E1-E41 to the subject.
- [0053] E43. A method for treating a disease or disorder in a subject responsive to antagonizing or inverse agonizing the CB1 receptor, the method comprises administering to the subject the antibody of any one of E1-E41.
- [0054] E44. The method of E42 or E43, wherein the administration results in one or more of reduced body weight, reduced appetite, improved metabolic parameters, reduced blood glucose levels, reduced insulin levels, reduced triglyceride levels, reduced kidney injury, reduced kidney fibrosis, reduced kidney inflammation, and improved kidney function.
- [0055] E45. The method of E43, wherein the disease or disorder is selected from obesity, diabetes, dyslipidemia, metabolic diseases, liver disease, fibrosis, non-alcoholic steatohepatitis (NASH), primary biliary cirrhosis, renal disease, kidney fibrosis, chronic kidney disease, IgA nephropathy, osteoporosis, atherosclerosis, cardiovascular disease, cancer, and inflammatory disease.
- [0056] E46. The antibody of any one of E1-E41 for use in the treatment of a disease or disorder in a subject, wherein the disease or disorder is selected from obesity, diabetes, dyslipidemia, metabolic diseases, fibrosis, liver disease, NASH, primary biliary cirrhosis, renal disease, kidney fibrosis, chronic kidney disease, IgA nephropathy, osteoporosis, atherosclerosis, cardiovascular disease, cancer, and inflammatory disease.
- [0057] E47. The antibody according to E46, wherein the use results in one or more of reduced body weight, reduced appetite, improved metabolic parameters, reduced blood glucose levels, reduced insulin levels, reduced triglyceride levels, reduced kidney injury, reduced kidney fibrosis, reduced kidney inflammation, and improved kidney function.
- [0058] E48. The method of E45 or the antibody of E46, wherein the disease or disorder is obesity, optionally, the subject has a BMI of at least 27 kg/m² or the subject has a BMI of at least 27 kg/m².
- [0059] E49. The method of any one of E42-45 or the antibody of E47 or E48, wherein the subject is a human.
- [0060] E50. An isolated polynucleotide that encodes the antibody of any one of E1-E41.
- [0061] E51. An expression vector comprising the polynucleotide of E50.
- [0062] E52. A host cell comprising the expression vector of E51.
- [0063] E53. The antibody of E5 or E15, wherein the amino acid at position 32 according to AHO numbering in the heavy chain is an R.

BRIEF DESCRIPTION OF THE FIGURES

[0064] FIGS. 1A and 1B show the binding profile of top affinity-matured cognate variants.

[0065] FIGS. 2A and 2B show the binding profile of top affinity-matured LC-swapped variants.

[0066] FIG. 3A shows HV CDR and FR3 sequence alignment of top cognate affinity-matured variants. FIG. 3B shows LV CDR and FR3 sequence alignment of top cognate affinity-matured variants.

[0067] FIG. 4A shows HV CDR and FR3 sequence alignment of top LC-swapped affinity-matured variants. FIG. 4B shows LV CDR sequence alignment of top LC-swapped affinity-matured variants.

[0068] FIG. 5A shows antagonist of exemplary affinity-matured antibodies. FIG. 5B shows inverse agonist activity of exemplary affinity-matured antibodies.

[0069] FIGS. 6A and 6B show reduction of body weight and food intake by an exemplary affinity-matured antibody in obese human CB1 knock in mice. FIG. 6C shows reduction of fat mass, liver triglyceride and insulin levels of the antibody. FIG. 6D shows impact on kidney injury, fibrosis and inflammatory markers of an exemplary affinity-matured antibody.

[0070] FIG. 7 shows HV and LV CDR sequence alignment of top further LC-swapped affinity-matured variants.

[0071] FIG. 8A shows HV CDR and FR3 sequence alignment of top further cognate affinity-matured variants. FIG. 8B shows LV CDR sequence alignment of top further cognate affinity-matured variants.

[0072] FIG. 9 shows antagonist activity of antibodies isolated from mouse antibody campaigns.

[0073] FIG. 10A shows reduction of body weight by exemplary further affinity-matured antibodies in obese human CB1 knock in mice. FIG. 10B shows the impact on food intake of the antibodies. FIG. 10C shows the impact on fat mass of the antibodies and FIG. 10D shows the impact on liver weight of the antibodies.

[0074] FIG. 11A shows treatment with an exemplary anti-CB1 antibody increases energy expenditure and decreases respiratory exchange ratio in an animal model (mice). FIG. 11B shows treatment with an exemplary anti-CB1 antibody increases insulin sensitivity in an animal model.

DETAILED DESCRIPTION

[0075] Cannabinoid receptor 1 (CB1) is a G-protein coupled receptor for endogenous cannabinoids such as 2-arachidonoylglycerol (2-AG) and the receptor mediates many cannabinoid-induced effects (e.g., food intake). Activation of the receptor leads to a decrease of intracellular cyclic AMP (cAMP) concentration and an increase of mitogen-activated protein kinase (MAP kinase) concentration. (Gerard C. et al., *Biochem. J.* 279:129-134(1991)). The human CB1 (huCB1) receptor is a polypeptide of 472 amino acids (UniProtKB/Swiss-Prot: P21554) and is encoded by the CNR1 gene. The amino acid sequence of the huCB1 is listed below (SEQ ID NO: 559). huCB1 is a membrane protein that has four extracellular regions (amino acid residues 1-116, 176-187, 256-273, and 366-377 of SEQ ID NO: 559), four cytoplasmic regions (amino acid residues 143-154, 213-232, 300-344, and 400-472 of SEQ ID NO: 559) and seven transmembrane domains (amino acid residues 117-142, 155-175, 188-212, 233-255, 274-299, 345-365, and 378-399 of SEQ ID NO: 559).

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MKSILDGLAD TTFRTITDDL LYVGSNDIQY EDIKGDMASK LGYFPQKFPL
      60          70          80          90         100

TSFRGSPFQE KMTAGDNPQL VPADQVNITE FYNKSLSEK ENEENIQCGE
      110        120        130        140        150

NFMIDIECFMV LNPSQQLAIA VLSLTLTGFT VLENLLVLCV ILHSRSLRCR
      160        170        180        190        200

PSYHFIGSLA VADLLGSVIF VYSFIDFHV FHRKDSRNVFL FKLGGVTASF
      210        220        230        240        250

TASVGSGLFLT AIDRYISIH RPLAYKRIVTR PKAVVAFCLM WTIAIVIAVL
      260        270        280        290        300

PLLGWNCEKL QSVCSDFPH IDETYLMEFWI GVTSVLLLFV VYAYMYILWK
      310        320        330        340        350

AHSHAVRMIQ RGTQKSIIH TSEDGKVQVT RPDQARMDIR LAKTIVLILV
      360        370        380        390        400

VLIICWGPLL AIMVYDVFGK MNKLIKTVFA FCSMLCLLNS TVNPIIYALR
      410        420        430        440        450

SKDLRHAFPS MFPSCGTAQ PLDNSMGDSD CLHKHANNA SVHRAAESCI
      460        470

KSTVKIAKVT MSVSTDTSAE AL

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[0076] CB1 receptors are expressed in the CNS and peripheral systems and are involved in regulating energy homeostasis. The receptor has been a target for the development of treatments for obesity and its comorbidities. In view of the experiences with rimonabant, the field has focused on developing peripherally acting CB1 receptor antagonists or inverse agonists to confer the beneficial effects on metabolism without the CNS-related side effects.

[0077] Antibodies against the CB1 receptor can serve as peripherally acting antagonists or inverse agonists because they do not appreciably penetrate the CNS. Efforts have been made to identify antagonist or inverse agonist anti-CB1 antibodies that have rimonabant-like potency to achieve therapeutic effects without the side effect profile. The 10D10 antibody was a lead antagonist anti-CB1 antibody isolated from Xenomouse antibody campaigns but its potency was significantly lower than that of rimonabant. (WO 2014/210205, the disclosure of which is incorporated by reference in its entirety). Multiple rounds of affinity maturation were carried out to improve the potency of the antibody. Although several 10D10 affinity matured variants isolated from the earlier affinity maturation campaigns had improved potency compared to 10D10, none of the variants met the design goal of rimonabant-like potency. The lack of stable, high-quality preparations of conformationally-relevant CB1 protein was believed to have contributed to the inability in isolating anti-CB1 antibodies with rimonabant-like potency. In addition, the 10D10 antibody and its earlier affinity matured variants had suboptimal manufacturability (e.g., low expression levels and instability).

[0078] The evolution of technical sophistication in generating high-quality, soluble preparations of complex membrane proteins culminated with the ability to formulate huCB1 into nanodisc (ND) or styrene maleic acid lipoparticles (SMALP) (Luna V. et al., European Polymer Journal 109 (2018) 483-488; Lavington S. and Watts A, Biophysical Reviews, vol. 12 (2020)1287-1302). In contrast to detergent-solubilized cell lysates prepared from huCB1-express-

ing cells, both CB1-ND and -SMALP are purified reagents that are stable after multiple freeze/thaw cycles and compatible with long-term storage. Furthermore, they can be added to binding experiments at user-defined concentrations of huCB1. The availability of CB1-ND and -SMALP enabled the isolation of high potency antagonist or inverse agonist anti-CB1 antibodies disclosed herein.

[0079] The present application relates to antagonist or inverse agonist anti-CB1 antibodies and antigen binding fragments, in particular anti-huCB1 antibodies and antigen binding fragments, with potencies that are comparable to or better than that of rimonabant. Antibodies that inhibit CB1 cause an increase of cAMP, which can be measured using in vitro cAMP assays such as a cell-based cAMP assay. As exemplified in the examples, potencies of the antibodies and antigen binding fragments disclosed herein are in the single digit nanomolar range, e.g., they have an IC₅₀ of less than about 10 nM in a cell-based cAMP assay (see Examples 5, 8 and 9). As shown in the examples, the high potency antibodies and antigen binding fragments disclosed herein reduced body weight, food intake and multiple metabolic parameters in obese animal models (e.g., reduce body fat, insulin level and liver triglyceride levels, Example 7), and increased energy expenditure, lipid oxidation and insulin sensitivity in animal models while had minimal brain exposure, indicating that they can enable safe and effective therapeutic treatment of various diseases and disorders (e.g., obesity and its comorbidities, chronic kidney disease, NAFLD, NASH). In contrast, antagonist or inverse agonist antibodies with potency values lower than that of rimonabant are not considered to be adequate for such therapeutic purposes. Furthermore, the anti-CB1 antibodies and antigen binding fragments disclosed herein are stable and express well for manufacture purpose. These high potency anti-CB1 antibodies and antigen binding fragments are described and exemplified herein.

Definitions

[0080] The present invention provides antagonist or inverse agonist antibodies against human CB1. An “anti-

body” is a protein that comprises an antigen-binding fragment that specifically binds to an antigen, and a scaffold or framework portion that allows the antigen-binding fragment to adopt a conformation that promotes binding of the antibody to the antigen. As used herein, the term “antibody” generally refers to a tetrameric immunoglobulin protein comprising two light chain polypeptides (about 25 kDa each) and two heavy chain polypeptides (about 50-70 kDa each). The term “light chain” or “immunoglobulin light chain” refers to a polypeptide comprising, from amino terminus to carboxyl terminus, a single immunoglobulin light chain variable region (VL or LV) and a single immunoglobulin light chain constant domain (CL or LC). The immunoglobulin light chain constant domain (CL) can be a human kappa (κ) or human lambda (λ) constant domain. The term “heavy chain” or “immunoglobulin heavy chain” refers to a polypeptide comprising, from amino terminus to carboxyl terminus, a single immunoglobulin heavy chain variable region (VH or HV), an immunoglobulin heavy chain constant domain 1 (CH1), an immunoglobulin hinge region, an immunoglobulin heavy chain constant domain 2 (CH2), an immunoglobulin heavy chain constant domain 3 (CH3), and optionally an immunoglobulin heavy chain constant domain 4 (CH4). Heavy chains are classified as mu (μ), delta (δ), gamma (γ), alpha (α), and epsilon (ϵ), and define the antibody’s isotype as IgM, IgD, IgG, IgA, and IgE, respectively. The IgG-class and IgA-class antibodies are further divided into subclasses, namely, IgG1, IgG2, IgG3, and IgG4, and IgA1 and IgA2, respectively. The heavy chains in IgG, IgA, and IgD antibodies have three domains (CH1, CH2, and CH3), whereas the heavy chains in IgM and IgE antibodies have four domains (CH1, CH2, CH3, and CH4). The immunoglobulin heavy chain constant domains can be from any immunoglobulin isotype, including subtypes. The antibody chains are linked together via inter-polypeptide disulfide bonds between the CL domain and the CH1 domain (i.e., between the light and heavy chain) and between the hinge regions of the two antibody heavy chains.

[0081] In a human antibody, CH1 means a region having the amino acid sequence at positions 118 to 215 of the EU index or EU numbering system, which is based on the sequential numbering of the first human IgG1 sequenced (i.e., the “EU antibody”) (Edelman et al., *Proc Natl Acad Sci USA*, 63(1): 78-85 (1969)). A highly flexible amino acid region called a “hinge region” exists between CH1 and CH2. CH2 represents a region having the amino acid sequence at positions 231 to 340 of the EU index, and CH3 represents a region having the amino acid sequence at positions 341 to 446 of the EU index.

[0082] “CL” represents a constant region of a light chain. In the case of a κ chain of a human antibody, CL represents a region having the amino acid sequence at positions 108 to 214 of the EU index. In a λ chain, CL represents a region having the amino acid sequence at positions 108 to 215.

[0083] A “variable domain” refers to the variable region of the antibody light chain (VL or LV) or the variable region of the antibody heavy chain (VH or HV), either alone or in combination. As known in the art, the variable regions of the heavy and light chains each consist of four framework regions (FR) connected by three complementarity determining regions (CDRs), and contribute to the formation of the antigen-binding site of antibodies. From N-terminus to C-terminus, naturally-occurring light and heavy chain vari-

able regions both typically conform with the following order of these elements: FR1, CDR1, FR2, CDR2, FR3, CDR3 and FR4.

[0084] The “Complementarity Determining Regions” (CDRs) of an antibody can be defined according to Kabat, Chothia, the accumulation of both Kabat and Chothia, AbM, contact, North, and/or conformational definitions or any method of CDR determination well known in the art. See, e.g., Kabat et al., 1991, *Sequences of Proteins of Immunological Interest*, 5th ed. (hypervariable regions); Chothia et al., 1989, *Nature* 342:877-883 (structural loop structures). AbM definition of CDRs is a compromise between Kabat and Chothia and uses Oxford Molecular’s AbM antibody modeling software (Accelrys®). The identity of the amino acid residues in a particular antibody that make up a CDR can be determined using methods well known in the art.

[0085] The term “antigen-binding fragment” refers to a molecule that derived from an antibody and retains the ability to specifically bind to an antigen (preferably with substantially the same binding affinity). Examples of an antigen-binding fragment includes (i) a Fab fragment, a monovalent fragment consisting of the VL, VH, CL and CH1 domains; (ii) a F(ab')₂ fragment, a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region; (iii) a Fd fragment consisting of the VH and CH1 domains; (iv) a Fv fragment consisting of the VL and VH domains of a single arm of an antibody, and (v) a dAb fragment (Ward et al., 1989 *Nature* 341:544-546), which consists of a VH domain. Furthermore, although the two domains of the Fv fragment, VL and VH, are coded for by separate genes, they can be joined, using recombinant methods, by a synthetic linker that enables them to be made as a single protein chain in which the VL and VH regions pair to form monovalent molecules (known as single chain Fv (scFv); see e.g., Bird et al. *Science* 242:423-426 (1988) and Huston et al., 1988, *Proc. Natl. Acad. Sci. USA* 85:5879-5883).

[0086] “About” or “approximately,” when used in connection with a measurable numerical variable, refers to the indicated value of the variable and to all values of the variable that are within the experimental error of the indicated value (e.g. within the 95% confidence interval for the mean) or $\pm 10\%$ of the indicated value, whichever is greater. Numeric ranges are inclusive of the numbers defining the range.

[0087] The term “isolated molecule” (where the molecule is, for example, a polypeptide, a polynucleotide, an antibody, or antigen-binding fragment) is a molecule that by virtue of its origin or source of derivation (1) is not associated with naturally associated components that accompany it in its native state, (2) is substantially free of other molecules from the same species (3) is expressed by a cell from a different species, or (4) does not occur in nature. Thus, a molecule that is chemically synthesized, or expressed in a cellular system different from the cell from which it naturally originates, will be “isolated” from its naturally associated components. A molecule also may be rendered substantially free of naturally associated components by isolation, using purification techniques well known in the art. Molecule purity or homogeneity may be assayed by a number of means well known in the art. For example, the purity of a polypeptide sample may be assayed using polyacrylamide gel electrophoresis and staining of the gel to visualize the polypeptide using techniques well known in the art. For

certain purposes, higher resolution may be provided by using HPLC or other means well known in the art for purification.

[0088] Unless indicated otherwise, throughout the present specification and claims, the numbering of the amino acid residues in an immunoglobulin heavy chain or light chain is according to the AHO numbering scheme as described in Honegger and Pluckthun, J. Mol. Biol. 309(3):657-670; 2001. In some embodiments, the EU numbering scheme as described in Edelman et al., Proc. Natl. Acad. USA, Vol. 63: 78-85 (1969) is used when referring to the position of an amino acid with an immunoglobulin constant region. Both numbering schemes are well known in the art.

[0089] An “antagonist” refers to an agent that binds to a protein (e.g., a receptor) and inter alia, partially or totally inhibits, blocks, decreases, prevents, delays, inactivates, desensitizes, or down regulates the signaling activity of a ligand of the protein (e.g., agonist of the receptor). An inverse agonist is an agent that causes the opposite effect of an agonist, for example, it decreases basal activity of a receptor.

Anti-CB1 Antibodies and Antigen Binding Fragments

[0090] Disclosed herein are antibodies and antigen binding fragments against huCB1 (SEQ ID NO: 559) and are antagonist and/or inverse agonist of huCB1. Potencies of the antibodies and antigen binding fragments are comparable to or better than that of rimonabant, which can enable them to achieve therapeutic effects (e.g., treating obesity, chronic kidney disease, liver disease) without CNS-related side effects. In some embodiments, the antibodies and antigen binding fragments have an IC50 in the single digit nanomolar range as measured by a cell-based cAMP assay. For example, in some embodiments, the IC50 of the antibodies and antigen binding fragments is less than about 10 nM as measured by a cell-based cAMP assay. In some embodiments, the IC50 of the antibodies and antigen binding fragments is less than about 5 nM as measured by a cell-based cAMP assay. In some embodiments, the IC50 of the antibodies and antigen binding fragments is less than about 3 nM or less than about 1 nM as measured by a cell-based cAMP assay.

[0091] Anti-CB1 antibodies and antigen binding fragments disclosed herein were identified from affinity maturation of 10D10 (e.g., Examples 1-4 and 9). The 10D10 antibody was isolated from previous antibody campaigns (WO2014/210205) and the heavy and light chain CDR sequences and heavy and light chain variable region sequences of 10D10 are listed in Tables 7 and 8. The yeast display library designs for affinity maturation were guided by next generation sequencing wherein amino acids at specific HV and LV positions of 10D10 were optimized to improve binding affinity and potency. The 10D10 LC belongs to the VK2 germline and as disclosed herein, only the HC of the antibody binds to the CB1 receptor (Examples 1 and 14). LC-swapped affinity matured antibodies (i.e., swapping the 10D10 LC with light chains of VK1 or VK3 germline that have improved manufacturability) were also isolated. Heavy and light chain consensus CDR sequences of the affinity matured antibodies and antigen binding fragments are summarized in Table 35. Thus, in one aspect, disclosed herein is an antibody or antigen binding fragment

thereof that binds to huCB1 and comprises a heavy chain variable region (HV) and a light chain variable region (LV), wherein

[0092] a) the HV comprises HCDR1, HCDR2 and HCDR3, wherein HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A, S, or G; HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is H, Y, or Q; X4 is E, T, or S; X5 is K, S, or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprises the amino acid sequence of X11YDX12X13X14GX15SYYYYGMVDV (SEQ ID NO: 532), wherein X11 is D, E, or N; X12 is A, I, P, T, or V; X13 is F, L, or V; X14 is S or T; and X15 is H, N, or Y; and

[0093] b) the LV is of or derived from a VK1/O2/JK4, a VK2/A19/JK1, or a VK3/A27/JK1 germline.

[0094] As used herein, a variable region sequence (such as a variable region or CDR amino acid sequence) “of a germline” or “derived from a germline” refers to that sequence having structural features that are characteristic of the specified germline, rather than another germline. In order for a host organism to generate an antibody that binds to an antigen, rearranged immunoglobulin germline genes (of a B cell of the host organism) must initially encode an initial antibody with a three-dimensional structure and biochemical properties capable of interacting with that antigen. Each host organism has a defined repertoire of immunoglobulin germline genes that may be rearranged to produce initial antibodies. Through the process of somatic hypermutation, the host organism can select for a mutation or collection of mutations in the genes encoding the antibody that further enhance its interaction with the antigen. Accordingly, an antibody sequence “of” or “derived from” a specified germline may have variable region sequences that contain differences compared to that germline. However, the antibody sequence will contain structural features, such as particular amino acids, CDR lengths and/or conformation characteristic of a specified germline, indicating that the specified germline is the closest germline for that sequence to be associated with. This interaction can be analogized to a hand (antigen) being fitted by a glove (germline antibody). The initial structures defined by the germline genes must be sufficient for the glove to fit the hand in the first place, which may then be tailored to fit with even more precision. Since the specific combination of rearranged immunoglobulin germline genes that encoded the antibody must have specified a three-dimensional structure (glove) that interacted with the antigen (hand), the germline identity of an antibody will be understood to define structural features of paratope that interact with the antigen. As light chain of the CB1 antibodies disclosed herein does not make direct contact with the antigen, the light chain germline identity will be understood to define structural features of the light chains for pairing with the heavy chains for paratope interaction with the antigen.

[0095] Repertoires of germline genes that are typically utilized for generating antibodies, including *Homo sapiens* and *Mus musculus* are sequences, and are known and may be accessed, for example, from IMGT/GENE-DB (Giudicelli V., et al. “IMGT/GENE-DB: a comprehensive database for human and mouse immunoglobulin and T cell receptor

genes.” *Nucleic Acids Res.*, 33: D256-D261 (2005)). A sequence of an antibody may be aligned against germline gene repertoires, for example using an algorithm such as Basic Local Alignment Search Tool (BLAST) under default settings, to determine which host organism germline gene segment (or segments) that antibody is of.

[0096] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A or S; HCDR2 comprising the sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is Y or Q; X4 is E, T, or S; X5 is S or Q; X6 is A or K; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N, and HCDR3 comprising the sequence of X11YDX12X13X14GX15SYYYGMDV (SEQ ID NO: 532), wherein X11 is D, E, or N; X12 is A, I, P, or V; X13 is L or V; X14 is S or T; and X15 is H or Y. In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or S; HCDR2 comprising the sequence SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is E, T, or S; X5 is S or Q; X6 is A or K; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N, and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D, E, or N; X12 is A, I, P, or V; X13 is L or V; X14 is T; and X15 is Y. In various embodiments, wherein the HCDR1, HCDR2 and HCDR3 comprise the sequence defined above, the LV is of or derived from a VK1/O2/JK4, a VK2/A19/JK1, or a VK3/A27/JK1 germline. In some embodiments, the LV is of or derived from a VK2/A19/JK1 germline.

[0097] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or S; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is E, T, or S; X5 is S or Q; X6 is A or K; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N, and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D, E, or N; X12 is A, I, P, or V; X13 is L or V; X14 is S or T; and X15 is H or Y, and b) the LV comprises LCDR2 and LCDR3, wherein LCDR1 comprises the amino acid sequence RSSQSLX16SX17GX18NYX19D (SEQ ID NO: 533), wherein X16 is H, S, or T; X17 is S, T, or Y; X18 is A, N, I or Y; and X19 is L or V; LCDR2 comprises the amino acid sequence of X20GSNRA (SEQ ID NO: 534), wherein X20 is L or Q; and LCDR3 comprises the amino acid sequence of X21QAX22X23X24PRT (SEQ ID NO: 535), wherein X21 is M or R; X22 is L, I, R, or V; X23 is Q, E, T, A, or G; and X24 is T, I, L, or Q.

[0098] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or S; HCDR2 comprising the sequence SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is E, T, or S; X5 is S or Q; X6 is A or K; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N, and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D, E, or N; X12 is A, I, P, or V; X13 is L or V; X14 is T; and X15 is Y, and b) LCDR1 comprises the sequence SEQ ID NO: 533, wherein X16 is H, S, or T; X17 is S, T, or Y; X18 is A, N, I or Y; and X19 is L or V; LCDR2 comprises the sequence of SEQ ID NO: 534, wherein X20 is L or Q; and LCDR3

comprises the sequence of SEQ ID NO: 535, wherein X21 is M or R; X22 is L, I, R, or V; X23 is Q, E, T, A, or G; and X24 is T, I, L, or Q.

[0099] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or S; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is E, T, or S; X5 is S or Q; X6 is A or K; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N, and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D, E, or N; X12 is A, I, P, or V; X13 is L or V; X14 is S or T; and X15 is H or Y, and b) LCDR1 comprises the sequence SEQ ID NO: 533, wherein X16 is H, S, or T; X17 is S, T, or Y; X18 is A, N, I or Y; and X19 is L or V; LCDR2 comprises the sequence of SEQ ID NO: 534, wherein X20 is L or Q; and LCDR3 comprises the sequence of SEQ ID NO: 535, wherein X21 is M or R; X22 is L, I, R, or V; X23 is Q, E, T, A, or G; and X24 is L.

[0100] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or S; HCDR2 comprising the sequence SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is E, T, or S; X5 is S or Q; X6 is A or K; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N, and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D, E, or N; X12 is A, I, P, or V; X13 is L or V; X14 is T; and X15 is Y, and b) LCDR1 comprises the amino acid sequence SEQ ID NO: 533, wherein X16 is H, S, or T; X17 is S, T, or Y; X18 is A, N, I or Y; and X19 is L or V; LCDR2 comprises the amino acid sequence of SEQ ID NO: 534, wherein X20 is L or Q; and LCDR3 comprises the amino acid sequence of SEQ ID NO: 535, wherein X21 is M or R; X22 is L, I, R, or V; X23 is Q, E, T, A, or G; and X24 is L.

[0101] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A, G or S; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is H, Y, or Q; X4 is T or S; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N, and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is S or T; and X15 is H, N or Y. In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or G; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is T; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is T; and X15 is Y. In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or G; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is T; X5 is S; X6 is K or N; X7 is S or R; X8 is F; X9 is K; and X10 is G or D; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is N; X12 is T; X13 is L or V; X14 is T; and X15 is Y. In various embodiments, wherein the HCDR1, HCDR2 and HCDR3 comprise the sequence defined above, the LV is of or derived from a VK1/O2/JK4 or a VK3/A27/

JK1 germline. In some embodiments, the LV is of or derived from a VK1/O2/JK4 germline.

[0102] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A, G or S; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is H, Y, or Q; X4 is T or S; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is S or T; and X15 is H, N or Y, and b) the LV is of or derived from a VK1/O2/JK4 or a VK3/A27/JK1 germline, preferably, the LV is of or derived from a VK1/O2/JK4 germline.

[0103] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or G; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is T; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is T; and X15 is Y, and b) the LV is of or derived from a VK1/O2/JK4 germline.

[0104] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A, G or S; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is H, Y, or Q; X4 is T or S; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is S or T; and X15 is H, N or Y, and b) the LV is of or derived from a VK1/O2/JK4 and comprises LCDR1, LCDR2, and LCDR3, wherein LCDR1 comprises the amino acid sequence of RASQSIISNYLN (SEQ ID NO: 132), RASQSIISYLN (SEQ ID NO: 150), or RASQSISSYLN (SEQ ID NO: 186); LCDR2 comprises the amino acid sequence of AASSLHS (SEQ ID NO: 133) or AASSLRS (SEQ ID NO: 151); and LCDR3 comprises the amino acid sequence of QQYQSYPLT (SEQ ID NO: 134) or QQYSNYPLT (SEQ ID NO: 152); or the VL is of a VK3/A27/JK1 germline and comprises light chain LCDR1, LCDR2 and LCDR3, and wherein LCDR1 comprises the amino acid sequence of RASQSVSSYLG (SEQ ID NO: 168); LCDR2 comprises the amino acid sequence of GASSRAT (SEQ ID NO: 169); and LCDR3 comprises the amino acid sequence of QQYGSSPRT (SEQ ID NO: 170).

[0105] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A, G or S; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is H, Y, or Q; X4 is T or S; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is S or T; and X15 is H, N or Y, and b) LCDR1 comprising the sequence of RASQSIISNYLN (SEQ ID NO: 132), RASQSIISYLN (SEQ ID NO: 150), or RASQSISSYLN (SEQ ID NO: 186); LCDR2 comprising the sequence of AASSLHS (SEQ ID NO: 133) or AASSLRS (SEQ ID NO: 151); and LCDR3 comprising the sequence of QQYQSY-

PLT (SEQ ID NO: 134) or QQYSNYPLT (SEQ ID NO: 152); preferably, b) LCDR1 comprising the sequence of RASQSIISNYLN (SEQ ID NO: 132); LCDR2 comprising the sequence of AASSLHS (SEQ ID NO: 133); and LCDR3 comprising the sequence of QQYQSYPLT (SEQ ID NO: 134).

[0106] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or G; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is T; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F; X9 is E or K; and X10 is G, D, S, or N; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is T; and X15 is Y, and b) the LV is of or derived from a VK1/O2/JK4 germline and comprises LCDR1, LCDR2, and LCDR3, wherein LCDR1 comprises the amino acid sequence of RASQSIISNYLN (SEQ ID NO: 132), RASQSIISYLN (SEQ ID NO: 150), or RASQSISSYLN (SEQ ID NO: 186); LCDR2 comprises the amino acid sequence of AASSLHS (SEQ ID NO: 133) or AASSLRS (SEQ ID NO: 151); and LCDR3 comprises the amino acid sequence of QQYQSYPLT (SEQ ID NO: 134) or QQYSNYPLT (SEQ ID NO: 152); preferably, b) LCDR1 comprising the sequence of RASQSIISNYLN (SEQ ID NO: 132); LCDR2 comprising the sequence of AASSLHS (SEQ ID NO: 133); and LCDR3 comprising the sequence of QQYQSYPLT (SEQ ID NO: 134).

[0107] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or G; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is T; X5 is S; X6 is K or N; X7 is S or R; X8 is F; X9 is K; and X10 is G or D; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is N; X12 is T; X13 is L or V; X14 is T; and X15 is Y, and b) the LV is of or derived from a VK1/O2/JK4 germline. In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the amino acid sequence of RGGDYWA (SEQ ID NO: 414) or RGGDYWG (SEQ ID NO: 408); HCDR2 comprising the amino acid sequence of HVYYTGSTKYNPSFKD (SEQ ID NO: 427), HVYYTGSTNYPNPFKD (SEQ ID NO: 445), HIYQTGSTNYPNPFKG (SEQ ID NO: 415), or HVYQTGSTKYNPSFKD (SEQ ID NO: 409); and HCDR3 comprising the amino acid sequence of NYDTLTGYSYYYYGMDV (SEQ ID NO: 410) or NYDTVTGYSYYYYGMDV (SEQ ID NO: 446, and b) the LV is of or derived from a VK1/O2/JK4 germline.

[0108] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the sequence of SEQ ID NO: 530, wherein X1 is A or G; HCDR2 comprising the sequence of SEQ ID NO: 531, wherein X2 is I or V; X3 is Y or Q; X4 is T; X5 is S; X6 is K or N; X7 is S or R; X8 is F; X9 is K; and X10 is G or D; and HCDR3 comprising the sequence of SEQ ID NO: 532, wherein X11 is N; X12 is T; X13 is L or V; X14 is T; and X15 is Y, and b) the LV is of or derived from a VK1/O2/JK4 germline comprising LCDR1, LCDR2 and LCDR3, wherein LCDR1 comprises the sequence of RASQSISSYLN (SEQ ID NO: 405); LCDR2 comprises the sequence of X39ARX40LX41S (SEQ ID NO: 536), wherein X39 is N, S, K or G; X40 is R, K, L or A; and X41 is A, G or S; and LCDR3 comprises the sequence of

QQX42X43X44X45PX46T (SEQ ID NO: 537), wherein X42 is Y or F, X43 is R, A, S, G, or Y; X44 is S, K, R or H; X45 is S, L, F, Y, P, or M; and X46 is L, I or V. In some embodiments, X42 is Y and X1-X15, X39-X41, and X43-X44 are as defined above.

[0109] In some embodiments, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the amino acid sequence of RGGDYWA (SEQ ID NO: 414) or RGGDYWG (SEQ ID NO: 408); HCDR2 comprising the amino acid sequence of HVYYTGSTKYNPSFKD (SEQ ID NO: 427), HVYYTGSTNYPNPRFKD (SEQ ID NO: 445), HIYQTGSTNYPNPRFKG (SEQ ID NO: 415), or HVYQTGSTKYNPSFKD (SEQ ID NO: 409); and HCDR3 comprising the amino acid sequence of NYDTLTGYSYYYYGMDV (SEQ ID NO: 410) or NYDTVTGYSYYYYGMDV (SEQ ID NO: 446, and b) LCDR1 comprises the sequence of RASQSISSYLN (SEQ ID NO: 405); LCDR2 comprises the sequence of X39ARX40LX41S (SEQ ID NO: 536), wherein X39 is N, S, K or G; X40 is R, K, L or A; and X41 is A, G or S; and LCDR3 comprises the sequence of QQX42X43X44X45PX46T (SEQ ID NO: 537), wherein X42 is Y or F, X43 is R, A, S, G, or Y; X44 is S, K, R or H; X45 is S, L, F, Y, P, or M; and X46 is L, I or V. In some embodiments, X42 is Y and X39-X41 and X43-X44 are as defined above.

[0110] In certain embodiments, the anti-CB1 antibodies or antigen binding fragments disclosed herein comprise a heavy chain variable region comprising a HCDR1, HCDR2, and HCDR3, and a light chain variable region comprising a LCDR1, LCDR2, and LCDR3 from any of the anti-CB1 antibodies described herein. Thus, in some embodiments, described herein is an anti-CB1 antibody or antigen binding fragment thereof comprising a HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, or LCDR3 of any one of LIBC523596-1, LIBC527906-1, LIBC523603-1, LIBC527814-1, LIBC523661-1, LIBC527997-1, LIBC523670-1, LIBC528141-1, LIBC523748-1, LIBC527912-1, LIBC523760-1, LIBC528116-1, LIBC523797-1, LIBC527879-1, LIBC523813-1, LIBC527919-1, LIBC523815-1, LIBC527984-1, LIBC523844-1, LIBC527968-1, LIBC523857-1, LIBC528169-1, LIBC523862-1, LIBC528131-1, LIBC523868-1, LIBC527827-1, LIBC523969-1, LIBC527869-1, LIBC524049-1, LIBC528148-1, LIBC680562-1, LIBC680574-1, LIBC680773-1, LIBC680800-1, LIBC680812-1, LIBC681593-1, LIBC681594-1, and LIBC681737-1. In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HCDR1, HCDR2, and HCDR3, and a LCDR1, LCDR2, and LCDR3 of any one of these antibodies. In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HCDR1, HCDR2, and HCDR3, and a LCDR1, LCDR2, and LCDR3 of any one of LIBC523661-1, LIBC523797-1, LIBC524049-1, LIBC527814-1, LIBC527997-1, LIBC528116, LIBC527919-1, and LIBC680574-1. HV and LV CDR sequences of these antibodies are listed in Tables 7, 10 and 26.

[0111] In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HV or a LV of any one of LIBC523596-1, LIBC527906-1, LIBC523603-1, LIBC527814-1, LIBC523661-1, LIBC527997-1, LIBC523670-1, LIBC528141-1, LIBC523748-1, LIBC527912-1, LIBC523760-1, LIBC528116-1,

LIBC523797-1, LIBC527879-1, LIBC523813-1, LIBC527919-1, LIBC523815-1, LIBC527984-1, LIBC523844-1, LIBC527968-1, LIBC523857-1, LIBC528169-1, LIBC523862-1, LIBC528131-1, LIBC523868-1, LIBC527827-1, LIBC523969-1, LIBC527869-1, LIBC524049-1, LIBC528148-1, LIBC680562-1, LIBC680574-1, LIBC680773-1, LIBC680800-1, LIBC680812-1, LIBC681593-1, LIBC681594-1, and LIBC681737-1. In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HV and a LV of any one of LIBC523596-1, LIBC527906-1, LIBC523603-1, LIBC527814-1, LIBC523661-1, LIBC527997-1, LIBC523670-1, LIBC528141-1, LIBC523748-1, LIBC527912-1, LIBC523760-1, LIBC528116-1, LIBC523797-1, LIBC527879-1, LIBC523813-1, LIBC527919-1, LIBC523815-1, LIBC527984-1, LIBC523844-1, LIBC527968-1, LIBC523857-1, LIBC528169-1, LIBC523862-1, LIBC528131-1, LIBC523868-1, LIBC527827-1, LIBC523969-1, LIBC527869-1, LIBC524049-1, LIBC528148-1, LIBC680562-1, LIBC680574-1, LIBC680773-1, LIBC680800-1, LIBC680812-1, LIBC681593-1, LIBC681594-1, and LIBC681737-1. HV and LV sequences of these antibodies are listed in Tables 8, 11 and 27.

[0112] In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HV or LV of any one of LIBC523661-1, LIBC523797-1, LIBC524049-1, LIBC527814-1, LIBC527997-1, LIBC528116, LIBC527919-1, and LIBC680574-1. In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HV and LV of any one of LIBC523661-1, LIBC523797-1, LIBC524049-1, LIBC527814-1, LIBC527997-1, LIBC528116, LIBC527919-1, and LIBC680574-1.

[0113] Certain mutations in HC_FR3 de-loop of 10D10 that are associated with improved binding to CB1 were included in the affinity maturation library design (Example 1). In addition, all tested sequences identified after affinity maturation included the R94S germline reversion of a LC FR3 non-consensus mutation (Examples 2 and 3). Further, R32 (Aho numbering) that is adjacent to HCDR1 is also involved in epitope interaction. Thus, in some embodiments of the antibodies and antigen binding fragments described above, the amino acid at HV position 83 (Aho numbering scheme) is N and the amino acid at HV position 85 (Aho numbering scheme) is Y. In some embodiments of the antibodies and antigen binding fragments described above, the amino acid at HV position 83 is N, the amino acid at HV position 85 is Y, and the amino acid at LV position 94 is S. In some embodiments, of the antibodies and antigen binding fragments described above the amino acid at HV position 32 is R.

[0114] In various embodiments, the antibodies and antigen binding fragments disclosed herein are antagonist and/or inverse agonist of huCB1, for example, the antibodies and binding fragments inhibit signaling of huCB1. Potencies of the antibodies and antigen binding fragments are in the nanomolar or subnanomolar range, for example, the antibodies and antigen binding fragments have an IC50 of less than about 10 nM, less than about 5 nM, less than about 3 nM or less than about 1 nM, as measured in a cell-based cAMP assay. In some embodiments, potencies of the antibodies and antigen binding fragments disclosed herein are

comparable to or better than that of rimonabant (e.g., at least two-fold, three-fold, five-fold, eight-fold, or ten-fold better than that of rimonabant).

[0115] In another aspect, disclosed herein is an antibody or antigen binding fragment thereof that binds to huCB1 and comprises a heavy chain variable region (HV) and a light chain variable region (LV), wherein

[0116] a) the VH comprises HCDR1, HCDH2 and HCDR3, wherein CDRH1 comprises the amino acid sequence of RGGDYWS (SEQ ID NO: 1); CDRH2 comprises the sequence of HX25YX26X27GX28TX29YNPX30X31X32X33 (SEQ ID NO: 538), wherein X25 is I or V; X26 is Y or Q; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L, X32 is K or R; and X33 is G, S, or N; and CDRH3 comprises the sequence of X34YDX35X36X37GYSYYYYGX38DV (SEQ ID NO: 539), wherein X34 is D or G; X35 is A, I, T, or V; X36 is L or absent; X37 is S or T; and X38 is M or L; and

[0117] b) the VL is of or derived from a VK2/A19/JK1 germline.

[0118] In some embodiments, the antibody or antigen binding fragment thereof comprises a) CDRH1 comprising the sequence of RGGDYWS (SEQ ID NO: 1); CDRH2 comprising the sequence of HX25YX26X27GX28TX29YNPX30X31X32X33 (SEQ ID NO: 538), wherein X25 is I or V; X26 is Y; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L, X32 is K or R; and X33 is G, S, or N; and CDRH3 comprising the sequence of X34YDX35X36X37GYSYYYYGX38DV (SEQ ID NO: 539), wherein X34 is G; X35 is A, I, T, or V; X36 is absent; X37 is S; and X38 is M or L, and b) the VL is of or derived from a VK2/A19/JK1 germline.

[0119] In some embodiments, the antibody or antigen binding fragment thereof comprises a) CDRH1 comprising the sequence of RGGDYWS (SEQ ID NO: 1); CDRH2 comprising the sequence of HX25YX26X27GX28TX29YNPX30X31X32X33 (SEQ ID NO: 538), wherein X25 is I or V; X26 is Y or Q; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L, X32 is K or R; and X33 is G, S, or N; and CDRH3 comprising the sequence of X34YDX35X36X37GYSYYYYGX38DV (SEQ ID NO: 539), wherein X34 is D or G; X35 is A, I, T, or V; X36 is L or absent; X37 is S or T; and X38 is M or L, and b) the VL is of or derived from a VK2/A19/JK1 germline and comprises LCDR1, LCDR2, and LCDR3, wherein LCDR1 comprises the sequence of RSSQSLLHRSGYNYLD (SEQ ID NO: 257); LCDR2 comprises the sequence of X20GSNRA (SEQ ID NO: 534), wherein X20 is L or Q; and LCDR3 comprises X47QX48X49X50X51PRT (SEQ ID NO: 540), wherein X47 is M or R; X48 is A or S; X49 is L, R, or V; X50 is Q or A; and X51 is T or L.

[0120] In some embodiments, the antibody or antigen binding fragment thereof comprises a) CDRH1 comprising the sequence of RGGDYWS (SEQ ID NO: 1); CDRH2 comprising the sequence of HX25YX26X27GX28TX29YNPX30X31X32X33 (SEQ ID NO: 538), wherein X25 is I or V; X26 is Y; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L, X32 is K or R; and X33 is G, S, or N; and CDRH3

comprising the sequence of X34YDX35X36X37GYSYYYYGX38DV (SEQ ID NO: 539), wherein X34 is G; X35 is A, I, T, or V; X36 is absent; X37 is S; and X38 is M or L, and b) LCDR1 comprising the sequence of RSSQSLLHRSGYNYLD (SEQ ID NO: 257); LCDR2 comprising the sequence of X20GSNRA (SEQ ID NO: 534), wherein X20 is L or Q; and LCDR3 comprising X47QX48X49X50X51PRT (SEQ ID NO: 540), wherein X47 is M or R; X48 is A or S; X49 is L, R, or V; X50 is Q or A; and X51 is T or L.

[0121] In some embodiments, the antibody or antigen binding fragment thereof comprises a) CDRH1 comprising the sequence of RGGDYWS (SEQ ID NO: 1); CDRH2 comprising the sequence of HX25YX26X27GX28TX29YNPX30X31X32X33 (SEQ ID NO: 538), wherein X25 is I or V; X26 is Y or Q; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L, X32 is K or R; and X33 is G, S, or N; and CDRH3 comprising the sequence of X34YDX35X36X37GYSYYYYGX38DV (SEQ ID NO: 539), wherein X34 is D or G; X35 is A, I, T, or V; X36 is L or absent; X37 is S or T; and X38 is M or L, and b) LCDR1 comprising the amino acid sequence of RSSQSLLHRSGYNYLD (SEQ ID NO: 257); LCDR2 comprising the amino acid sequence of LGSNRRAS (SEQ ID NO: 258) or QGSNRRAS (SEQ ID NO: 264); and LCDR3 comprising the amino acid sequence of MQSLQTPRT (SEQ ID NO: 259), RQSVALPRT (SEQ ID NO: 271), or RQARALPRT (SEQ ID NO: 265).

[0122] In some embodiments, the antibody or antigen binding fragment thereof comprises a) CDRH1 comprising the sequence of RGGDYWS (SEQ ID NO: 1); CDRH2 comprising the sequence of HX25YX26X27GX28TX29YNPX30X31X32X33 (SEQ ID NO: 538), wherein X25 is I or V; X26 is Y; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L, X32 is K or R; and X33 is G, S, or N; and CDRH3 comprising the sequence of X34YDX35X36X37GYSYYYYGX38DV (SEQ ID NO: 539), wherein X34 is G; X35 is A, I, T, or V; X36 is absent; X37 is S; and X38 is M or L, and b) LCDR1 comprising the amino acid sequence of RSSQSLLHRSGYNYLD (SEQ ID NO: 257); LCDR2 comprising the amino acid sequence of LGSNRRAS (SEQ ID NO: 258) or QGSNRRAS (SEQ ID NO: 264); and LCDR3 comprising the amino acid sequence of MQSLQTPRT (SEQ ID NO: 259), RQSVALPRT (SEQ ID NO: 271), or RQARALPRT (SEQ ID NO: 265).

[0123] In certain embodiments, the anti-CB1 antibodies or antigen binding fragments disclosed herein comprise a heavy chain variable region comprising a HCDR1, HCDR2, and HCDR3, and a light chain variable region comprising a LCDR1, LCDR2, and LCDR3 from any of the anti-CB1 antibodies described herein. For example, in some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, or LCDR3 of any one of LIBC673948-1, LIBC673952-1, LIBC673965-1, LIBC673982-1, LIBC673972-1, LIBC674024-1, LIBC674035-1, LIBC674043-1, LIBC674090-1, LIBC674002-1, LIBC674153-1, LIBC674200-1, LIBC674214-1, LIBC674216-1, LIBC674229-1, LIBC674235-1, LIBC674257-1, and LIBC674276-1. In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HCDR1, HCDR2, and HCDR3, and a LCDR1,

LCDR2, and LCDR3 of any one of LIBC673948-1, LIBC673952-1, LIBC673965-1, LIBC673982-1, LIBC673972-1, LIBC674024-1, LIBC674035-1, LIBC674043-1, LIBC674090-1, LIBC674002-1, LIBC674153-1, LIBC674200-1, LIBC674214-1, LIBC674216-1, LIBC674229-1, LIBC674235-1, LIBC674257-1, and LIBC674276-1. HV and LV CDR sequences of these antibodies are listed in Table 30.

[0124] In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HV or a LV of any one of LIBC673948-1, LIBC673952-1, LIBC673965-1, LIBC673982-1, LIBC673972-1, LIBC674024-1, LIBC674035-1, LIBC674043-1, LIBC674090-1, LIBC674002-1, LIBC674153-1, LIBC674200-1, LIBC674214-1, LIBC674216-1, LIBC674229-1, LIBC674235-1, LIBC674257-1, and LIBC674276-1. In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HV and a LV of any one of LIBC673948-1, LIBC673952-1, LIBC673965-1, LIBC673982-1, LIBC673972-1, LIBC674024-1, LIBC674035-1, LIBC674043-1, LIBC674090-1, LIBC674002-1, LIBC674153-1, LIBC674200-1, LIBC674214-1, LIBC674216-1, LIBC674229-1, LIBC674235-1, LIBC674257-1, and LIBC674276-1. HV and LV sequences of these antibodies are listed in Table 31.

[0125] In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, or LCDR3 of any one of LIBC529593-1, LIBC560340-1 and LIBC560657-1. In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HCDR1, HCDR2, and HCDR3, and a LCDR1, LCDR2, and LCDR3 of any one of LIBC529593-1, LIBC560340-1 and LIBC560657-1. HV and LV CDR sequences of the antibodies are listed in Table 19. For example, the antibody or antigen binding fragment thereof comprises a) HCDR1 comprising the amino acid sequence of RGGDYWS (SEQ ID NO: 474) or RGGDYWN (SEQ ID NO: 480); HCDR2 comprising the amino acid sequence of HIYYSGSTNYPNPSLRS (SEQ ID NO: 475), HIYYSGSKNYPNPSLKS (SEQ ID NO: 481), or HIYYTGTKYYPNPSLKS (SEQ ID NO: 487), and HCDR3 comprising the amino acid sequence of DYDILYGYSSYYYGLDV (SEQ ID NO: 476), or GYDSSGYSSYYYGMDV (SEQ ID NO: 475), and b) LCDR1 comprising the amino acid sequence of RSSQSLHRSQYNYLD (SEQ ID NO: 471), RSSQSLLYSNGHNFLD (SEQ ID NO: 477), or RSSQSLLYSNGHNFLD (SEQ ID NO: 483), LCDR2 comprising the amino acid sequence of LGSNRAS (SEQ ID NO: 472) or LGSNRAP (SEQ ID NO: 484), and LCDR3 comprising the amino acid sequence of MQSLQTPRT (SEQ ID NO: 473) or MQALQTPRT (SEQ ID NO: 479).

[0126] In some embodiments, the antibody or antigen binding fragment thereof comprises a HV or LV of any one of LIBC529593-1, LIBC560340-1 and LIBC560657-1. In some embodiments, the antibody or antigen binding fragment thereof comprises a HV and a LV of any one of LIBC529593-1, LIBC560340-1 and LIBC560657-1. HV and LV sequences of the antibodies are listed in Table 20.

[0127] In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises amino acid mutations in the HV CDRs compared to a reference antibody. For example, in some embodiments, the antibody or antigen binding fragment thereof comprises a) a HV com-

prising a HCDR1, HCDR2 and HCDR3, wherein HCDR2 of the antibody comprises 5 or fewer mutations compared to HVYYTGSTNYPNPRFKD (SEQ ID NO: 136). In some embodiments, HCDR2 of the antibody or antigen binding fragment thereof comprises 4 or fewer mutations compared to HVYYTGSTNYPNPRFKD (SEQ ID NO: 136). In some embodiments, HCDR2 of the antibody or antigen binding fragment thereof comprises 5 or fewer mutations compared to HVYYTGSTNYPNPRFKD (SEQ ID NO: 136) and HCDR3 of the antibody or antigen binding fragment thereof comprises 3 or fewer mutations compared to NYDTVTGYSSYYYGMDV (SEQ ID NO: 137). In some embodiments, HCDR2 of the antibody or antigen binding fragment thereof comprises 4 or fewer mutations compared to HVYYTGSTNYPNPRFKD (SEQ ID NO: 136) and HCDR3 of the antibody or antigen binding fragment thereof comprises 3 or fewer mutations compared to NYDTVTGYSSYYYGMDV (SEQ ID NO: 137). In such embodiments, the HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A, S, or G, preferably A or S, and the LV of the antibody or antigen binding fragment thereof is of or derived from a VK1/O2/JK4, a VK2/A19/JK1, or a VK3/A27/JK1 germline. Preferably the LV is of or derived from a VK1/O2/JK4 or a VK2/A19/JK1 germline, more preferably a VK1/O2/JK4 germline. Exemplary sequences of LV sequences of or derived from the germlines are described above and in Tables 8, 11, 27 and 31.

[0128] In some embodiments, the antibody or antigen binding fragment thereof comprises a) a HV comprising a HCDR1, HCDR2 and HCDR3, wherein HCDR2 of the antibody or antigen binding fragment thereof comprises 5 or fewer mutations compared to HVYYTGSTKYNPNFKG (SEQ ID NO: 8). In some embodiments, HCDR2 of the antibody or antigen binding fragment thereof comprises 4 or fewer mutations compared to HVYYTGSTKYNPNFKG (SEQ ID NO: 8). In some embodiments, HCDR2 of the antibody or antigen binding fragment thereof comprises 5 or fewer mutations compared to HVYYTGSTKYNPNFKG (SEQ ID NO: 8) and HCDR3 of the antibody or antigen binding fragment thereof comprises 3 or fewer mutations compared to DYDILTGYSSYYYGMDV (SEQ ID NO: 9). In some embodiments, HCDR2 of the antibody or antigen binding fragment thereof comprises 4 or fewer mutations compared to HVYYTGSTKYNPNFKG (SEQ ID NO: 8) and HCDR3 of the antibody or antigen binding fragment thereof comprises 3 or fewer mutations compared to DYDILTGYSSYYYGMDV (SEQ ID NO: 9). In such embodiments, the HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A, S, or G, preferably A or S, and the LV of the antibody or antigen binding fragment thereof is of or derived from a VK1/O2/JK4 or a VK2/A19/JK1 germline, preferably a VK2/A19/JK1 germline. Exemplary sequences of LV sequences of or derived from the germlines are described above and in Tables 8, 11, 20, 27 and 31.

[0129] In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a heavy chain variable region comprising a sequence that is at least 90% identical, or at least 95% identical (e.g., at least 96%, at least 97%, or at least 98% identical) to the sequence of SEQ ID NO: 228. The LV of the antibody or antigen binding fragment thereof is of or derived from a VK1/O2/JK4, a VK2/A19/JK1, or a VK3/A27/JK1 germline. Preferably the

LV is of or derived from a VK1/O2/JK4 or a VK2/A19/JK1 germline, more preferably a VK1/O2/JK4 germline. In some embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a heavy chain variable region comprising a sequence that is at least 90% identical, or at least 95% identical (e.g., at least 96%, at least 97%, or at least 98% identical) to the sequence of SEQ ID NO: 100. The LV of the antibody or antigen binding fragment thereof is of or derived from a VK1/O2/JK4 or a VK2/A19/JK1 germline, preferably a VK2/A19/JK1 germline. Exemplary sequences of LV sequences of or derived from the germlines are described above and in Tables 8, 11, 20, 27 and 31.

[0130] The term “identity,” as used herein, refers to a relationship between the sequences of two or more polypeptide molecules or two or more nucleic acid molecules, as determined by aligning and comparing the sequences. “Percent identity,” as used herein, means the percent of identical residues between the amino acids or nucleotides in the compared molecules and is calculated based on the size of the smallest of the molecules being compared. For these calculations, gaps in alignments (if any) must be addressed by a particular mathematical model or computer program (i.e., an “algorithm”). Methods that can be used to calculate the identity of the aligned nucleic acids or polypeptides include those described in Computational Molecular Biology, (Lesk, A. M., ed.), 1988, New York: Oxford University Press; Biocomputing Informatics and Genome Projects, (Smith, D. W., ed.), 1993, New York: Academic Press; Computer Analysis of Sequence Data, Part I, (Griffin, A. M., and Griffin, H. G., eds.), 1994, New Jersey: Humana Press; von Heinje, G., 1987, Sequence Analysis in Molecular Biology, New York: Academic Press; Sequence Analysis Primer, (Gribskov, M. and Devereux, J., eds.), 1991, New York: M. Stockton Press; and Carillo et al., 1988, SIAM J. Applied Math. 48:1073. For example, sequence identity can be determined by standard methods that are commonly used to compare the similarity in position of the amino acids of two polypeptides. Using a computer program such as BLAST or FASTA, two polypeptide or two polynucleotide sequences are aligned for optimal matching of their respective residues (either along the full length of one or both sequences, or along a pre-determined portion of one or both sequences). The programs provide a default opening penalty and a default gap penalty, and a scoring matrix such as PAM 250 (Dayhoff et al., in Atlas of Protein Sequence and Structure, vol. 5, supp. 3, 1978) or BLOSUM62 (Henikoff et al., 1992, Proc. Natl. Acad. Sci. U.S.A. 89:10915-10919) can be used in conjunction with the computer program. For example, the percent identity can then be calculated as: the total number of identical matches multiplied by 100 and then divided by the sum of the length of the longer sequence within the matched span and the number of gaps introduced into the longer sequences in order to align the two sequences. In calculating percent identity, the sequences being compared are aligned in a way that gives the largest match between the sequences.

[0131] The GCG program package is a computer program that can be used to determine percent identity, which package includes GAP (Devereux et al., 1984, Nucl. Acid Res. 12:387; Genetics Computer Group, University of Wisconsin, Madison, WI). The computer algorithm GAP is used to align the two polypeptides or two polynucleotides for which the percent sequence identity is to be determined. The sequences are aligned for optimal matching of their respec-

tive amino acid or nucleotide (the “matched span,” as determined by the algorithm). A gap opening penalty (which is calculated as 3× the average diagonal, wherein the “average diagonal” is the average of the diagonal of the comparison matrix being used; the “diagonal” is the score or number assigned to each perfect amino acid match by the particular comparison matrix) and a gap extension penalty (which is usually $\frac{1}{10}$ times the gap opening penalty), as well as a comparison matrix such as PAM 250 or BLOSUM 62 are used in conjunction with the algorithm. In certain embodiments, a standard comparison matrix (see, Dayhoff et al., 1978, Atlas of Protein Sequence and Structure 5:345-352 for the PAM 250 comparison matrix; Henikoff et al., 1992, Proc. Natl. Acad. Sci. U.S.A. 89:10915-10919 for the BLOSUM 62 comparison matrix) is also used by the algorithm.

[0132] Recommended parameters for determining percent identity for polypeptides or nucleotide sequences using the GAP program include the following:

[0133] Algorithm: Needleman et al. 1970, J. Mol. Biol. 48:443-453;

[0134] Comparison matrix: BLOSUM 62 from Henikoff et al., 1992, supra;

[0135] Gap Penalty: 12 (but with no penalty for end gaps)

[0136] Gap Length Penalty: 4

[0137] Threshold of Similarity: 0

[0138] Certain alignment schemes for aligning two amino acid sequences may result in matching of only a short region of the two sequences, and this small aligned region may have very high sequence identity even though there is no significant relationship between the two full-length sequences. Accordingly, the selected alignment method (GAP program) can be adjusted if so desired to result in an alignment that spans at least 50 contiguous amino acids of the target polypeptide.

[0139] As described above, the anti-CB1 antibody or antigen binding fragment thereof disclosed herein can contain mutations in HC_FR3 region compared to 10D10. For antibody or antigen binding fragment thereof that comprises HCDR1 of SEQ ID NO: 1, HCDR2 comprising SEQ ID NO: 538 and HCDR3 of SEQ ID NO: 539, in addition to mutations at HV position 83 and/or 85, the amino acid at HV position 79 may be R. In some embodiments, the amino acid at HV position 83 of the antibody is N and the amino acid at HV position 85 of the antibody is Y. In some embodiments, the amino acid at position 79 of the antibody or antigen binding fragment thereof is R, the amino acid at HV position 83 of the antibody or antigen binding fragment thereof is N and the amino acid at HV position 85 of the antibody or antigen binding fragment thereof is Y. In some embodiments of the antibodies described above, the amino acid at HV position of the antibody or antigen binding fragment thereof 83 is N, the amino acid at HV position 85 of the antibody or antigen binding fragment thereof is Y, and the amino acid at LV position 94 of the antibody or antigen binding fragment thereof is S. In some embodiments, the amino acid at position 79 of the antibody or antigen binding fragment thereof is R, the amino acid at HV position 83 of the antibody or antigen binding fragment thereof is N and the amino acid at HV position 85 of the antibody or antigen binding fragment thereof is Y, and the amino acid at LV position 94 of the antibody or antigen binding fragment thereof is S.

[0140] In certain preferred embodiments, the anti-CB1 antibody or antigen binding fragment thereof comprises a HCDR1, HCDR2, HCDR3, LCDR1, LCDR2 or LCDR3 of any one of LIBC528116-1, LIBC680574-1, LIBC673952-1, LIBC523797-1, LIBC527814-1, LIBC527997-1, LIBC527919-1, LIBC523661-1 and LIBC524049-1. For example, the anti-CB1 antibody or antigen binding fragment thereof comprises a set of heavy and light chain CDRs of any one of the following:

[0141] (a) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 135-137, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 138-140, respectively;

[0142] (b) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 426-428, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 423-425, respectively;

[0143] (c) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 266-268, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 263-235, respectively;

[0144] (d) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 129-131, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 132-134, respectively;

[0145] (e) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 159-161, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 162-164, respectively;

[0146] (f) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 207-209, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 210-212, respectively;

[0147] (g) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 1-3, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 4-6, respectively; and

[0148] (h) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 55-57, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 58-60, respectively.

[0149] In some preferred embodiments, the antibody or antigen binding fragment thereof comprises a HV comprising an amino acid sequence selected from SEQ ID NO: 228, SEQ ID NO: 460, SEQ ID NO: 368, SEQ ID NO: 100, SEQ ID NO: 236, SEQ ID NO: 226, SEQ ID NO: 252, SEQ ID NO: 98, and SEQ ID NO: 116. In some preferred embodiments, the antibody or antigen binding fragment thereof comprises a LV comprising an amino acid sequence selected from SEQ ID NO: 227, SEQ ID NO: 459, SEQ ID NO: 367, SEQ ID NO: 99, SEQ ID NO: 235, SEQ ID NO: 225, SEQ ID NO: 251, SEQ ID NO: 97, and SEQ ID NO: 115. In some preferred embodiments, the antibody or antigen binding fragment thereof comprises a HV and LV selected from any of the following: i) a HV comprising the amino acid sequence of SEQ ID NO: 228, a LV comprising the amino acid sequence of SEQ ID NO: 227; ii) a HV comprising the

amino acid sequence of SEQ ID NO: 460, a LV comprising the amino acid sequence of SEQ ID NO: 459; iii) a HV comprising the amino acid sequence of SEQ ID NO: 368, a LV comprising the amino acid sequence of SEQ ID NO: 367; iv) a HV comprising the amino acid sequence of SEQ ID NO: 100, a LV comprising the amino acid sequence of SEQ ID NO: 99; v) a HV comprising the amino acid sequence of SEQ ID NO: 236, a LV comprising the amino acid sequence of SEQ ID NO: 235; vi) a HV comprising the amino acid sequence of SEQ ID NO: 226, a LV comprising the amino acid sequence of SEQ ID NO: 225; vii) a HV comprising the amino acid sequence of SEQ ID NO: 252, a LV comprising the amino acid sequence of SEQ ID NO: 251; viii) a HV comprising the amino acid sequence of SEQ ID NO: 98, a LV comprising the amino acid sequence of SEQ ID NO: 97; and ix) a HV comprising the amino acid sequence of SEQ ID NO: 116, a LV comprising the amino acid sequence of SEQ ID NO: 115.

[0150] In various embodiments, the antibodies or antigen binding fragments thereof disclosed herein have a higher binding affinity for CB1 (e.g., huCB1) compared to 10D10 LC N35Y (i.e., 10D10 that contains N to Y substitution at LC amino acid position 35). In some embodiments, the antibodies or antigen binding fragments thereof have a binding affinity for CB1 that is at least three times higher compared to 10D10 LC N35Y. In some embodiments, the antibodies or antigen binding fragments thereof have a binding affinity for CB1 that is at least five times higher (e.g., at least eight times higher, at least ten times higher, at least twelve times higher, at least fifteen times higher, or at least twenty times higher) compared to 10D10 LC N35Y.

[0151] Affinity is determined using a variety of techniques, an example of which is an affinity ELISA assay. In various embodiments, affinity is determined by a surface plasmon resonance assay (e.g., BIAcore®-based assay). Using this methodology, the association rate constant (k_a in $M^{-1}s^{-1}$) and the dissociation rate constant (k_d in s^{-1}) can be measured. The equilibrium dissociation constant (K_D in M) can then be calculated from the ratio of the kinetic rate constants (k_d/k_a). In some embodiments, affinity is determined by a kinetic method, such as a Kinetic Exclusion Assay (KinExA) as described in Rathanaswami et al. Analytical Biochemistry, Vol. 373:52-60, 2008. Using a KinExA assay, the equilibrium dissociation constant (K_D in M) and the association rate constant (k_a in $M^{-1}s^{-1}$) can be measured. The dissociation rate constant (k_d in s^{-1}) can be calculated from these values ($K_D \times k_a$). In other embodiments, affinity is determined by a bio-layer interferometry method, such as that described in Kumaraswamy et al., Methods Mol. Biol., Vol. 1278:165-82, 2015 and employed in Octet® systems (Pall ForteBio). The kinetic (k_a and k_d) and affinity (K_D) constants can be calculated in real-time using the bio-layer interferometry method. Binding to CB1 (e.g., huCB1) can be measured using these techniques and an antigen source (e.g., CB1-ND, CB1-SMALP, or CB1 expressing cells (e.g., 293T cells).

[0152] Antibody or antigen binding fragment that specifically bind an antigen may have an equilibrium dissociation constant (K_D) $\leq 1 \times 10^{-6}$ M. The antibody or antigen binding fragment specifically binds antigen with "high affinity" when the K_D is $\leq 3 \times 10^{-8}$ M. In some embodiments, the anti-CB1 antibodies or antigen binding fragments bind to target antigen (e.g., huCB1) with a K_D of ≤ 100 nM (e.g., less than about 90 nM, 70 nM, 50 nM, 30 nM, 20 nM, 10 nM,

or a range defined by any two of the foregoing values). In some embodiments, the anti-CB1 antibodies or binding fragments bind to CB1 (e.g., huCB1 such as huCB1-ND) with a dissociation constant (K_D) of less than about 70 nM-10 nM (e.g., about 60 nM-20 nM, or about 40 nM-10 nM), as determined using Octet® systems and huCB1-ND. In some embodiments, the antibodies or antigen binding fragments bind to CB1 (e.g., hu-CB1) with a K_D of $\leq 1 \times 10^{-8}$ M (e.g., $\leq 1 \times 10^{-9}$ M, $\leq 5 \times 10^{-1}$ M, or $\leq 1 \times 10^{-10}$ M) as determined using a KinExA assay and huCB1-SMAP or CB1 expressing cells.

[0153] In various embodiments, the anti-CB1 antibodies and antigen binding fragments disclosed herein are antagonist and/or inverse agonist anti-huCB1 antibodies and antigen binding fragments, for example, they inhibit signaling of huCB1. Potencies of the antibodies and antigen binding fragments are in single digit nanomolar range or subnanomolar range, for example, they have an IC50 of less than about 10 nM, less than about 5 nM, less than about 3 nM or less than about 1 nM as measured in a cell-based cAMP assay (e.g., a cell-based cAMP assay in the presence of CP 55,940). In some embodiments, the anti-CB1 antibodies and antigen binding fragments disclosed herein have a potency comparable to that of rimonabant. In some embodiments, the anti-CB1 antibodies and antigen binding fragments disclosed herein have a potency greater than that of rimonabant

SEQ ID NO: 559) (WO2014210205). The anti-CB1 antibodies and antigen binding fragments disclosed herein bind to the same region on huCB1 as they are affinity matured variants of the 10D10 antibody. Thus, in some embodiments, the anti-CB1 antibodies and antigen binding fragments disclosed herein bind to the extracellular loop 2 region of huCB1 and have an IC50 of less than about 10 nM, less than about 5 nM, less than about 3 nM, or less than about 1 nM, as measured in a cell-based cAMP assay (e.g., a cell-based cAMP assay in the presence of CP 55,940).

[0155] The anti-CB1 antibodies of the invention can comprise any immunoglobulin constant region. The term “constant region,” used interchangeably herein with “constant domain” refers to all domains of an antibody other than the variable region. The constant region is not involved directly in binding of an antigen, but exhibits various effector functions. As described above, antibodies are divided into particular isotypes (IgA, IgD, IgE, IgG, and IgM) and subtypes (IgG1, IgG2, IgG3, IgG4, IgA1, IgA2) depending on the amino acid sequence of the constant region of their heavy chains. The light chain constant region can be, for example, a kappa- or lambda-type light chain constant region, e.g., a human kappa- or lambda-type light chain constant region, which are found in all five antibody isotypes. Examples of human immunoglobulin light chain constant region sequences are shown in Table 1:

TABLE 1		
Exemplary Human Immunoglobulin Light Chain Constant Regions		
Designation	SEQ ID NO:	CL Domain Amino Acid Sequence
Human lambda v1	513	GQPKANPTVTLPFPPSSEELQANKATLVCLISDFYPGAVTVAWKADGSPVKAGVETTTKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTECS
Human lambda v2	514	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSPVKAGVETTTKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTECS
Human lambda v3	515	QPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSPVKAGVETTTKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTECS
Human lambda v4	516	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSPVKAGVETTTKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTECS
Human lambda v5	517	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADGSPVKAGVETTTKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTECS
Human kappa v1	518	TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSPVTKSFNRGEC
Human kappa v2	519	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSPVTKSFNRGEC

(e.g., at least two-fold, three-fold, five-fold, eight-fold, or ten-fold greater than that of rimonabant). In various embodiments, potency can be measured using a cell-based cAMP assay.

[0154] The 10D10 antibody binds to the extracellular loop 2 region of huCB1 (i.e., amino acid residues 256 to 273 of

[0156] The heavy chain constant region of the anti-CB1 antibodies of the invention can be, for example, an alpha-, delta-, epsilon-, gamma-, or mu-type heavy chain constant region, e.g., a human alpha-, delta-, epsilon-, gamma-, or mu-type heavy chain constant region. In some embodiments, the anti-CB1 antibodies comprise a heavy chain constant region from an IgG1, IgG2, IgG3, or IgG4 immunoglobulin,

such as a human IgG1, IgG2, IgG3, or IgG4 immunoglobulin. In one embodiment, the anti-CB1 antibody comprises a heavy chain constant region from a human IgG1 immunoglobulin. In such embodiments, the human IgG1 immunoglobulin constant region may comprise one or more mutations to prevent glycosylation and/or half life of the antibody as described in more detail herein. In another embodiment,

the anti-CB1 antibody comprises a heavy chain constant region from a human IgG2 immunoglobulin. In yet another embodiment, the anti-CB1 antibody comprises a heavy chain constant region from a human IgG4 immunoglobulin. Examples of human IgG1, IgG2, and IgG4 heavy chain constant region sequences are shown below in Table 2.

TABLE 2

Exemplary Human Immunoglobulin Heavy Chain Constant Regions		
Ig isotype	SEQ ID NO:	Heavy Chain Constant Region Amino Acid Sequence
Human IgG1z	520	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVH NAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPP VLDSGGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK
Human IgG1za	521	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVH NAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSDGGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK
Human IgG1f	522	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVH NAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPP VLDSGGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK
Human IgG1fa	523	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVH NAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSDGGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK
Human IgG1z aglycosylated v1	524	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVH NAKTKPREEQYGSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPP VLDSGGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK
Human IgG1z aglycosylated v2	525	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVH NAKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKG QPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPP VLDSGGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK
Human IgG1z aglycosylated v2_half life extended	526	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP APPELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWYVDGVEVH AKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQ PREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVL DSDGGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK
huIgG1z_half life extended	527	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP APPELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWYVDGVEVH AKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQ PREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVL DSDGGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK
Human IgG2	528	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVL QSSGLYSLSSVVTVPSSNFGTQTYTCNVNHPKPSNTKVDKTKVERKCCVECPCPAPP VAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVQFNWYVDGVEVHNAKT KPREEQYNSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPRE PQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPMLDS DGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK

TABLE 2-continued

Exemplary Human Immunoglobulin Heavy Chain Constant Regions			
Ig isotype	SEQ ID NO:	Heavy Chain Constant Region Amino Acid Sequence	
Human IgG4	529	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTWNSGALTSGLVHTFPAVL QSSGLYSLSSVVTVPSSSLGKTKYTCNVDPKPSNTKVDKRVESKYGPCCPSCPAPEFL GGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSQEDPEVQFNWYVDGVEVHNAKTK PREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQPREP QVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSRLTVDKSRWQEGNVFSCVMHEALHNYHTQKSLSLSLGK	

[0157] Each of the light chain variable regions and each of the heavy chain variable regions disclosed herein (Tables 8, 11, 20, 27 and 31) may be attached to the above light chain constant regions (Table 1) and heavy chain constant regions (Table 2) to form complete antibody light and heavy chains, respectively. In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in Tables 8, 11, 20, 27 and 31 are attached to SEQ ID NO: 518 or 519 and SEQ ID NO: 525, 526, or 527 to form complete antibody light and heavy chains, respectively. In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in Tables 8, 11, 20, 27 and 31 are attached to SEQ ID NO: 518 and SEQ ID NO: 525, 526, or 527 to form complete antibody light and heavy chains, respectively. In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in Tables 8, 11, 20, 27 and 31 are attached SEQ ID NO: 518 and SEQ ID NO: 525 to form complete antibody light and heavy chains, respectively. In some embodiments, the monoclonal antibody comprises a heavy chain having the amino acid sequence of SEQ ID NO: 562 and a light chain having the amino acid sequence of SEQ ID NO: 563. In some embodiments, the monoclonal antibody comprises a heavy chain having the amino acid sequence of SEQ ID NO: 562 with the amino acid residue K at the C-terminal end deleted and a light chain having the amino acid sequence of SEQ ID NO: 563. Further, each of the so generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed above.

[0158] The anti-CB1 antibodies or antigen binding fragments of the invention can be monoclonal antibodies, polyclonal antibodies, recombinant antibodies, human antibodies, humanized antibodies, chimeric antibodies, or multispecific antibodies or antigen-binding fragments thereof. In certain embodiments, the anti-CB1 antibody is a monoclonal antibody. In such embodiments, the anti-CB1 antibody may be a chimeric antibody, a humanized antibody, or a fully human antibody having a human immunoglobulin constant domain. In these and other embodiments, the anti-CB1 antibody is a human IgG1 (e.g., IgG1z), IgG2, IgG3, or IgG4 antibody. Thus, the anti-CB1 antibody may, in some embodiments, have a human IgG1, IgG2, IgG3, or IgG4 constant domain. In one embodiment, the anti-CB1 antibody is a monoclonal human IgG1 antibody, preferably, a monoclonal human IgG1z antibody. In another embodi-

ment, the anti-CB1 antibody is a monoclonal human IgG2 antibody. In yet another embodiment, the anti-CB1 antibody is a monoclonal human IgG4 antibody.

[0159] The term “monoclonal antibody” (or “mAb”) as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies, i.e., the individual antibodies comprising the population are identical except for possible naturally occurring mutations that may be present in minor amounts. Monoclonal antibodies are highly specific, being directed against an individual antigenic site or epitope, in contrast to polyclonal antibody preparations that typically include different antibodies directed against different epitopes. Monoclonal antibodies may be produced using any technique known in the art, e.g., by immortalizing spleen cells harvested from an animal after completion of the immunization schedule. The spleen cells can be immortalized using any technique known in the art, e.g., by fusing them with myeloma cells to produce hybridomas. See, for example, *Antibodies*; Harlow and Lane, Cold Spring Harbor Laboratory Press, 1st Edition, e.g. from 1988, or 2nd Edition, e.g. from 2014. Myeloma cells for use in hybridoma-producing fusion procedures preferably are non-antibody-producing, have high fusion efficiency, and enzyme deficiencies that render them incapable of growing in certain selective media, which support the growth of only the desired fused cells (hybridomas). Examples of suitable cell lines for use in fusions with mouse cells include, but are not limited to, Sp-20, P3-X63/Ag8, P3-X63-Ag8.653, NS1/1. Ag 4 1, Sp210-Ag14, FO, NSO/U, MPC-11, MPC11-X45-GTG 1.7 and 5194/5XXO Bul. Examples of suitable cell lines used for fusions with rat cells include, but are not limited to, R210.RCY3, Y3-Ag 1.2.3, IR983F and 4B210. Other cell lines useful for cell fusions are U-266, GM1500-GRG2, LICR-LON-HMy2 and UC729-6. Additional exemplary methods for isolating monoclonal antibodies include screening plasma B cells from an animal after completion of the immunization schedule. (See e.g., Pedrioli A. and Oxenius A., *Trends in Immunology*, 2021, 42(12):1148-1158). Properties of monoclonal antibodies isolated by these methods can be optimized (e.g., to improve binding) using in vitro affinity maturation techniques such as yeast display-based affinity maturation known in the art. (see e.g., Cherf, G. M. and Cochran, J. R., *Yeast Surface Display: Methods, Protocols, and Applications*. 2015, 155-175).

[0160] In some embodiments, the anti-CB1 antibodies or antigen-binding fragments of the invention are chimeric or humanized antibodies or antigen-binding fragments thereof based upon the CDR and variable region sequences of the antibodies described herein. A chimeric antibody is an antibody composed of protein segments from different anti-

bodies that are covalently joined to produce functional immunoglobulin light or heavy chains or binding fragments thereof. Generally, a portion of the heavy chain and/or light chain is identical with or homologous to a corresponding sequence in antibodies derived from a particular species or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is/are identical with or homologous to a corresponding sequence in antibodies derived from another species or belonging to another antibody class or subclass. For methods relating to chimeric antibodies, see, for example, U.S. Pat. No. 4,816,567 and Morrison et al., 1985, *Proc. Natl. Acad. Sci. USA* 81:6851-6855, both of which are hereby incorporated by reference in their entireties.

[0161] Generally, the goal of making a chimeric antibody is to create a chimera in which the number of amino acids from the intended species is maximized. One example is the “CDR-grafted” antibody, in which the antibody comprises one or more CDRs from a particular species or belonging to a particular antibody class or subclass, while the remainder of the antibody chain(s) is/are identical with or homologous to a corresponding sequence in antibodies derived from another species or belonging to another antibody class or subclass. CDR grafting is described, for example, in U.S. Pat. Nos. 6,180,370, 5,693,762, 5,693,761, 5,585,089, and 5,530,101. For use in humans, the variable region or selected CDRs from a rodent or rabbit antibody often are grafted into a human antibody, replacing the naturally-occurring variable regions or CDRs of the human antibody.

[0162] One useful type of chimeric antibody is a “humanized” antibody. Generally, a humanized antibody is produced from a monoclonal antibody raised initially in a non-human animal, such as a rodent or rabbit. Certain amino acid residues in this monoclonal antibody, typically from non-antigen recognizing portions of the antibody, are modified to be homologous to corresponding residues in a human antibody of corresponding isotype. Humanization can be performed, for example, using various methods by substituting at least a portion of a rodent or rabbit variable region for the corresponding regions of a human antibody (see, e.g., U.S. Pat. Nos. 5,585,089, and 5,693,762; Jones et al., 1986, *Nature* 321:522-525; Riechmann et al., 1988, *Nature* 332:323-27; and Verhoeyen et al., 1988, *Science* 239:1534-1536).

[0163] In one aspect, the CDRs of the light and heavy chain variable regions of the antibodies provided herein (see, Tables 7, 10, 19, 26 and 30) are grafted to framework regions (FRs) from antibodies from the same, or a different, phylogenetic species. For example, the CDRs of the heavy and light chain variable regions listed in Tables 7, 10, 19, 26 and 30 can be grafted to consensus human FRs. To create consensus human FRs, FRs from several human heavy chain or light chain amino acid sequences may be aligned to identify a consensus amino acid sequence. Alternatively, the grafted variable regions from the one heavy or light chain may be used with a constant region that is different from the constant region of that particular heavy or light chain as disclosed herein.

[0164] In certain embodiments, the anti-CB1 antibodies or antigen-binding fragments of the invention are fully human antibodies or antigen-binding fragments thereof. A “fully human antibody” is an antibody that comprises variable and constant regions derived from or indicative of human germline immunoglobulin sequences. One specific means pro-

vided for implementing the production of fully human antibodies is the “humanization” of the mouse humoral immune system. Introduction of human immunoglobulin (Ig) loci into mice in which the endogenous Ig genes have been inactivated is one means of producing fully human monoclonal antibodies (mAbs) in mouse, an animal that can be immunized with any desirable antigen. Using fully human antibodies can minimize the immunogenic and allergic responses that can sometimes be caused by administering mouse or mouse-derived mAbs to humans as therapeutic agents.

[0165] The transgenic mice described above, referred to as “HuMab” mice, contain a human immunoglobulin gene minilocus that encodes unrearranged human heavy (mu and gamma) and kappa light chain immunoglobulin sequences, together with targeted mutations that inactivate the endogenous mu and kappa chain loci (Lonberg et al., 1994, *Nature* 368:856-859). Accordingly, the mice exhibit reduced expression of mouse IgM and kappa proteins and in response to immunization, the introduced human heavy and light chain transgenes undergo class switching and somatic mutation to generate high affinity human IgG kappa monoclonal antibodies (Lonberg and Huszar, 1995, *Intern. Rev. Immunol.* 13: 65-93; Harding and Lonberg, 1995, *Ann. N.Y. Acad. Sci.* 764:536-546). The preparation of HuMab mice is described in detail in Taylor et al., 1992, *Nucleic Acids Research* 20:6287-6295; Chen et al., 1993, *International Immunology* 5:647-656; Tuailon et al., 1994, *J. Immunol.* 152:2912-2920; Lonberg et al., 1994, *Nature* 368:856-859; Lonberg, 1994, *Hand book of Exp. Pharmacology* 113:49-101; Taylor et al., 1994, *International Immunology* 6:579-591; Lonberg and Huszar, 1995, *Intern. Rev. Immunol.* 13:65-93; Harding and Lonberg, 1995, *Ann. N.Y. Acad. Sci.* 764:536-546; Fishwild et al., 1996, *Nature Biotechnology* 14:845-851; the foregoing references are hereby incorporated by reference in their entireties. One particular transgenic mouse line suitable for generation of fully human anti-CB1 antibodies is the XenoMouse® transgenic mouse line described in U.S. Pat. Nos. 6,114,598; 6,162,963; 6,833,268; 7,049,426; 7,064,244; Green et al., 1994, *Nature Genetics* 7:13-21; Mendez et al., 1997, *Nature Genetics* 15:146-156; Green and Jakobovitis, 1998, *J. Ex. Med.* 188:483-495; Green, 1999, *Journal of Immunological Methods* 231:11-23; Kellerman and Green, 2002, *Current Opinion in Biotechnology* 13, 593-597, all of which are hereby incorporated by reference in their entireties. Human-derived antibodies can also be generated using phage display techniques. Phage display is described in e.g., Dower et al., WO 91/17271, McCafferty et al., WO 92/01047, and Caton and Koprowski, 1990, *Proc. Natl. Acad. Sci. USA*, 87:6450-6454, each of which is incorporated herein by reference in its entirety.

[0166] In certain embodiments, the anti-CB1 antibodies and antigen-binding fragments of the invention may comprise one or more mutations or modifications to a constant region. For example, the heavy chain constant regions or the Fc regions of the anti-CB1 antibodies may comprise one or more amino acid substitutions that affect the glycosylation, effector function, and/or Fcγ receptor binding of the antibody. The term “Fc region” refers to the C-terminal region of an immunoglobulin heavy chain, which may be generated by papain digestion of an intact antibody. The Fc region of an immunoglobulin generally comprises two constant domains, a CH2 domain and a CH3 domain, and optionally

comprises a CH4 domain. In certain embodiments, the Fc region is an Fc region from an IgG1, IgG2, IgG3, or IgG4 immunoglobulin. In some embodiments, the Fc region comprises CH2 and CH3 domains from a human IgG1 or human IgG2 immunoglobulin. The Fc region may retain effector function, such as C1q binding, complement-dependent cytotoxicity (CDC), Fc receptor binding, antibody-dependent cell-mediated cytotoxicity (ADCC), and phagocytosis. In other embodiments, the Fc region may be modified to reduce or eliminate effector function and/or improve half-life as described in further detail below.

[0167] In some embodiments, the anti-CB1 antibodies of the invention comprise one or more amino acid substitutions in the Fc region to reduce effector function. An amino acid substitution in an amino acid sequence is typically designated herein with a one-letter abbreviation for the amino acid residue in a particular position, followed by the numerical amino acid position relative to an original sequence of interest, which is then followed by the one-letter abbreviation for the amino acid residue substituted in. For example, “C220S” symbolizes a substitution of a cysteine residue by a serine residue at amino acid position 220, relative to the original sequence of interest. Exemplary amino acid substitutions (according to the EU numbering scheme) that can reduce effector function include, but are not limited to, C220S, C226S, C229S, E233P, L234A, L234V, V234A, L234F, L235A, L235E, G237A, P238S, S267E, H268Q, N297A, N297G, V309L, E318A, L328F, A330S, A331S, P331S, or combinations of any of the foregoing.

[0168] Glycosylation can contribute to the effector function of antibodies, particularly IgG1 antibodies. Thus, in some embodiments, the anti-CB1 antibodies of the invention may comprise one or more amino acid substitutions that affect the level or type of glycosylation of the antibodies. Glycosylation of polypeptides is typically either N-linked or O-linked. N-linked refers to the attachment of the carbohydrate moiety to the side chain of an asparagine residue. The tri-peptide sequences asparagine-X-serine and asparagine-X-threonine, where X is any amino acid except proline, are the recognition sequences for enzymatic attachment of the carbohydrate moiety to the asparagine side chain. Thus, the presence of either of these tri-peptide sequences in a polypeptide creates a potential glycosylation site. O-linked glycosylation refers to the attachment of one of the sugars N-acetylgalactosamine, galactose, or xylose, to a hydroxyamino acid, most commonly serine or threonine, although 5-hydroxyproline or 5-hydroxylysine may also be used.

[0169] In some embodiments, glycosylation of the anti-CB1 antibodies described herein is decreased or eliminated by removing one or more glycosylation sites, e.g., from the Fc region of the antibody. In some embodiments, the anti-CB1 antibody is an aglycosylated human monoclonal antibody, e.g., an aglycosylated human IgG1 monoclonal antibody. Amino acid substitutions that eliminate or alter N-linked glycosylation sites can reduce or eliminate N-linked glycosylation of the antibody. In certain embodiments, the anti-CB1 antibodies described herein comprise a heavy chain mutation at position N297 (according to the EU numbering scheme), such as N297Q, N297A, or N297G. In some embodiments, the anti-CB1 antibodies of the invention comprise an Fc region from a human IgG1 antibody with a mutation at position N297. In one particular embodiment, the anti-CB1 antibodies of the invention comprise an Fc

region from a human IgG1 antibody with a N297G mutation. For instance, in some embodiments, the anti-CB1 antibodies of the invention comprise a heavy chain constant region comprising the sequence of SEQ ID NO: 524.

[0170] To improve the stability of molecules comprising a N297 mutation, the Fc region of the anti-CB1 antibodies may be further engineered. For instance, in some embodiments, one or more amino acids in the Fc region are substituted with cysteine to promote disulfide bond formation in the dimeric state. Residues corresponding to V259, A287, R292, V302, L306, V323, or I332 (according to the EU numbering scheme) of an IgG1 Fc region may thus be substituted with cysteine. Preferably, specific pairs of residues are substituted with cysteine such that they preferentially form a disulfide bond with each other, thus limiting or preventing disulfide bond scrambling. Preferred pairs include, but are not limited to, A287C and L306C, V259C and L306C, R292C and V302C, and V323C and I332C. In certain embodiments, the anti-CB1 antibodies described herein comprise an Fc region from a human IgG1 antibody with mutations R292C and V302C. In such embodiments, the Fc region may also comprise a N297 mutation, such as a N297G mutation. In some embodiments, the anti-CB1 antibodies of the invention comprise a heavy chain constant region comprising the sequence of SEQ ID NO: 525.

[0171] Modifications of the anti-CB1 antibodies of the invention to increase serum half-life (e.g., in vivo half-life) also may be desirable. One approach for achieving increased serum half-life is by amino acid mutations in certain positions in the constant domain such as those described in WO2002060919. Such mutations increase the affinity of the antibody for the FcRn and its serum half-life. In some embodiments, the anti-CB1 antibodies described herein comprise an Fc region from a human IgG1 antibody with one or more mutations at amino acid positions 251-256 (according to the EU numbering scheme). In some embodiments, the Fc region comprises one or more mutations at positions 252, 254 and 256. In some embodiments, the Fc region comprises M252Y, S254T, and T256E (according to the EU numbering scheme). In some embodiments, the anti-CB1 antibodies of the invention comprise a heavy chain constant region comprising the sequence of SEQ ID NO: 527.

[0172] In some embodiments, Fc region of the anti-CB1 antibodies described herein may comprise modifications that alter glycosylation, stability and serum half-life of the antibodies. For example, in some embodiments, the anti-CB1 antibodies described herein comprise an Fc region from a human IgG1 antibody with mutations R292C and V302C, a N297 mutation, such as a N297G mutation (according to the EU numbering scheme), and may also comprise one or more mutations at positions 252, 254 and 256, such as M252Y, S254T, and T256E (according to the EU numbering scheme). In some embodiments, the anti-CB1 antibodies of the invention comprise a heavy chain constant region comprising the sequence of SEQ ID NO: 526.

[0173] It is envisioned that the anti-CB1 antibodies and antigen binding fragments described herein do not cross-react with (i.e., essentially do not recognize or bind to) human cannabinoid receptor type 2 (CB2). Human CB2 is a G protein-coupled receptor from the cannabinoid receptor family and is encoded by the CNR2 gene. Human CB2 contains 360 amino acids (UniProtKB—P34972, SEQ ID NO: 560) and shares about 44% sequence similarity with

huCB1. The anti-CB1 antibodies and antigen binding fragments described herein also do not cross-react with mouse CB1 protein (UniProtKB—P47746, SEQ ID NO: 561).

[0174] The present invention includes one or more polynucleotides or nucleic acids encoding the anti-CB1 antibodies or antigen-binding fragments described herein. In addition, the present invention encompasses vectors comprising the nucleic acids, host cells or cell lines comprising the nucleic acids, and methods of making the anti-CB1 antibodies and antigen-binding fragments of the invention. The nucleic acids comprise, for example, polynucleotides that encode all or part of an antibody or antigen-binding fragment, for example, one or both chains of an antibody of the invention, or a fragment, derivative, or variant thereof, PCR primers or sequencing primers for identifying, analyzing, mutating or amplifying a polynucleotide encoding a polypeptide, anti-sense oligonucleotides for inhibiting expression of a polynucleotide, and complementary sequences of the foregoing. The nucleic acids can be any length as appropriate for the desired use or function, and can comprise one or more additional sequences, for example, regulatory sequences, and/or be part of a larger nucleic acid, for example, a vector. Nucleic acid molecules of the invention include DNA and RNA in both single-stranded and double-stranded form, as well as the corresponding complementary sequences. DNA includes, for example, cDNA, genomic DNA, chemically synthesized DNA, DNA amplified by PCR, and combinations thereof. The nucleic acid molecules of the invention include full-length genes or cDNA molecules as well as a combination of fragments thereof. The nucleic acids of the invention can be derived from human sources as well as non-human species.

[0175] Relevant amino acid sequences from an immunoglobulin or region thereof (e.g., variable region, Fc region, etc.) or polypeptide of interest may be determined by direct protein sequencing, and suitable encoding nucleotide sequences can be designed according to a universal codon table. Alternatively, genomic or cDNA encoding monoclonal antibodies or binding fragments thereof of the invention can be isolated and sequenced from cells producing such antibodies using conventional procedures (e.g., by using oligonucleotide probes that are capable of binding specifically to genes encoding the heavy and light chains of the monoclonal antibodies).

[0176] The present invention also includes vectors comprising one or more nucleic acids encoding one or more components of the antibodies or antigen-binding fragments of the invention (e.g., variable regions, light chains, and heavy chains). The term “vector” refers to any molecule or entity (e.g., nucleic acid, plasmid, bacteriophage or virus) used to transfer protein coding information into a host cell. Examples of vectors include, but are not limited to, plasmids, viral vectors, non-episomal mammalian vectors and expression vectors, for example, recombinant expression vectors. The term “expression vector” or “expression construct” as used herein refers to a recombinant DNA molecule containing a desired coding sequence and appropriate nucleic acid control sequences necessary for the expression of the operably linked coding sequence in a particular host cell. An expression vector can include, but is not limited to, sequences that affect or control transcription, translation, and, if introns are present, affect RNA splicing of a coding region operably linked thereto. Nucleic acid sequences necessary for expression in prokaryotes include a promoter,

optionally an operator sequence, a ribosome binding site and possibly other sequences. Eukaryotic cells are known to utilize promoters, enhancers, and termination and polyadenylation signals.

[0177] A secretory signal peptide sequence can also, optionally, be encoded by the expression vector, operably linked to the coding sequence of interest, so that the expressed polypeptide can be secreted by the recombinant host cell, for more facile isolation of the polypeptide of interest from the cell, if desired. For instance, in some embodiments, signal peptide sequences may be appended/fused to the amino terminus of any of the variable region polypeptide sequences listed in Tables 8, 11, 20, 27, and 31. In certain embodiments, a signal peptide having the amino acid sequence of MDMRVPAQLGLLLLWLRGARC (SEQ ID NO: 541) is fused to the amino terminus of any of the variable region polypeptide sequences in Tables 8, 11, 20, 27, and 31. In other embodiments, a signal peptide having the amino acid sequence of MAWALLLLTLTQGTGSWA (SEQ ID NO: 542) is fused to the amino terminus of any of the variable region polypeptide sequences in Tables 8, 11, 20, 27, and 31. In still other embodiments, a signal peptide having the amino acid sequence of MTCSPLLLLTLIHCTGSWA (SEQ ID NO: 543) is fused to the amino terminus of any of the variable region polypeptide sequences in Tables 8, 11, 20, 27, and 31. Other suitable signal peptide sequences that can be fused to the amino terminus of the variable region polypeptide sequences described herein include: MEAPAQLFLLLLWLPDPTTG (SEQ ID NO: 544), MEWTWRVFLVAAATGAHS (SEQ ID NO: 545), METPAQLFLLLLWLPDPTTG (SEQ ID NO: 546), METPAQLFLLLLWLPDPTTG (SEQ ID NO: 547), MKHLWFFLLLVAAPRWVLS (SEQ ID NO: 548), MEWSWVFLFFLSVTTGVHS (SEQ ID NO: 549), MDIRAPTQLLGLLLLWLPGAKC (SEQ ID NO: 550), MDIRAPTQLLGLLLLWLPGARC (SEQ ID NO: 551), MDTRAPTQLLGLLLLWLPGATF (SEQ ID NO: 552), MDTRAPTQLLGLLLLWLPGARC (SEQ ID NO: 553), METGLRWLLLVAVLKGVQC (SEQ ID NO: 554), METGLRWLLLVAVLKGVQCQE (SEQ ID NO: 555), MDMRAPTQLLGLLLLWLPGARC (SEQ ID NO: 556), MKILILGIFLFLCSTPAWA (SEQ ID NO: 557), and MRT-LAILAAILLVALQAQA (SEQ ID NO: 558). Other signal or secretory peptides are known to those of skill in the art and may be fused to any of the variable region polypeptide chains listed in Tables 8, 11, 20, 27, and 31, for example, to facilitate or optimize expression in particular host cells.

[0178] Typically, expression vectors used in the host cells to produce the anti-CB1 antibodies and antigen-binding fragments of the invention will contain sequences for plasmid maintenance and for cloning and expression of exogenous nucleotide sequences encoding the components of the antibodies and antigen-binding fragments. Such sequences, collectively referred to as “flanking sequences,” in certain embodiments will typically include one or more of the following nucleotide sequences: a promoter (e.g., a promoter suitable for a host cell), one or more enhancer sequences (e.g., a viral enhancer sequence), an origin of replication (e.g., plasmid origin for bacteria or viral origins for mammalian system), a transcriptional termination sequence, a complete intron sequence containing a donor and acceptor splice site, a sequence encoding a signal sequence for polypeptide secretion, a ribosome binding site (e.g., a Shine-Dalgarno sequence (prokaryotes) or a Kozak

sequence (eukaryotes)), a polyadenylation sequence, a polylinker region for inserting the nucleic acid encoding the polypeptide to be expressed, and a selectable marker element (e.g., sequences that encode proteins conferring resistance to antibiotics or complement auxotrophic deficiencies of host cells). Franking sequences and vectors are well known in the art. For example, expression vectors are commercially available. In addition, desired modifications may be made to commercially available expression vectors or constructed using sequences known in the art. Nucleic acids encoding the different components of the anti-CB1 antibodies or binding fragments (e.g., HV and/or LV) may be inserted into proper sites of the same vector or different vectors for expression in a host cell.

[0179] A “host cell” refers to a cell that has been transformed, or is capable of being transformed, with a nucleic acid and thereby expresses a gene of interest. The term includes the progeny of the parent cell, whether or not the progeny is identical in morphology or in genetic make-up to the original parent cell, so long as the gene of interest is present. A host cell that comprises an isolated nucleic acid of the invention, preferably operably linked to at least one expression control sequence (e.g. promoter or enhancer), is a “recombinant host cell.” Methods for transforming or transfecting vectors into a host cell is well known in the art.

[0180] Suitable host cells include prokaryotic or eukaryotic cells, and also include but are not limited to bacteria, yeast cells, fungi cells, plant cells, and animal cells such as insect cells and mammalian cells, e.g., murine, rat, macaque or human.

[0181] Prokaryotic host cells include eubacteria, such as Gram-negative or Gram-positive organisms, for example, Enterobacteriaceae such as *Escherichia*, e.g., *E. coli*, *Enterobacter*, *Erwinia*, *Klebsiella*, *Proteus*, *Salmonella*, e.g., *Salmonella typhimurium*, *Serratia*, e.g., *Serratia marcescans*, and *Shigella*, as well as *Bacillus*, such as *B. subtilis* and *B. licheniformis*, *Pseudomonas*, and *Streptomyces*. Eukaryotic microbes such as filamentous fungi or yeast are suitable cloning or expression hosts for recombinant polypeptides. *Saccharomyces cerevisiae*, or common baker's yeast, is the most commonly used among lower eukaryotic host microorganisms. However, a number of other genera, species, and strains are commonly available and useful herein, such as *Pichia*, e.g. *P. pastoris*, *Schizosaccharomyces pombe*; *Kluyveromyces*, *Yarrowia*; *Candida*; *Trichoderma reesia*; *Neurospora crassa*; *Schwanniomyces*, such as *Schwanniomyces occidentalis*; and filamentous fungi, such as, e.g., *Neurospora*, *Penicillium*, *Tolypocladium*, and *Aspergillus* hosts such as *A. nidulans* and *A. niger*.

[0182] Host cells for the expression of glycosylated antibodies and antigen-binding fragments can be derived from multicellular organisms. Examples of invertebrate cells include plant and insect cells. Numerous baculoviral strains and variants and corresponding permissive insect host cells from hosts such as *Spodoptera frugiperda* (caterpillar), *Aedes aegypti* (mosquito), *Aedes albopictus* (mosquito), *Drosophila melanogaster* (fruitfly), and *Bombyx mori* have been identified. A variety of viral strains for transfection of such cells are publicly available, e.g., the L-1 variant of *Autographa californica* NPV and the Bm-5 strain of *Bombyx mori* NPV.

[0183] Vertebrate host cells are also suitable hosts, and recombinant production of antibodies and antigen-binding fragments from such cells has become routine procedure.

Mammalian cell lines available as hosts for expression are well known in the art and include, but are not limited to, immortalized cell lines available from the American Type Culture Collection (ATCC), including but not limited to Chinese hamster ovary (CHO) cells, including CHOK1 cells (ATCC CCL61), DXB-11, DG-44, and Chinese hamster ovary cells/-DHFR (CHO, Urlaub et al., Proc. Natl. Acad. Sci. USA 77: 4216, 1980); monkey kidney CV1 line transformed by SV40 (COS-7, ATCC CRL 1651); human embryonic kidney line (293 or 293 cells subcloned for growth in suspension culture, (Graham et al., J. Gen Virol. 36: 59, 1977); baby hamster kidney cells (BHK, ATCC CCL 10); mouse sertoli cells (TM4, Mather, Biol. Reprod. 23: 243-251, 1980); monkey kidney cells (CV1 ATCC CCL 70); African green monkey kidney cells (VERO-76, ATCC CRL-1587); human cervical carcinoma cells (HELA, ATCC CCL 2); canine kidney cells (MDCK, ATCC CCL 34); buffalo rat liver cells (BRL 3A, ATCC CRL 1442); human lung cells (W138, ATCC CCL 75); human hepatoma cells (Hep G2, HB 8065); mouse mammary tumor (MMT 060562, ATCC CCL51); TRI cells (Mather et al., Annals N.Y Acad. Sci. 383: 44-68, 1982); MRC 5 cells or FS4 cells; mammalian myeloma cells, and a number of other cell lines. CHO cells are preferred host cells in some embodiments for expressing the anti-CB1 antibodies and antigen-binding fragments of the invention.

[0184] Host cells are transformed or transfected with the above-described nucleic acids or vectors for production of anti-CB1 antibodies or antigen-binding fragments and are cultured in conventional nutrient media modified as appropriate for inducing promoters, selecting transformants, or amplifying the genes encoding the desired sequences. Thus, the present invention also provides a method for producing an anti-CB1 antibody or antigen-binding fragment described herein comprising culturing a host cell comprising one or more expression vectors described herein in a culture medium under conditions permitting expression of the antibody or antigen-binding fragment encoded by the one or more expression vectors; and recovering the antibody or antigen-binding fragment from the culture medium or host cell.

[0185] Upon culturing the host cells, the antibody or antigen-binding fragment can be produced intracellularly, in the periplasmic space, or directly secreted into the medium. If the antibody or antigen-binding fragment is produced intracellularly, as a first step, the particulate debris, either host cells or lysed fragments, is removed, for example, by centrifugation or ultrafiltration. The antibody or antigen-binding fragment can be purified using methods known in the art, for example, hydroxyapatite chromatography, cation or anion exchange chromatography, or preferably affinity chromatography, using the antigen(s) of interest or protein A or protein G as an affinity ligand. Protein A can be used to purify proteins that include polypeptides that are based on human $\gamma 1$, $\gamma 2$, or $\gamma 4$ heavy chains (Lindmark et al., J. Immunol. Meth. 62: 1-13, 1983). Protein G is recommended for all mouse isotypes and for human 73 (Guss et al., EMBO J. 5: 1567-1575, 1986). The matrix to which the affinity ligand is attached is most often agarose, but other matrices are available. Mechanically stable matrices such as controlled pore glass or poly(styrenedivinyl)benzene allow for faster flow rates and shorter processing times than can be achieved with agarose. Where the protein comprises a CH3 domain, the Bakerbond ABX™ resin (J. T. Baker, Phillips-

burg, N.J.) is useful for purification. Other techniques for protein purification such as ethanol precipitation, Reverse Phase HPLC, chromatofocusing, SDS-PAGE, and ammonium sulfate precipitation are also possible depending on the particular antibody or antigen-binding fragment to be recovered.

Compositions

[0186] Also disclosed herein are compositions comprising the anti-CB1 antibodies or antigen binding fragments. Suitable compositions include those that comprise the anti-CB1 antibodies or antigen binding fragments described and exemplified herein and one or more pharmaceutically acceptable excipients. "Pharmaceutically-acceptable" refers to molecules, and compounds that are non-toxic to human recipients at the dosages and concentrations employed and/or do not produce allergic or adverse reactions when administered to humans. Pharmaceutical compositions of the invention include, but are not limited to, liquid, frozen, and lyophilized compositions.

[0187] In some embodiments, the pharmaceutical composition may contain materials for modifying, maintaining or preserving, for example, the pH, osmolarity, viscosity, clarity, color, isotonicity, odor, sterility, stability, rate of dissolution or release, adsorption or penetration of the composition. In such embodiments, suitable formulation materials include, but are not limited to, buffers (such as acetate buffer, Tris-HCl, citrate buffer, phosphate buffer or buffers containing other organic acids); bulking agents or fillers such as monosaccharides; disaccharides (e.g., sucrose); and other carbohydrates; proteins (such as serum albumin, gelatin or immunoglobulins); amino acids (e.g., glycine, glutamine, asparagine, histidine, arginine, or lysine), coloring and diluting agents; emulsifying agents; hydrophilic polymers (such as polyvinylpyrrolidone); chelating agents (such as ethylenediamine tetraacetic acid (EDTA)); complexing agents (such as caffeine, polyvinylpyrrolidone, beta-cyclodextrin or hydroxypropyl-beta-cyclodextrin); low molecular weight polypeptides; salt-forming counterions (such as sodium); preservatives (such as benzalkonium chloride, benzoic acid, salicylic acid, thimerosal, phenethyl alcohol, methylparaben, propylparaben, chlorhexidine, sorbic acid or hydrogen peroxide); solvents (such as glycerin, propylene glycol or polyethylene glycol); suspending agents; surfactants or wetting agents (such as pluronics, PEG, sorbitan esters, polysorbates such as polysorbate 20, polysorbate 80, triton, tromethamine, lecithin, cholesterol, tyloxapal); stability enhancing agents (such as sucrose or sorbitol); tonicity enhancing agents (such as alkali metal halides, preferably sodium or potassium chloride, mannitol, sorbitol); delivery vehicles; diluents; excipients and/or pharmaceutical adjuvants. Methods and suitable materials for formulating molecules for therapeutic use are known in the pharmaceutical arts, and are described, for example, in REMINGTON'S PHARMACEUTICAL SCIENCES, 18th Edition, (A. R. Genrmo, ed.), 1990, Mack Publishing Company.

[0188] In some embodiments, the composition disclosed herein is a liquid composition such as an aqueous solution. In some embodiments, the composition comprises a buffering agent, a tonicity or bulking agent, a surfactant, and has a pH in the range of from about 4.5 to about 7.5. Optionally, the composition further comprises a salt and/or a preservative.

[0189] In some embodiments, the pH of the composition is in the range of from about 4.5 to about 7.0, for example, from about 4.5 to about 6.5, or from about 4.8 to about 5.5, or alternatively about 5.0. Examples of buffers that are suitable for a pH within this range include acetate (e.g., sodium acetate), succinate (such as sodium succinate), gluconate, histidine, citrate and other organic acid buffers. The buffer concentration can be from about 1 mM to about 200 mM, or from about 10 mM to about 60 mM, depending, for example, on the buffer and the desired isotonicity of the composition.

[0190] A tonicity agent, which may also stabilize the antibody or antigen-binding fragment, may be included in the composition. Exemplary tonicity agents include polyols, such as sucrose, mannitol, or trehalose. Preferably the aqueous formulation is isotonic, although hypertonic or hypotonic solutions may be suitable. Exemplary concentrations of the polyol in the formulation may range from about 1% to about 15% w/v.

[0191] A surfactant may also be added to the formulation to reduce aggregation of the formulated antibody or antigen-binding fragment and/or minimize the formation of particulates in the formulation and/or reduce adsorption. Exemplary surfactants include nonionic surfactants such as polysorbates (e.g. polysorbate 20 or polysorbate 80) or poloxamers (e.g. poloxamer 188). Exemplary concentrations of surfactant may range from about 0.001% to about 0.5%, or from about 0.005% to about 0.2%, or alternatively from about 0.004% to about 0.01% w/v.

[0192] In one embodiment, the composition contains the above-identified agents (i.e. antibody or antigen-binding fragment, buffer, polyol and surfactant) and is essentially free of one or more preservatives, such as benzyl alcohol, phenol, m-cresol, chlorobutanol and benzethonium chloride. In another embodiment, a preservative may be included in the formulation, e.g., at concentrations ranging from about 0.1% to about 2%, or alternatively from about 0.5% to about 1%. In one embodiment, the composition may further comprise a salt such as sodium chloride or one or more other pharmaceutically acceptable excipients. One or more other pharmaceutically acceptable carriers, excipients or stabilizers such as those described in REMINGTON'S PHARMACEUTICAL SCIENCES, 18th Edition, (A. R. Genrmo, ed.), 1990, Mack Publishing Company, may be included in the formulation provided that they do not adversely affect the desired characteristics of the formulation.

[0193] The formulations to be used for in vivo administration must be sterile. The compositions of the invention may be sterilized by conventional, well-known sterilization techniques. For example, sterilization is readily accomplished by filtration through sterile filtration membranes. The resulting solutions may be packaged for use or filtered under aseptic conditions and lyophilized, the lyophilized preparation being combined with a sterile solution prior to administration.

[0194] In some embodiments, the composition comprising the anti-CB1 antibodies or antigen binding fragment and one or more pharmaceutically acceptable excipients is a solid composition such as a lyophilized cake. The process of lyophilization is often employed to stabilize polypeptides for long-term storage, particularly when the polypeptide is relatively unstable in liquid compositions. A lyophilization cycle is usually composed of three steps: freezing, primary drying, and secondary drying (see Williams and Polli, Jour-

nal of Parenteral Science and Technology, Volume 38, Number 2, pages 48-59, 1984). In the freezing step, the solution is cooled until it is adequately frozen. Bulk water in the solution forms ice at this stage. The ice sublimates in the primary drying stage, which is conducted by reducing chamber pressure below the vapor pressure of the ice, using a vacuum. Finally, sorbed or bound water is removed at the secondary drying stage under reduced chamber pressure and an elevated shelf temperature. The process produces a material known as a lyophilized cake. Thereafter the cake can be reconstituted prior to use. The standard reconstitution practice for lyophilized material is to add back a volume of pure water (typically equivalent to the volume removed during lyophilization), although dilute solutions of antibacterial agents are sometimes used in the production of pharmaceuticals for parenteral administration (see Chen, Drug Development and Industrial Pharmacy, Volume 18: 1311-1354, 1992).

[0195] Excipients have been noted in some cases to act as stabilizers for freeze-dried products (see Carpenter et al., Volume 74: 225-239, 1991). For example, known excipients include polyols (including mannitol, sorbitol and glycerol); sugars (including glucose and sucrose); and amino acids (including alanine, glycine and glutamic acid). In addition, polyols and sugars are also often used to protect polypeptides from freezing- and drying-induced damage and to enhance the stability during storage in the dried state. In general, sugars, in particular disaccharides, are effective in both the freeze-drying process and during storage. Other classes of molecules, including mono- and di-saccharides and polymers such as PVP, have also been reported as stabilizers of lyophilized products.

[0196] The formulations of the invention may be designed to be short-acting, fast-releasing, long-acting, or sustained-releasing as described herein. Thus, the pharmaceutical formulations may also be formulated for controlled release or for slow release. Suitable examples of sustained-release preparations include semipermeable matrices of solid hydrophobic polymers containing the antibody or antigen-binding fragment, which matrices are in the form of shaped articles, e.g., films, or microcapsule. Examples of sustained-release matrices include polyesters, hydrogels (for example, poly(2-hydroxyethyl-methacrylate), or poly(vinylalcohol)), polylactides (U.S. Pat. No. 3,773,919), copolymers of L-glutamic acid and ethyl-L-glutamate, non-degradable ethylene-vinyl acetate, degradable lactic acid-glycolic acid copolymers such as the Lupron DepotTM (injectable microspheres composed of lactic acid-glycolic acid copolymer and leuprolide acetate), and poly-D-(-)-3-hydroxybutyric acid. While polymers such as ethylene-vinyl acetate and lactic acid-glycolic acid enable release of molecules for over 100 days, certain hydrogels release proteins for shorter time periods. When encapsulated polypeptides remain in the body for a long time, they may denature or aggregate as a result of exposure to moisture at 37° C., resulting in a loss of biological activity and possible changes in immunogenicity. Rational strategies can be devised for stabilization depending on the mechanism involved. For example, if the aggregation mechanism is discovered to be intermolecular S—S bond formation through thio-disulfide interchange, stabilization may be achieved by modifying sulfhydryl residues, lyophilizing from acidic solutions, controlling moisture content, using appropriate additives, and developing specific polymer matrix compositions. Excipients and meth-

ods for making sustained-release compositions are described in e.g., REMINGTON'S PHARMACEUTICAL SCIENCES, 18th Edition, (A. R. Genrmo, ed.), 1990.

[0197] Compositions comprising the anti-CB1 antibodies or antigen binding fragments and one or more pharmaceutically acceptable excipients described above can be administered by any suitable means, including parenteral administration. Parenteral administration includes intravenous, intraarterial, intraperitoneal, intramuscular, intradermal or subcutaneous administration. Preferably, the dosing is given by injections, most preferably subcutaneous, intramuscular, or intravenous injections. Other administration methods are contemplated, including topical, particularly transdermal, transmucosal, rectal, oral or local administration e.g., through a catheter placed close to the desired site.

Use

[0198] Disclosed herein also includes the use of the anti-CB1 antibodies or antigen binding fragments or compositions comprising the same for inhibiting, reducing or neutralizing the amount, activity or signaling of the CB1 receptor (e.g., huCB1). The antibodies and antigen binding fragments of the invention are useful for the treatment, prevention and/or amelioration of any disease or disorder associated with or mediated by CB1 expression or activity, or treatable by blocking the interaction between CB1 and a CB1 ligand (e.g., a cannabinoid) or otherwise inhibiting CB1 activity and/or signaling, and/or promoting receptor internalization and/or decreasing cell surface receptor number. In various embodiments, the CB1 is a peripheral CB1.

[0199] Exemplary disorders, diseases and conditions that may be treated by the anti-CB1 antibodies or antigen-binding fragments described herein include obesity and comorbidities such as syndromic obesities including Prader-Willi syndrome, Alstrom syndrome, Bardet-Biedel syndrome (BBS), Albright Hereditary Osteodystrophy (AHO), and SIM1 deletion syndrome; diabetes and related complications (e.g., abnormal plasma glucose, insulin, and/or resistin levels); dyslipidemia (e.g., abnormal HDL, plasma cholesterol and/or triglyceride levels); liver diseases such as, for example, non-alcoholic steatohepatitis (NASH), non-alcoholic fatty liver disease (NAFLD), and primary biliary cirrhosis; fibrosis, for example, kidney fibrosis; chronic kidney disease (CKD) such as CKD driven by metabolic derangements; IgA nephropathy; renal disease; metabolic diseases, osteoporosis, atherosclerosis, inflammatory disease, cardiovascular disease, cancer, pain, systemic sclerosis, multiple sclerosis spasticity, glaucoma, and nicotine addiction.

[0200] Obesity refers to abnormal or excessive fat accumulation that may impair health of an individual and may be measured or classified by body mass index (BMI). BMI is defined as a person's weight (e.g., kilograms) divided by the square of his height in meters (e.g., kg/m²).

[0201] Thus, in one aspect, disclosed herein is a method for treating a disease or disorder in a subject responsive of antagonizing or inverse agonizing the CB1 receptor (e.g., huCB1), or in a subject in need of antagonizing or inverse agonizing the CB1 receptor, the method comprises administering the antibodies or antigen binding fragments described above. In some embodiments, the antibodies or antigen binding fragments are potent antagonist and/or inverse agonist of the CB1 receptor, for example they have a potency (IC₅₀) of less than about 10 nM as measured in

a cell-based cAMP assay (e.g., less than about 8 nM, less than about 5 nM, less than about 3 nM, or less than about 1 nM). In some embodiments, the antibodies or antigen binding fragments have an IC₅₀ less than about 10 nM (e.g., less than about 5 nM, less than about 3 nM, or less than about 1 nM) as measured in a cell-based cAMP assay (e.g., a cell-based cAMP assay in the presence of CP 55,940). In some embodiments, the antibodies or antigen binding fragments comprise a heavy and/or a light chain CDR as described above. CDR sequences of various exemplary antibodies are listed in Tables 7, 10, 19, 26, and 30. In some embodiments, the antibodies or antigen binding fragments comprise a heavy and/or light chain variable region as described above. In other embodiments, the antibodies comprise a heavy and/or light chain as described above. Heavy and light chain variable region sequences of exemplary antibodies are listed in Tables 8, 11, 20, 27 and 31. Exemplary heavy and light chain constant region sequences are listed in Tables 1 and 2. In various embodiments, the antibodies or antigen binding fragments is administered in an effective amount for treating the disorder or disease (e.g., cause a potential benefit or therapeutic effect). In various embodiments, the CB1 receptor is a peripheral CB1 receptor.

[0202] The terms “subject in need” or those “in need of treatment” includes those suffering from the disorder or disease, as well as those in which the disorder or disease has not yet clinically manifested. The “subject” includes human and other mammalian subjects that receive either prophylactic or therapeutic treatment. In preferred embodiments, the subject is human.

[0203] In some embodiments, the disorder or disease is obesity, diabetes, dyslipidemia, metabolic diseases, fibrosis, liver disease such as non-alcoholic steatohepatitis (NASH) or NAFLD, primary biliary cirrhosis, renal disease, kidney fibrosis, chronic kidney disease (e.g. chronic kidney disease driven by metabolic derangements), IgA nephropathy, osteoporosis, atherosclerosis, cardiovascular disease, cancer, or inflammatory disease. In some embodiments, the disorder or disease is obesity. In some embodiments, the subject being treated has chronic kidney disease such as chronic kidney disease driven by metabolic derangements or kidney injury. In some embodiments, the subject being treated has obesity with a BMI of 27 kg/m² or higher. In some embodiments, the subject being treated has obesity with a BMI of 30 kg/m² or higher. In other embodiments, the disorder or disease is a liver disease (e.g., NASH).

[0204] In some embodiments, the administration or treatment results in one or more of the following in the subject: reduced body weight, reduced appetite, improved metabolic parameters, reduced blood glucose levels (e.g., reduced fasting blood glucose levels), reduced insulin levels, reduced HbA1c levels, reduced blood lipid levels (e.g., reduced HDL, cholesterol and/or triglyceride levels), reduced kidney injury, fibrosis, inflammation, and improved kidney function.

Kits

[0205] In some embodiments, disclosed herein is a kit comprising compositions comprising the anti-CB1 antibodies or antigen binding fragments and one or more pharmaceutically acceptable excipients described above. For example, such a kit includes a composition comprising anti-CB1 antibodies or antigen binding fragments described

herein such as an aqueous or lyophilized composition described above, packaged in a container such as a sealed bottle, vessel, single-use or multi-use vial, prefilled syringe, or prefilled injection device. A suitable vehicle or carrier may be provided, such as water for injection or physiological saline solution, for lyophilized compositions. Other materials common in formulations for parenteral administration may also be included. In one aspect, the composition is packaged in a unit dosage form. The kit may further include a device suitable for administering the formulation according to a specific route of administration. Preferably, the kit further contains a label that describes use of the antibodies or antigen binding fragments or compositions comprising the same.

[0206] All of the references cited herein, including patents, patent applications, literature publications, and the like, are hereby incorporated in their entireties by reference.

[0207] The following examples are given merely to illustrate the present invention and not in any way to limit its scope.

EXAMPLES

Example 1. Deep Mutational Scan and Design of Affinity Maturation Libraries

[0208] Materials and Methods: CB1-ND preparation. For nanodisc preparation, CB1 construct described in Tian H. et al., *Cell*, 167(3) (2016) 750-762 was expressed with a histidine purification tag in Sf9 cells and purified using affinity and size exclusion chromatography in HEPES buffer containing 0.05% DDM; 0.01% CHS and 10 uM ligand-1981202 #5. POPC lipid was solubilized with the help of 100 mM sodium cholate and mixed with MSP1D1 scaffold protein and CB1 at the molar ratio of 480:8:1. Nanodisc formation was achieved by incubation with SM2 BioBeads for 1 h at 4° C.

[0209] Binding to CB1-ND and empty-ND. The propagation and induction of yeast cells displaying antibodies for binding experiments have been described by Luna V. et al. (Luna V. et al., *European Polymer Journal* 109 (2018) 483-488). At least 2×10⁵ yeast cells displaying IgGs were removed and washed with 1× phosphate-buffered saline (no Mg²⁺, no Ca²⁺) supplemented with 5 mg/mL bovine serum albumin (PBS-B). His-tagged huCB1-ND or empty-ND was added to the desired concentration and incubated with the yeast cells at 25° C. for 1 hour. The cells were washed with PBS-B and then incubated with an allophycocyanin (APC)-conjugated anti-His antibody to detect binding and a phycoerythrin (PE)-conjugated anti-huFc antibody to detect IgG surface display. The cells were washed with PBS-B before flow cytometry analysis using a FACS-Canto (BD Biosciences). APC and PE median fluorescence intensities (MFI) were recorded.

[0210] FACS library sorting. At least 10⁷ cells were incubated with antigen as described above. Following fluorescent antibody staining, the cells were sorted using a FACS Aria (BD Biosciences) using the appropriate settings for allophycocyanin and phycoerythrin detection.

[0211] The anti-CB1 antibody 10D10 was previously generated by immunizing Xenomouse with HEK293 cells expressing human CB1. (See WO2014210205). Although affinity maturation is a common strategy employed to improve the potency of antagonist antibodies, previous efforts of affinity maturing 10D10 failed to make antibodies

with sufficient potency (e.g., potency in the single digit nM range in a cell-based cAMP assay) to produce desired therapeutic effects (e.g., reduce body weight) without CNS-related side effects. A lack of high-quality, soluble preparations of complex membrane proteins such as CB1 was considered to be a major contributor of previous failures. The development of CB1 nanodisc (ND) and CB1 styrene maleic acid lipoparticles (SMALP) greatly improved the quality of soluble preparations of CB1. (Luna V. et al., European Polymer Journal 109 (2018) 483-488). Both CB1-ND and -SMALP are purified reagents that are stable after multiple freeze/thaw cycles and compatible with long-term storage. Furthermore, these reagents can be added to binding experiments at user-defined concentrations of huCB1. The availability of these reagents enabled additional affinity maturation efforts.

[0212] Deep mutational scanning of the antibody was carried out to inform the design of affinity maturation libraries. Deep mutational scanning entails generating comprehensive libraries of point mutations, performing binding selections under different conditions, and evaluating changes in mutation frequencies after selections with high-throughput sequencing. (See e.g., Araya, C. L. and Fowler, D. M., Trends Biotechnol. 2011 September; 29(9): 435-442). Identification of the subset of mutations associated with improved and specific huCB1 binding enables focused combinatorial library designs for affinity maturation without the benefit of a high-resolution antibody-antigen co-crystal structure.

[0213] Fifty-three and thirty-seven two positions within the 10D10 heavy chain (HC) and light chain (LC), respectively, were chosen for single-site saturation mutagenesis. Point mutation IgG-yeast libraries were generated wherein the comprehensive set of 10D10 point mutations in the heavy and light chain CDRs and the heavy chain (HC) and light chain FR3 de-loops were split into four libraries (2 each for heavy and light chain). Each library was sorted using FACS for surface expression, and separately against huCB1-ND at two different stringencies and against empty-ND to assess non-specific binding. The lower-stringency huCB1-ND binding gate collected all cells exhibiting at least 10D10-like binding (10-20% of the population), whereas the higher-stringency gate collected only the top binders (1.5-6% of the population). After FACS, the plasmids from the sorted pools were harvested using a ZymoPrep II kit (Zymo Research) and amplicons were generated by minimal PCR amplification of the recovered plasmids.

[0214] Illumina MiSeq was used to rank 1710 point mutations by comprehensively assessing changes in the frequencies of mutations within the sort gates vs their starting frequencies (freq) within an expression-sorted reference pool. Binding enrichment (EF) factors for each mutation were calculated based on NGS analysis and using the expression-sorted pools as reference samples ($EF = \frac{freq_{bind}}{freq_{reference} (expression)}$). The enrichment factors were then used to generate sequence-binding fitness maps for each mutation in the CDRs and FR3 de loop. Mutations that resulted in better, comparable, and worse binding to CB1 compared to the 10D10 antibody were referred to as beneficial, neutral, and deleterious mutations, respectively. Based on the fitness map, many changes within HCDR1 and HCDR3 were deleterious, indicating that these regions already seem well-optimized and may have less need and/or potential for optimization. A number of mutations in

HCDR2 and HC_FR3 de loop appeared to drive improved binding. The data from the library containing LCDR1 and LCDR2 mutations indicated a lack of beneficial mutations in these regions. Overall, binding fitness maps suggested that the HCDRs were more promising areas to diversify for affinity maturation to improve binding.

[0215] Based on experience, an ideal position to diversify for affinity maturation typically contains changes to both deleterious and beneficial mutations. To identify the most promising mutations at a single position to be included in combinatorial library designs, each neutral or beneficial mutation for huCB1-ND binding was also assessed for binding to empty-ND. The most promising mutations were enriched for huCB1-ND binding and depleted for empty ND binding (vs the 10D10 parent). Mutations exhibiting positive enrichment factors against both huCB1-ND and empty-ND were discarded from consideration for library designs.

[0216] Two NGS-guided HC yeast libraries were designed and constructed for affinity maturation. Each library was designed to explore separate CDR loop regions and to keep the combinatorial theoretical diversities manageable for coverage by a routine yeast library transformation. Tables 3-4 below summarize diversification strategies for the HC libraries. Combinations of the desired mutations were fully explored within discrete regions of 10D10 HC by introducing multiple degenerate codons during library gene synthesis to construct an initial set of two yeast-displayed IgG HC variant libraries paired with the parent LC of 10D10. Each degenerate codon at a specific position was chosen to maximize the representation of the desired mutations within the encoded theoretical diversity

TABLE 3

Design of 10D10 H1H3 library 1*		
10D10 residue	Location	Diversification strategy
S30	Extended HCDR1	S, N
R32	Extended HCDR1	R, K
S42	Extended HCDR1	S, A, G, T
D109	HCDR3	D, E, N, K
I112	HCDR3	I, T, V, P, Q, A
L113	HCDR3	L, F, V
T114	HCDR3	T, S
Y116	HCDR3	Y, H, N

*Theoretical combinatorial diversity: 6.9E3 (paired with 10D10 LC)

TABLE 4

Design of 10D10 H2/de loop library 2*		
10D10 residue	location	Diversification strategy
Y57	HCDR2	Y, H
I58	HCDR2	I, V
Y60	HCDR2	Y, H, Q, stop
S65	HCDR2	S, E, T, A, K, stop
S67	HCDR2	S, Q
N69	HCDR2	N, A, K
S73	HCDR2	S, K, N, R
L74	HCDR2	L, F
K75	HCDR2	K, E
S76	HCDR2	S, G, N, D
D83	de loop (HC_FR3)	D, K, E, N
D85	de loop (HC_FR3)	S, Y

*Theoretical combinatorial diversity: 3.0E5 (paired with 10D10 LC)

[0217] One NGS-guided yeast library was designed and constructed for LC optimization to improve manufacturability of 10D10. It was believed that the long HCDR3 and LCDR1 may contribute to the poor expression and aggregation propensities of earlier 10D10 mutants, and the design of LC libraries incorporated strategies to improve these properties (e.g., reducing hydrophobicity at key solvent-exposed LC residues in LCDR1). As the fitness maps of the LC mutations showed a lack of beneficial mutations, the LC library design included mutations exhibiting 10D10-like binding enrichment against CB1-ND and depletion against empty ND. Table 5 below summarizes diversification strategies for the LC optimization library. Combinations of the desired mutations were fully explored within discrete regions of 10D10 LC by introducing multiple degenerate codons during library gene synthesis to construct a yeast-displayed IgG LC variant library paired with the parent 10D10 HC. Each degenerate codon at a specific position was chosen to maximize the representation of the desired set of mutations within the encoded theoretical diversity. This library was used in the cognate affinity maturation described below (Example 2).

TABLE 5		
Design of 10D10 LC optimization library**		
10D10 residue	Location	Diversification strategy
H33	LCDR1	H, S, T
N35	LCDR1	N, S*, T*, Y#
Y38	LCDR1	Y, A*, E*, I*, N*, F
L41	LCDR1	L, V
L58	LCDR2	L, Q*
R94	LC FR3	R, S
M107	LCDR3	M, R
L110	LCDR3	L, F, I, R, V, C, G, S

TABLE 5-continued		
Design of 10D10 LC optimization library**		
10D10 residue	Location	Diversification strategy
Q111	LCDR3	Q, E, K, R, T, A, G, P
T135	LCDR3	T, I, L, Q, K, P

*Amino acids with reduced hydrophobicity vs corresponding 10D10 residue
**Theoretical combinatorial diversity: 5.0E5 (paired with 10D10 HC)

[0218] The 10D10 LC belongs to the VK2 germline. It was known in the art that LCs of germlines such as VK1 or VK3 confer better biophysical properties and better manufacturability (see e.g., Ewert S. et al., J. Mol. Biol., (2003) 325(3):531-53). In a second strategy for LC optimization, swapping out the 10D10 LC with LCs of a different germline (e.g., VK1 or VK3) predicted to have more favorable biophysical properties was explored. Both low-resolution X-ray and CryoEM structures of mAb/antigen complex showed that only the HC loops of 10D10 interact with the CB1 receptor. These results indicated that it may indeed be possible to swap out the 10D10 VK2 LC for another LC belonging to a germline (e.g., VK1 or VK3) predicted to have superior biophysical properties to improve manufacturability. To this end, the 10D10 HC was paired with a diverse human LC library, screened and sorted for binders that retained equivalent binding as 10D10. Next, those that possess LCs related to the 10D10 LC optimization design described above were removed from the positive clones. The process resulted in yeast clones that exhibit equivalent CB1 binding as 10D10 but contain LCs from VK1 and VK3 germlines, which demonstrated the feasibility of a LC swap strategy. The top 6 clones obtained from the LC swap are listed in Table 6 below. The six LCs were then expanded to a set of 22 LCs to remedy germline derivations within the FR regions of four of the clones.

TABLE 6							
Six new LCs that can replace 10D10 LC without impacting CB1 binding							
NAME	SEQ NO.	SEQ LCDR1	SEQ NO.	SEQ LCDR2	SEQ NO.	SEQ LCDR3	LC Germline
LIBC488095-1	495	RASQSVSSSYLA	496	GASSRAT	497	QQGYSNPRT	VK3 A27/JK3
LIBC488333-1	498	RASQSIISYLN	499	AASSLRS	500	QQYSNYPLT	VK1 O2/JK4
LIBC488370-1	501	RASQSVSNYLA	502	DASNRAT	503	QQYGSLPRT	VK3 L6/JK2
LIBC488554-1	504	RASQSVSSYLG	505	GASSRAT	506	QQYGSSPRT	VK3 A27/JK1
LIBC488452-1	507	RASQSVSSYLA	508	DASNRAT	509	QQYGSLPRT	VK3 A11/JK2
LIBC488537-1	510	RASQSIISYLN	511	AASSLHS	512	QQYQSYPLT	VK1 O2/JK4
10D10	94	RSSQSLHSHNGYN YLD	95	LGSNRAS	96	MQALQTPRT	VK2 A19/JK1

Example 2. Cognate Affinity Maturation

[0219] Materials and methods: GFP-fused huCB1-SMALP was prepared as described in Luna V. et al., European Polymer Journal 109 (2018) 483-488.

[0220] The preparation of yeast cells for binding experiments and measurement of binding to GFP-fused huCB1-SMALP by flow cytometry were carried out as described in Luna V. et al., European Polymer Journal 109 (2018) 483-488, except an AlexaFluor647-conjugated anti-huFab antibody was used to detect IgG surface display. The median fluorescence intensities (MFI) for GFP (binding) and AlexaFluor647 (display) were recorded.

[0221] For the nanodisc binding experiments, a binding/display ratio for each tested yeast clone was calculated by dividing the APC MFI by the PE MFI. The binding/display ratio of each clone was further normalized to that of 10D10 LC N35Y control to calculate a “relative binding, -fold change” ratio enabling comparison of yeast clones. Specific binding to huCB1-ND over empty-ND was assessed by dividing the huCB1-ND binding/display ratio by the empty-ND binding/display ratio. For SMALP binding experiments, each clone’s GFP MFI after background subtraction was normalized to that of 10D10 LC N35Y control to compare binding of yeast clones.

[0222] The two HC libraries and the one LC optimization library designed and constructed in Example 1 were used in cognate affinity maturation of 10D10. The two HC libraries were separately enriched by FACS for improved binders relative to 10D10 following established methods (see e.g., Boder, E.T. and K. D. Wittrup, Yeast surface display for directed evolution of protein expression, affinity, and stability. Methods Enzymol, 2000. 328: p. 430-44; and Chao, G., et al., Isolating and engineering human antibodies using yeast surface display. Nat Protoc, 2006. 1(2): p. 755-68). The LC optimization library was enriched for cells exhibiting 10D10-like binding signals. The surviving HC and LC mutations after enrichment were recovered from the yeast pools by zymoprep and PCR, and pools of HCs and LCs

were mixed by design during yeast transformation to construct a new set of shuffled HC/LC libraries (mutH1H3/mutLC, mutH2/mutLC and mutH1H2H3/mutLC). Higher-stringency FACS enrichments utilizing successively lower concentrations of huCB1-ND were implemented on the chain-shuffled libraries. Clones from the enriched shuffled libraries were screened in two rounds for top clone selection. In the first round, approximately 600 clones from the three libraries were screened for clones that bind to huCB1-ND better than the 10D10 LC N35Y control (10D10 with N35Y in the LC).

[0223] In the second round of screening, the 200 best binding clones from the first round were re-tested for higher-stringency binding to 200 μ M huCB1-ND, 100 nM GFP-tagged huCB1-SMALP and against empty-ND for nonspecific binding. After ranking positive clones by CB1-specific binding, those with N-linked glycosylation and unpaired Cys Tier 1 hotspots were removed and those with predicted Tier 1.5 deamidation and isomerization sites other than DT were de-prioritized. In addition, positive clones with R94S fixes for the LC FR3 consensus violation were prioritized. Positive clones possessing the HC D83K FR3 mutation believed to be associated with enhanced anti-CB1 function were also prioritized if they satisfied the minimum threshold for binding improvement. Furthermore, clones with a reduction in hydrophobicity at key residues in LCDR1 and LCDR2 were prioritized. Tables 7 and 8 below summarize the sequence information of the top 15 cognate affinity matured 10D10 variants. FIGS. 1A and 1B show the flow cytometry binding profiles of the 15 variants; binding data of the variants are shown in in Table 9 below. All top clones exhibited significantly improved binding to huCB1-ND and huCB1-SMALP while maintaining minimal binding to empty-ND (Table 9). In addition, Y57H within HCDR2, and D83N and S85Y within the HC FR3 de loop were observed in all top clones (FIG. 3A). None of the top 15 binders contain predicted hotspots that pose the highest risk for chemical modification during therapeutic antibody manufacturing and storage.

TABLE 7

CDR sequence of top 15 affinity-matured cognate 10D10 variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC523661-1_HC huCB1 HV	1	RGGDYWS	2	HIYYEGSTKYNPSFKG	3	DYDPVTGHSYYYYGMDV
LIBC523661-1_LC huCB1 LV	4	RSSQSLLTSSGYNYVD	5	QGSNRAS	6	RQARAIPRT
LIBC523797-1_HC huCB1 HV	7	RGGDYWS	8	HVYYTGSTKYNPNFKG	9	DYDILTGYSYYYYGMDV
LIBC523797-1_LC huCB1 LV	10	RSSQSLLTSGYINVD	11	QGSNRAS	12	RQAVGLPRT
LIBC523969-1_HC huCB1 HV	13	RGGDYWS	14	HIYQTGQTKYNPNLKG	15	DYDILTGYSYYYYGMDV
LIBC523969-1_LC huCB1 LV	16	RSSQSLLSSTGANYLD	17	QGSNRAS	18	RQAVGLPRT
LIBC523748-1_HC huCB1 HV	19	RGGDYWS	20	HVYYTGSTKYNPSFEG	21	DYDILTGYSYYYYGMDV
LIBC523748-1_LC huCB1 LV	22	RSSQSLLSYGANLYD	23	LGSNRAS	24	RQARALPRT

TABLE 7-continued

CDR sequence of top 15 affinity-matured cognate 10D10 variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC523857-1_HC huCB1 HV	25	RGGDYWS	26	HIYQTGQTAYNPRLEG	27	DYDILTGYSYYYYGMDV
LIBC523857-1_LC huCB1 LV	28	RSSQSLLSSTGANYLD	29	LGSNRAS	30	RQAVELPRT
LIBC523868-1_HC huCB1 HV	31	RGGDYWS	32	HVYVEGSTAYNPSFED	33	DYDILTGYSYYYYGMDV
LIBC523868-1_LC huCB1 LV	34	RSSQSLLTSYGINYVD	35	LGSNRAS	36	MQARQIPRT
LIBC523844-1_HC huCB1 HV	37	RGGDYWS	38	HVYYTGSTKYNPSFKD	39	DYDILTGYSYYYYGMDV
LIBC523844-1_LC huCB1 LV	40	RSSQSLLSSSGANYVD	41	QGSNRAS	42	RQARALPRT
LIBC523760-1_HC huCB1 HV	43	RGGDYWS	44	HVYQTGSTKYNPKFKG	45	DYDILTGYSYYYYGMDV
LIBC523760-1_LC huCB1 LV	46	RSSQSLLTSSGANYVD	47	LGSNRAS	48	RQAVGTPRT
LIBC523862-1_HC huCB1 HV	49	RGGDYWS	50	HIYQTGSTAYNPSLKG	51	DYDILTGYSYYYYGMDV
LIBC523862-1_LC huCB1 LV	52	RSSQSLLHSTGANYVD	53	LGSNRAS	54	RQAIALPRT
LIBC524049-1_HC huCB1 HV	55	RGGDYWS	56	HIYYTGSTAYNPSLKS	57	DYDILTGYSYYYYGMDV
LIBC524049-1_LC huCB1 LV	58	RSSQSLLHSYGANYLD	59	LGSNRAS	60	RQARTLPRT
LIBC523813-1 + VK2 germline fix_HC huCB1 HV	61	RGGDYWS	62	HVYYTGQTAYNPKFKD	63	DYDILTGYSYYYYGMDV
LIBC523813-1 + VK2 germline fix_LC huCB1 LV	64	RSSQSLLSSSGANYVD	65	LGSNRAS	66	MQARGIPRT
LIBC523815-1_HC huCB1 HV	67	RGGDYWS	68	HIYQSGQTKYNPSLKD	69	DYDILTGYSYYYYGMDV
LIBC523815-1_LC huCB1 LV	70	RSSQSLLTSSGANYVD	71	LGSNRAS	72	RQAREQPRT
LIBC523603-1_HC huCB1 HV	73	RGGDYWA	74	HVYQSGQTAYNPNFEN	75	DYDAVSGYSYYYYGMDV
LIBC523603-1_LC huCB1 LV	76	RSSQSLLSSYGNNYVD	77	QGSNRAS	78	RQAVGLPRT
LIBC523596-1 HV	79	RGGDYWS	80	HVYQSGQTAYNPNFED	81	EYDVVSGYSYYYYGMDV
LIBC523596-1 LV	82	RSSQSLLTSSGYNYVD	83	QGSNRAS	84	MQAVKIPRT

TABLE 7-continued

CDR sequence of top 15 affinity-matured cognate 10D10 variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC523670-1 + VK2 germline fix_HV	85	RGGDYWA	86	HVYQTGQTAYNP NLES	87	NYDVLSGYSYYYYGMDV
LIBC523670-1 + VK2 germline fix_LV	88	RSSQSL L TSYGANYLD	89	QGSNRAS	90	MQALQIPRT
10D10 HV	91	RGGDYWS	92	YIYSGSTNYP NPSLKS	93	DYDILTGYSYYYYGMDV
10D10 LV	94	RSSQSL L HSNGYNYLD	95	LGSNRAS	96	MQALQTPRT

TABLE 8

Variable region sequence of top 15 cognate affinity-matured 10D10 variants

Name	Seq No.	Lc LV	Seq No.	Hc HV
LIBC523661-1	97	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} SSGNYVDW YLQKPGQSP _{QL} LIYQGSNRRASGV _{PD} RFSGSGSG _{TD} FTLKI SRVEAEDVGVYYCRQARALPRTFGQGT _K VEIKR	98	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHIYYEGSTKY _{NP} SPKGRATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{PT} GH _{TS} YYIGMDVWGQGT _{TT} TVSS
LIBC523797-1	99	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} SYGINVVDW YLQKPGQSP _{QL} LIYQGSNRRASGV _{PD} RFSGSGSG _{TD} FTLKI SRVEAEDVGVYYCRQAVGLPRTFGQGT _K VEIKR	100	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHVY _Y TGSTKY _{NP} NPKGRATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{IL} TGY _{SY} YYIGMDVWGQGT _{TT} TVSS
LIBC523969-1	101	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} SSGTGANYLDW YLQKPGQSP _{QL} LIYQGSNRRASGV _{PD} RFSGSGSG _{TD} FTLKI SRVEAEDVGVYYCRQAVGLPRTFGQGT _K VEIKR	102	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHIYQTGQTKY _{NP} NLKG _R ATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{IL} TGY _{SY} YYIGMDVWGQGT _{TT} TVSS
LIBC523748-1	103	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} HSYGANYLDW YLQKPGQSP _{QL} LIYLGSNRRASGV _{PD} RFSGSGSG _{TD} FTLKI SRVEAEDVGVYYCRQARALPRTFGQGT _K VEIKR	104	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHVY _Y TGSTKY _{NP} SPKGRATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{IL} TGY _{SY} YYIGMDVWGQGT _{TT} TVSS
LIBC523857-1	105	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} SSGTGANYLDW YLQKPGQSP _{QL} LIYLGSNRRASGV _{PD} RFSGSGSG _{TD} FTLKI SRVEAEDVGVYYCRQAVELPRTFGQGT _K VEIKR	106	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHIYQTGQTKY _{NP} PRLEGRATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{IL} TGY _{SY} YYIGMDVWGQGT _{TT} TVSS
LIBC523868-1	107	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} SYGINVVDW YLQKPGQSP _{QL} LIYLGSNRRASGV _{PD} RFSGSGSG _{TD} FTLKI SRVEAEDVGVYYCQWQARQIPRTFGQGT _K VEIKR	108	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHVY _Y EGSTAY _{NP} SPKDRATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{IL} TGY _{SY} YYIGMDVWGQGT _{TT} TVSS
LIBC523844-1	109	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} SSSGANYVDW YLQKPGQSP _{QL} LIYQGSNRRASGV _{PD} RFSGSGSG _{TD} FTLKI SRVEAEDVGVYYCRQARALPRTFGQGT _K VEIKR	110	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHVY _Y TGSTKY _{NP} SPKDRATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{IL} TGY _{SY} YYIGMDVWGQGT _{TT} TVSS
LIBC523760-1	111	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} SSGANYVDW YLQKPGQSP _{QL} LIYLGSNRRASGV _{PD} RFSGSGSG _{TD} FTLKI SRVEAEDVGVYYCRQAVGTPTRTFGQGT _K VEIKR	112	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHVY _Y TGSTKY _{NP} SPKGRATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{IL} TGY _{SY} YYIGMDVWGQGT _{TT} TVSS
LIBC523862-1	113	DIVMTQSP _{LS} LPVTPGEPASISCRSSQSL _{LT} HSGTGANYVD WY _L QKPGQSP _{QL} LIYLGSNRRASGV _{PD} RFSGSGSG _{TD} FTL KISRVEAEDVGVYYCRQAIALPRTFGQGT _K VEIKR	114	QVQLQESGPGLVKP _{SP} Q _{TL} SLTCTVSGGSI _{RR} GGDY _{WS} WIRQHPGK GLEWIGHIYQTGSTAY _{NP} SPKGRATI _{SV} NTYK _Q QFSLK _{LS} SVTAADT AVYYCARDYD _{IL} TGY _{SY} YYIGMDVWGQGT _{TT} TVSS

TABLE 8-continued

Variable region sequence of top 15 cognate affinity-matured 10D10 variants					
Name	Seq No.	LC LV	Seq No.	HC HV	
LIBC524049-1	115	DIVMTQSP	116	QVQLQESGPG	QVQLQESGPG
		PLSLPVTPG			
		EPASISCRSSQ			
LIBC523813-1 + VK2 germline fix	117	SLPVTGPEP	118	QVQLQESGPG	QVQLQESGPG
		ASISCRSSQSL			
		LSGANYVDW			
LIBC523815-1	119	YLPKPGSP	120	QVQLQESGPG	QVQLQESGPG
		QLLLIYLG			
		SNRSGVDP			
LIBC523603-1	121	DRFSGSG	122	QVQLQESGPG	QVQLQESGPG
		TDFTLKI			
		SRVEAEDV			
LIBC523596-1	123	YLPKPGSP	124	QVQLQESGPG	QVQLQESGPG
		QLLLIYQ			
		GNRSGVDP			
LIBC523670-1 + VK2 germline fix	125	DRFSGSG	126	QVQLQESGPG	QVQLQESGPG
		TDFTLKI			
		SRVEAEDV			
10D10	127	DIVMTQSP	128	QVQLQESGPG	QVQLQESGPG
		PLSLPVTPG			
		EPASISCRSSQ			

TABLE 9

Top 15 affinity-matured cognate 10D10 variants: binding information							
Name	0.2 nM- huCB1-ND (Relative to 10D10N35Y)	100 nM- huCB1- SMALP Binding (normalized to 10D10)	Specific huCB1 binding (CB1 ND/Empty ND ratio))	Name	0.2 nM- huCB1-ND (Relative to 10D10N35Y)	100 nM- huCB1- SMALP Binding (normalized to 10D10)	Specific huCB1 binding (CB1 ND/Empty ND ratio))
10D10	0.83	1.00	1.32	LIBC523760-1	4.19	5.70	7.21
N35Y	1.00	0.95	1.35	LIBC523862-1	5.07	3.97	13.55
benchmark							
LIBC523661-1	4.03	8.00	9.32	LIBC524049-1	5.01	5.51	9.50
LIBC523797-1	8.49	7.01	24.45	LIBC523813-1	3.38	6.32	5.79
LIBC523969-1	4.53	5.69	12.81	LIBC523815-1	6.15	4.55	14.78
LIBC523748-1	4.41	6.12	5.33	LIBC523603-1	6.05	4.05	19.35
LIBC523857-1	4.05	4.71	13.91	LIBC523596-1	4.92	4.19	20.63
LIBC523868-1	6.28	4.21	13.26	LIBC523670-1	7.46	5.42	23.04
LIBC523844-1	3.78	6.69	8.58				

Example 3. Light Chain Swapped Affinity Maturation

[0224] In a parallel arm of affinity maturation, the 22 VK1 and VK3 LCs obtained in Example 1 were transformed with 10D10 mutant HC genes or gene fragments extracted from yeast cells displaying improved binding variants obtained after FACS enrichment of H1H3 and H2/de-loop libraries (Example 2). The three LC-swapped libraries (mutH1H3/VK LCs, mutH2/VK LCs, and mutH1H2H3/VK LCs) for affinity maturation were enriched for the most improved binders using higher-stringency FACS enrichments. Clones isolated from the LC-swapped libraries were screened in two rounds for top clone selection. The screening and selection criteria for LC-swapped 10D10 affinity matured variants were the same as those for the cognate affinity maturation described in Example 2 except that prioritization of sequences with a reduction in hydrophobicity in LCDR1 and LCDR2 was not used. A total of 600 positive clones entered the first round of screening and 200 positive clones entered

the second round. The 200 positive clones were then ranked for CB1 specific binding to obtain the final 15 top LC-swapped binders. CDR and variable region sequence of the top 15 LC-swapped affinity-matured 10D10 variant are shown in Tables 10 and 11 below. Table 12 below summarizes the binding data of the top 15 LC-swapped affinity matured 10D10 variants. FIGS. 2A and 2B show the flow cytometry binding profiles of these variants. Similar to the cognate affinity matured variants, all LC-swapped top clones exhibit significantly improved binding to huCB1-ND and huCB1-SMALP while maintaining the minimal binding to empty-ND compared to 10D10 LC N35Y (Table 12). In addition, Y57H within HCDR2, and D83N and S85Y mutations within the HC FR3 de loop were observed in the top 15 LC-swapped 10D10 affinity maturation variants (FIG. 4A). Furthermore, VK1 LCs were heavily represented among the top LC-swapped 10D10 variants, with only one variant having a VK3 LC (Table 13). The R94S germline reversion was imposed in all VK LC sequences of the top variants if not present in the original sequence.

TABLE 10

CDR sequence of top 15 affinity-matured LC-swapped 10D10 variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC527997-1_HC huCB1 HV	129	RGGDYWA	130	HIYQTGSTNYNPRFKG	131	NYDTLTGYSYYYYGMDV
LIBC527997-1_LC huCB1 LV	132	RASQISISNYLN	133	AASSLHS	134	QQYQSYPLT
LIBC528116-1 + VK1 germline fix(es)_HC huCB1 HV	135	RGGDYWA	136	HVYYTGSTNYNPRFKD	137	NYDVTGTGYSYYYYGMDV
LIBC528116-1 + VK1 germline fix(es)_LC huCB1 LV	138	RASQISISNYLN	139	AASSLHS	140	QQYQSYPLT
LIBC528169-1 + VK1 germline fix(es)_HC huCB1 HV	141	RGGDYWS	142	HIYQSGSTKYNPSFED	143	DYDILTGYYYYYGMDV

TABLE 10-continued

CDR sequence of top 15 affinity-matured LC-swapped 10D10 variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC528169-1 + VK1 germline fix(es)_LC huCB1 LV	144	RASQSIISNYLN	145	AASSLHS	146	QQYQSYPLT
LIBC527879-1_HC huCB1 HV	147	RGGDYWG	148	HIYQSGSTNYNPKLKS	149	DYDILTGYSYYYYGMDV
LIBC527879-1_LC huCB1 LV	150	RASQSIISNYLN	151	AASSLRS	152	QQYSNYPLT
LIBC528141-1_HC huCB1 HV	153	RGGDYWS	154	HVYQTGSTKYNPFRKG	155	DYDILTGYSYYYYGMDV
LIBC528141-1_LC huCB1 LV	156	RASQSIISNYLN	157	AASSLHS	158	QQYQSYPLT
LIBC527814-1_HC huCB1 HV	159	RGGDYWG	160	HVYQTGSTKYNPSPFKD	161	NYDTLTGYSYYYYGMDV
LIBC527814-1_LC huCB1 LV	162	RASQSIISNYLN	163	AASSLHS	164	QQYQSYPLT
LIBC528131-1_HC huCB1 HV	165	RGGDYWS	166	HIYQTGQTAYNPNEG	167	DYDILTGYSYYYYGMDV
LIBC528131-1_LC huCB1 LV	168	RASQSVSSYLK	169	GASSRAT	170	QQYGSSPRT
LIBC527984-1 + VK1 germline fix(es)_HC huCB1 HV	171	RGGDYWS	172	HIYQTGKTNYNPNFKS	173	
LIBC527984-1 + VK1 germline fix(es)_LC huCB1 LV	174	RASQSIISNYLN	175	AASSLHS	176	QQYQSYPLT
LIBC527906-1 + VK1 germline fix(es)_HC huCB1 HV	177	RGGDYWG	178	HIYQTGSTNYNPSPFKD	179	DYDILTGYSYYYYGMDV
LIBC527906-1 + VK1 germline fix(es)_LC huCB1 LV	180	RASQSIISNYLN	181	AASSLHS	182	QQYQSYPLT
LIBC527912-1_HC huCB1 HV	183	RGGDYWS	184	YIYQTGQTKYNPNEFG	185	DYDILTGYSYYYYGMDV
LIBC527912-1_LC huCB1 LV	186	RASQSISSNYLN	187	AASSLRS	188	QQYSNYPLT
LIBC528148-1 + VK1 germline fix(es)_HC huCB1 HV	189	RGGDYWS	190	HIYQSGQTKYNPCLKG	191	DYDILTGYSYYYYGMDV
LIBC528148-1 + VK1 germline fix(es)_LC huCB1 LV	192	RASQSIISNYLN	193	AASSLHS	194	QQYQSYPLT

TABLE 10-continued

CDR sequence of top 15 affinity-matured LC-swapped 10D10 variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC527968-1 + VK1 germline fix(es)_HC huCB1 HV	195	RGGDYWS	196	HIYQSGSTKYNPSFKD	197	DYDILTGYSYYYYGMDV
LIBC527968-1 + VK1 germline fix(es)_LC huCB1 LV	198	RASQSIISNYLN	199	AASSLHS	200	QQYQSYPLT
LIBC527869-1 + VK1 germline fix(es)_HC huCB1 HV	201	RGGDYWA	202	HIYHTGKTNYNPSFKG	203	NYDTLSGNSYYYYGMDV
LIBC527869-1 + VK1 germline fix(es)_LC huCB1 LV	204	RASQSIISNYLN	205	AASSLHS	206	QQYQSYPLT
LIBC527919-1_HC huCB1 HV	207	RGGDYWA	208	HVYQTGSTAYNPFPKD	209	NYDALSGYSYYYYGMDV
LIBC527919-1_LC huCB1 LV	210	RASQSIISNYLN	211	AASSLRS	212	QQYSNYPLT
LIBC527827-1 + VK1 germline fix(es)_HC huCB1 HV	213	RGGDYWG	214	HVYQSGQTAYNPKFEN	215	NYDTLTGHSYYYYGMDV
LIBC527827-1 + VK1 germline fix(es)_LC huCB1 LV	216	RASQSIISNYLN	217	AASSLHS	218	QQYQSYPLT
10D10 HV	91	RGGDYWS	92	YIYYSGSTNYNPSLKS	93	DYDILTGYSYYYYGMDV
10D10 LV	94	RSSQSLLSNGYNYLD	95	LGSNRAS	96	MQALQTPRT
10D10N35Y HV	219	RGGDYWS	220	YIYYSGSTNYNPSLKS	221	DYDILTGYSYYYYGMDV
20D10N35Y LV	222	RSSQSLLSYGYNYLD	223	LGSNRAS	224	MQALQTPRT

TABLE 11

Variable region sequence of top 15 LC-swapped 10D10 variants					
Name	Seq No.	LC LV	Seq No.	HC HV	
LiBC527997-1	225	DIQWTQSPSTLSASVGRVTTITCRASQSI SNYLNWYQQK	226	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQTGSTNPNPKRATISVNTYKNQFSLKLS SVTAA DTAVYYCARNYDILTGYSYYYGMDVWGQGT TTVTVSS
		PGKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPE			
		DFASYVCOQYQSYPLTFGQGT KVEIKR			
LiBC528116-1 + VK1 germline fix(es)	227	DIQLTQSPSTLSASVGRVTTITCRASQSI SNYLNWYQQK	228	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQTGSTNPNPKRATISVNTYKNQFSLKLS SVTAA DTAVYYCARNYDVTGTGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPE			
		DFASYVCOQYQSYPLTFGQGT KVEIKR			
LiBC528169-1 + VK1 germline fix(es)	229	DIQLTQSPSTLSASVGRVTTITCRASQSI SNYLNWYQQK	230	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQSGSTKNPKRATISVNTYKNQFSLKLS SVTAA DTAVYYCARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPE			
		DFASYVCOQYQSYPLTFGQGT KVEIKR			
LiBC527879-1	231	DIQLTQSPSTLSASVGRVTTITCRASQSI IISYLNWYQQK	232	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQSGSTNPNPKRATISVNTYKNQFSLKLS SVTAA DTAVYYCARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPED			
		FATYYCQYSNYPLTFGGGT KVEIKR			
LiBC528141-1	233	DIQLTQSPSTLSASVGRVTTITCRASQSI SNYLNWYQQK	234	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQTGSTKNPKRATISVNTYKNQFSLKLS SVTAA DTAVYYCARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPE			
		DFASYVCOQYQSYPLTFGQGT KVEIKR			
LiBC527814-1	235	DIQMTQSPSTLSASVGRVTTITCRASQSI SNYLNWYQQK	236	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHVYQTGSTKNPKRATISVNTYKNQFSLKLS SVTAA DTAVYYCARNYDILTGYSYYYGMDVWGQGT TTVTVSS
		PGKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPE			
		DFASYVCOQYQSYPLTFGQGT KVEIKR			
LiBC528131-1	237	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLGWYQQK	238	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQTGQTAYNPNLEGRATISVNTYKNQFSLKLS SVTAA DTAVYYCARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		GQAPKLLIYGASSRATGIPDRFSGSGGTDFTLTISRLEPE			
		DFAVYYCQYQSGSPRTFGQGT KVEIKR			
LiBC527984-1 + VK1 germline fix(es)	239	DIQMTQSPSTLSASVGRVTTITCRASQSI SNYLNWYQQK	240	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQTGKTNPNPKRATISVNTYKNQFSLKLS SVTAA DTAVYYCARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		PGKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPE			
		DFASYVCOQYQSYPLTFGQGT KVEIKR			
LiBC527906-1 + VK1 germline fix(es)	241	DIQLTQSPSTLSASVGRVTTITCRASQSI SNYLNWYQQK	242	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQTGSTNPNPKRATISVNTYKNQFSLKLS SVTAA DTAVYYCARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPE			
		DFASYVCOQYQSYPLTFGQGT KVEIKR			
LiBC527912-1	243	DIQLTQSPSTLSASVGRVTTITCRASQSI IISYLNWYQQK	244	QVQLQESGPGILVKPSQTLSTLTCTVSGGSI	IRRGDYNWGI RQHHPG KGLEWIGHIYQTGQTKNPNPEGRATISVNTYKNQFSLKLS SVTAA DTAVYYCARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASLSHSGVPSRFSGSGGTDFTLTISSLQPED			
		FATYYCQYSNYPLTFGGGT KVEIKR			

TABLE 11-continued

Variable region sequence of top 15 LC-swapped 10D10 variants					
Name	Seq No.	LC LV	Seq No.	HC HV	
LIBC528148-1 + VK1 germline fix(es)	245	DIQLTQSPSTLSASVGRVVTITCRASQSI SNYLNWYQQKP	246	QVQLQESGPGLVKPSQTLSTCTVSGGSI RRGDYWMI RQHPG	KGLEWIGHIYQSGQTKYNPKLGRATISVDITYKNQPSLKLSSVTAA DTAVYICARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASSLHSGVPSRPSGSGGTDFTLTISLQPE			
		DFASYCQQYQSYPLTFGQGT KVEIKR			
LIBC527968-1 + VK1 germline fix(es)	247	DIQLTQSPSTLSASVGRVVTITCRASQSI SNYLNWYQQKP	248	QVQLQESGPGLVKPSQTLSTCTVSGGSI RRGDYWMI RQHPG	KGLEWIGHIYQSGSTKYNPFKDPKPARATISVNTYKNQPSLKLSSVTAA DTAVYICARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASSLHSGVPSRPSGSGGTDFTLTISLQPE			
		DFASYCQQYQSYPLTFGQGT KVEIKR			
LIBC527869-1 + VK1 germline fix(es)	249	DIQLTQSPSTLSASVGRVVTITCRASQSI SNYLNWYQQKP	250	QVQLQESGPGLVKPSQTLSTCTVSGGSI RRGDYWMI RQHPG	KGLEWIGHIYHTGKTNYNPSPFKGRATISVKTYKNQPSLKLSSVTA ADTAVYICARNYDILTSGNSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASSLHSGVPSRPSGSGGTDFTLTISLQPE			
		DFASYCQQYQSYPLTFGQGT KVEIKR			
LIBC527919-1	251	DIQLTQSPSTLSASVGRVVTITCRASQSI ISYLNWYQQKP	252	QVQLQESGPGLVKPSQTLSTCTVSGGSI RRGDYWMI RQHPG	KGLEWIGHVYQTGSTAYNPNPKDRATISVNTYKNQPSLKLSSVTA ADTAVYICARNYDALSGYSYYYGMDVWGQGT TTVTVSS
		GKAPKLLIYAASSLHSGVPSRPSGSGGTDFTLTISLQPED			
		PATYICQQYSNYPLTFGGT RVEIKR			
LIBC527827-1 + VK1 germline fix(es)	253	DIQMTQSPSTLSASVGRVVTITCRASQSI SNYLNWYQQK	254	QVQLQESGPGLVKPSQTLSTCTVSGGSI RRGDYWMI RQHPG	KGLEWIGHVYQSGQTAYNPKFENRATISVKTYKNQPSLKLSSVTA ADTAVYICARNYDILTGHSYYYYGMDVWGQGT TTVTVSS
		PGKAPKLLIYAASSLHSGVPSRPSGSGGTDFTLTISLQPE			
		DFASYCQQYQSYPLTFGQGT KVEIKR			
10D1	127	DIVMTQSPSLPVTTPGPASTISCRSSQSLHSGYNYLDW	128	QVQLQESGPGLVKPSQTLSTCTVSGGSI RRGDYWMI RQHPG	KGLEWIGIYYSGSTNPNPSLKRATISVDTSKNQPSLKLSSVTAAD TAVYICARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		YLQKPGQSPQLLIYLGSNRASGVPRFSGSGGTDFTLKI			
		RRVEAEDVGYYCMAQLQTPRTFGQGT KVEIKR			
10D10 N35Y	255	DIVMTQSPSLPVTTPGPASTISCRSSQSLHSGYNYLDW	256	QVQLQESGPGLVKPSQTLSTCTVSGGSI RRGDYWMI RQHPG	KGLEWIGIYYSGSTNPNPSLKRATISVDTSKNQPSLKLSSVTAAD TAVYICARDYDILTGYSYYYGMDVWGQGT TTVTVSS
		YLQKPGQSPQLLIYLGSNRASGVPRFSGSGGTDFTLKI			
		RRVEAEDVGYYCMAQLQTPRTFGQGT KVEIKR			

TABLE 12

Top 15 LC-swapped 10D10 variants: Binding Information			
Name	100 nM-huCB1-GFP-SMALP Binding (normalized to 10D10N35Y)	0.200 nM-huCB1-Nanodisc (Relative to 10D10N35Y)	50.0 nM-Empty Nanodisc (Relative to 10D10N35Y)
10D10	1.00	0.68	0.81
10D10 N35Y	1.00	1.00	1.00
LIBC527906-1	3.79	8.02	0.51
LIBC527814-1	6.44	6.49	0.61
LIBC527997-1	6.63	6.38	0.44
LIBC528141-1	8.31	6.20	0.49
LIBC527912-1	4.32	5.95	0.51
LIBC528116-1	5.80	5.91	0.77
LIBC527879-1	3.24	5.75	0.73
LIBC527919-1	4.15	5.22	0.64
LIBC527984-1	4.85	5.21	0.61
LIBC527968-1	4.65	4.86	0.64
LIBC528169-1	4.58	4.82	0.68
LIBC528131-1	3.68	4.47	0.63
LIBC527827-1	4.99	4.31	1.16
LIBC527869-1	3.61	4.15	0.97
LIBC528148-1	3.38	5.80	0.27

TABLE 13

Top 15 LC-swapped 10D10 variants germline information		
Name	VH germline	VL Germline
10D10	VH4 4-31/D3 3-9 RF2/JH6	VK2 A19/JK1
10D10 N35Y	VH4 4-31/D3 3-9 RF2/JH6	VK2 A19/JK1
LIBC527906-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC527814-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC527997-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC528141-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC527912-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC528116-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC527879-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC527919-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC527984-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC527968-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC528169-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC528131-1	VH4 4-31/D3 3-9 RF2/JH6	VK3 A27/JK1
LIBC527827-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC527869-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1
LIBC528148-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK1

Example 4 Cross Reactivity and Specificity of Affinity Matured Antibodies

[0225] The 10D10Dam antibody does not bind to the mouse CB (muCB1) or the human CB2 (huCB2) protein (WO2014/210205). Because all the antibodies isolated in Examples 2 and 3 are variants of 10D 10, they are not expected to bind to huCB1 or muCB1. Binding activity of exemplary antibodies against muCB1 and huCB2 confirms that there were no cross-reactivity against these two proteins (Table 14).

TABLE 14

No cross-reactivity against muCB1 and huCB2		
Name	Fold over mock	
	huCB2	muCB1
LIBC527814-1	0.9	1.2
LIBC527919-1	0.9	1.0
LIBC527997-1	0.8	1.0

TABLE 14-continued

No cross-reactivity against muCB1 and huCB2		
Name	Fold over mock	
	huCB2	muCB1
LIBC528116-1	0.9	0.8
LIBC523661-1	0.8	0.7
LIBC523797-1	0.9	0.8
LIBC524049-1	1.0	0.8
10D10 HC D83K/LC	0.7	0.7
N35Y (26970-1)		
10D10	0.9	0.9
Positive control	16.5	7.9
Negative control	1.1	1.1

Example 5 Characterization of Affinity Maturation Variants

[0226] The top clones isolated from Examples 2 and 3 above were cloned and expressed as IgG1 SEFL2 mAbs and then tested in a cAMP functional assay to measure antagonist activity or inverse agonist activity.

Antagonist and Inverse Agonist Activity

[0227] Antagonist activity. Stably transfected CB1 CHO cells were serum starved overnight before the assay at 37° C./5% CO₂ and using DMEM (Sigma Cat #D5671) with 0.5% FBS, 25 mM HEPES. On the day of assay, CB1 antibodies and Rimonabant were titrated 1 in 4 at 4 times final concentration using DMEM (Sigma Cat #D5671) with 0.1% BSA (from PE kit), 25 mM HEPES, the titrated antibodies were transferred to the 384 well assay plate at a volume of 7.5 µL per well. The cells were harvested using Accutase (Sigma) and neutralized with DMEM (Sigma Cat #D5671) with 0.5% FBS, 25 mM HEPES. These cells were pelleted and resuspended to 1.33×10⁶ cells/mL using DMEM (Sigma Cat #D5671) with 0.1% BSA (from PE kit), 25 mM HEPES. The anti cAMP antibody from the Lance

Ultra cAMP Detection kit (Perkin Elmer; TRF0263) was added to the cells at a 1 in 150 ratio of antibody to cell volume, 3-Isobutyl-1-methylxanthine (IBMX; Sigma) was also added to the cells at a final concentration of 1 mM. The cells were transferred to the assay plate at a volume of 15 μ L per well or 20 000 cells per well. Forskolin and CP55,940 cocktail was diluted to 4 times final concentration using DMEM (Sigma Cat #D5671) with 0.1% BSA (from PE kit), 25 mM HEPES and transferred to the assay plate at a volume of 7.5 μ L per well. The assay plate was incubated at 37° C./5% CO₂ for ½ hour before EU-cAMP tracer (Perkin Elmer; TRF0263) was added to each well and further incubated at room temperature protected from light for 1 hour. After the final incubation, the assay plates were read using Envision plate reader with EU615 and APC 665 Emission filters. The data was analysed using calculation of: TR-FRET(665 nm/615 nm)×10 000. Lower value (lower FRET) was interpreted as stronger inhibition of CB1 and the data was used to calculate IC₅₀. Antagonist activity of clones obtained from Examples 2 and 3 are shown in Table 15A (n=1 and n=2) and Table 15B (n=3).

TABLE 15A

IC50 values of cognate and LC-swapped affinity matured variants							
Clone	Source	n = 1		n = 2		Mean	
		IC50, -		IC50, -		IC50, -	
		fold vs		fold vs		fold vs	
		n = 1	Rimo	n = 2	Rimo	Mean	Rimo
		IC50	(<1	IC50	(<1	IC50	(<1
		(nM)	desired)	(nM)	desired)	(nM)	desired)
LIBC528116-1	LC swap	3.93	0.93	0.88	0.49	2.40	0.80
LIBC527814-1	LC swap	4.72	1.12	1.22	0.68	2.97	0.99
LIBC527984-1	LC swap	7.24	1.72	1.27	0.71	4.25	1.42
LIBC527997-1	LC swap	4.37	1.04	1.57	0.88	2.97	0.99
LIBC527879-1	LC swap	6.12	1.45	1.78	0.99	3.95	1.32
LIBC527919-1	LC swap	5.09	1.21	1.80	1.00	3.44	1.15
LIBC528131-1	LC swap	6.63	1.57	1.83	1.02	4.23	1.41
LIBC528141-1	LC swap	6.26	1.49	1.84	1.03	4.05	1.35
LIBC527968-1	LC swap	6.10	1.45	1.95	1.09	4.03	1.34
LIBC527912-1	LC swap	5.91	1.40	1.99	1.11	3.95	1.32
LIBC527906-1	LC swap	7.66	1.82	2.64	1.48	5.15	1.72
LIBC528148-1	LC swap	9.10	2.16	2.80	1.56	5.95	1.98
LIBC528169-1	LC swap	12.30	2.92	3.31	1.85	7.81	2.60
LIBC527827-1	LC swap	32.03	7.60	15.29	8.54	23.66	7.89
LIBC527869-1	LC swap	57.82	13.73	36.16	20.20	46.99	15.66
LIBC524049-1	cognate	4.41	1.05	1.34	0.75	2.87	0.96
LIBC523797-1	cognate	4.70	1.12	1.44	0.80	3.07	1.02
LIBC523661-1	cognate	5.24	1.24	1.35	0.75	3.30	1.10
LIBC523844-1	cognate	6.20	1.47	1.46	0.82	3.83	1.28
LIBC523969-1	cognate	6.45	1.53	1.26	0.70	3.86	1.29
LIBC523760-1	cognate	6.12	1.45	2.44	1.36	4.28	1.43
LIBC523603-1	cognate	7.29	1.73	2.16	1.20	4.72	1.57
LIBC523813-1	cognate	8.96	2.13	2.94	1.64	5.95	1.98
LIBC523670-1	cognate	8.72	2.07	3.52	1.97	6.12	2.04
LIBC523748-1	cognate	9.57	2.27	4.43	2.48	7.00	2.33
LIBC523862-1	cognate	11.20	2.66	3.42	1.91	7.31	2.44
LIBC523868-1	cognate	11.86	2.81	3.32	1.86	7.59	2.53
LIBC523857-1	cognate	13.40	3.18	2.48	1.39	7.94	2.65
LIBC523596-1	cognate	13.98	3.32	4.00	2.23	8.99	3.00
10D10 HC	control	10.86	2.58	9.81	5.48	10.33	3.44
D83K/LC	(previous						
N35Y (26970-1)	benchmark)						
10D10	controls	118.11	28.04	43.44	24.27	80.77	26.92
Rimonabant	controls	4.21	1.00	1.79	1.00	3.00	1.00

TABLE 15B

IC50 values of exemplary 10D10 variants	
Name	Mean IC50 (nM, n = 3)
LIBC528116-1	1.77
LIBC527814-1	2.07
LIBC527997-1	2.54
LIBC527919-1	3.01
LIBC523797-1	3.42
LIBC524049-1	3.71
LIBC523661-1	3.88
10D10 HC D83K/LC	14.07
N35Y (26970-1)	
10D10	120.28
rimonabant	2.83

[0228] In addition, antagonist activity of exemplary 10D10 variants and a benchmark anti-CB1 antibody was measured in the presence of endogenous huCB1 ligands anandamide and 2-Arachidonoylglycerol (2-AG). Specifically, CHOK1 human CB1 receptor cells were plated over-

night in DMEM with 0.5% FBS, P/S glutamine, 25 mM HEPES, NEAA, Na Pyruvate, and 400 µg/ml G418 (Invitrogen) at a density of 20,000 cells per well in a 96-well white flat bottom plate (Costar, catalog #3917) and placed in a humidified incubator maintained at 37° C. with 5% CO₂. Cells were treated sequentially with a dose response of antibodies, then CB1 receptor ligands at concentrations of 100 nM WIN 55,212, 10 µM anandamide or 500 nM 2-AG (Sigma), then lastly with 35 µM forskolin (Sigma) in the assay buffer of DMEM with 0.1% BSA. After a 30 minute incubation, d2-cAMP and Eu3+-Cryptate-cAMP antibody (Cisbio, catalog #62AM4PEC) were added, followed by incubation at room temperature for 1 hour. Wavelengths at 665 nm and 620 nm were measured using the EnVision Multilabel Reader and the 665/620 ratio was calculated. The graphs were generated by plotting the concentration of CB1 antibody on the x-axis, against the average of the two replicate values for cAMP levels (665/620 ratio). The data points were then fit using a log (agonist) versus response, variable slope (4 parameters) in GraphPad Prism 7.04, to obtain values for IC₅₀. The results are shown in Tables 15C and 15D and FIG. 5A.

[0229] Inverse agonist activity. Inverse agonist activity of exemplary 10D10 variants was also measured. CHOK1 human CB1 receptor cells were plated overnight in DMEM with 0.5% FBS, P/S glutamine, 25 mM HEPES, NEAA, Na Pyruvate, and 400 µg/ml G418 at a density of 20,000 cells per well in a 96-well white flat bottom plate and placed in a humidified incubator maintained at 37° C. with 5% CO₂. Cells were treated sequentially with a dose response of antibodies and then with 2 µM forskolin in the assay buffer of DMEM with 0.1% BSA. After a 30 minute incubation, d2-cAMP and Eu3+-Cryptate-cAMP antibody were added, followed by incubation at room temperature for 1 hour. Wavelengths at 665 nm and 620 nm were measured using the EnVision Multilabel Reader and the 665/620 ratio was calculated. The graphs were generated by plotting the concentration of CB1 antibody on the x-axis, against the average of the two replicate values for cAMP levels (665/620 ratio). The data points were then fit using a log (agonist) versus response, variable slope (4 parameters) in GraphPad Prism 7.04, to obtain values for IC₅₀. The results are shown in Table 15C and FIG. 5B.

TABLE 15C

IC50 values of exemplary variants in antagonist and inverse agonist assays				
Name	Antagonist assay IC50 (nM)			Inverse agonist assay IC50 (nM)
	CP 55,940	2-AG	Anandamide	—
Rimonabant	2.8	2.6	62.5	44.2
10D10	120.3	177.8	6044.0	294.9
LIBC528116-1	1.8	7.2	17.0	31.7
LIBC523797-1	3.4	2.9	10.4	14.1

TABLE 15D

IC50 of exemplary variants and a benchmark antibody an antagonist assay	
Name	IC50 (nM)
LIBC527814-1	3.8
LIBC528116-1	3.4

TABLE 15D-continued

IC50 of exemplary variants and a benchmark antibody an antagonist assay	
Name	IC50 (nM)
10D10	143.8
Benchmark mAb*	95.8

*a benchmark mAb in the clinic

[0230] Potency of the affinity matured antibodies (both cognate and LC-swapped affinity matured antibodies) were analyzed using in vitro cAMP assays and the results are summarized in Tables 15A-15D and FIGS. 5A and 5B. Most variants from the affinity maturation were more potent than 26970-1, a 10D10 variant that was obtained in previous efforts and used as a benchmark in the assays (Tables 15A and 15B). 26970-1 contains HC D83K/LC N35Y substitutions and was the best available benchmark antibody. Some affinity matured variants exhibited similar or slightly better potency than rimonabant in multiple runs of the functional assay. The most potent antibodies were about 30-66× and 3.6-8× more potent than 10D10 and 26970-1, respectively (Table 15A). The LIBC528116-1 antibody was consistently the best antagonist in the CP 55,940 assay (Tables 15A-15C), and the LIBC523797-1 antibody was the most potent in assays run in the presence of endogenous CB1 receptor agonists 2-AG and anandamide (Table 15C). In addition, the LIBC523797-1 antibody was the most superior inverse agonist (Table 15C), its potency was equivalent to rimonabant in the 2-AG assay and 3-4× more potent in the anandamide and inverse agonist assays (Table 15C). Finally, the most potent 10D10 variants were shown to be ~30× more potent than a benchmark anti-CB1 antibody in the clinic when the antibodies were tested side-by-side (Table 15D). The potency results demonstrated that the affinity maturation efforts achieved the lofty design goal of obtaining antibodies with potency in the single digit nM range, a goal that had remained elusive for many years.

Binding Affinity

[0231] Affinity assay using Octet. The Octet@HTX (Sartorius, Germany) instrument was used to determine affinities of anti-CB1 antibodies. Prior to experiment, Streptavidin (SA) biosensors (ForteBio, #18-5021) were pre-hydrated for 10 min in Octet assay buffer containing 10 mM Tris (Fisher, #BP152-1), 150 mM NaCl (Fisher, #5271-10), 1 mM CaCl₂ (Fisher, #BP510-500), 0.1 mg/mL BSA (BioShop, #ALB001.1), and 0.1% Triton-X 100 (Calbiochem, #9410-OP), pH7.4. SA biosensors were loaded with biotinylated anti-human Fc antibody (Invitrogen, #A18827) followed by one baseline step of 60 s in Octet assay buffer. Anti-CB1 antibodies were prepared in Octet assay buffer and captured by the anti-human Fc antibody prior to being submerged in wells containing different concentrations of CB1-nanodisc (#CB1-008, in-house) for 5 min, followed by 10 min dissociation. CB1-nanodisc was titrated 1 to 2 from 400 nM to 6.25 nM and prepared in Octet assay buffer. SA biosensors were used once without regeneration. For data evaluation, ForteBio Data Analysis v11.0 software was used. The kinetic rate constants, association rate constant (k_a, M⁻¹s⁻¹), dissociation rate constant (k_d, s⁻¹), and the equilibrium rate constant (K_D, M) were determined by using a 1:1 Langmuir model.

[0232] On-cell ranking using KinExA The KinExA® 3200 (Sapidyne, Boise, ID) instrument was used to rank binding of anti-CB1 antibodies to full length CB1 (DNA391968-1, in-house) transiently expressed on HEK 293T cells. Prior to the on-cell ranking experiment, on-cell affinity was determined to a benchmark anti-CB1 antibody, LIBC523661. The affinity experiment was performed with HEK 293T CB1 cells one day post transfection and titrated 1 to 3 from 2.5 million cells/mL to ~51 cells/mL. The HEK 293T CB1 cells were equilibrated with a final antibody concentration (binding arm concentration is two times the antibody concentration) of either 5 µM, 100 µM, or 1 nM of LIBC523661 in cell culture media containing FreeStyle 293T culture media (Gibco, #12338-018), 2% Ultra Low IgG FBS (Invitrogen, #16250-078), 50 µg/mL G418 (Sigma, #G8168), and 0.05% w/v/sodium azide (Sigma, #S2002), for 24 hours at RT. Cells were separated by centrifugation and the free antibody in the supernatant at equilibrium was collected and measured by KinExA® technology by passing the supernatant over PMMA beads (Sapidyne) pre-coated with goat anti-human Fc (Invitrogen, #31125) and detected with goat anti-human IgG (H+L) AlexFluor649 (Jackson ImmunoResearch, #109-605-088). The KD was obtained from nonlinear regression of the curves using a one site homogeneous binding model provided in the KinExA® Pro software Version 4.2.12. The KD for LIBC523661 was determined to be 64.0 µM with 95% CI 46.6-88.2 µM and receptor number per cell equal to 1.1 million copies/cell. It was determined using the KD-control curve that ~12 thousand cells/mL are required to achieve 50% KD for LIBC523661. Therefore, on-cell ranking of anti-CB1 antibodies was performed using 20 µM of each antibody with 12 thousand cells/mL. Anti-CB1 antibodies were prepared in 293 culture media without cells or with HEK 293T CB1 cells and incubated for 24 hours at RT. Similarly, to affinity experiment, cells were separated by centrifugation and free antibody in supernatant was collected and measured by KinExA® technology. The binding of each anti-CB1 antibody is reported as percent inhibited free fraction (% IFF) with is calculated as antibody signal obtained in presence of 12 thousand cells/mL minus signal from blank divided by antibody signal obtained in the absence of cells minus signal from blank (the lower the % IFF, the higher the affinity, the higher the % IFF, the lower the affinity). Relative binding of all anti-CB1 antibodies was compared to LIBC523661 by taking the % IFF of LIBC523661 divided by the % IFF of antibody.

[0233] Octet and Kinexa binding assays of the affinity matured 10D10 variants confirm that potency enhancements came with improvements in binding affinities to huCB1 (Table 16). Octet revealed that faster on-rates and slower off-rates contributed to 10-50× higher huCB1-ND binding affinities vs 10D10, whereas Kinexa validated binding improvements to wt huCB1 expressed on cells.

TABLE 16						
Octet and Kinexa of exemplary variants						
Name	Octet (nanodisc)		KD, nM	KinExA* (on cell) Relative binding*	IC50, nM	(CP agonist)
	k on, M ⁻¹ s ⁻¹	k off, s ⁻¹				
Benchmark#	9.46E+03	1.25E-02	1326	0.5	95	
10D10	1.64E+3	1.25E-02	755	0.5	121	
26970-1	7.33E+04	5.53E-03	76	0.5	14	
LIBC527814	5.66E+04	1.98E-03	35	1.9	2.1	

TABLE 16-continued						
Octet and Kinexa of exemplary variants						
Name	Octet (nanodisc)		KD, nM	KinExA* (on cell) Relative binding*	IC50, nM	(CP agonist)
	k on, M ⁻¹ s ⁻¹	k off, s ⁻¹				
LIBC528116	4.31E+04	1.23E-03	29	2.0	1.8	
LIBC523797	5.15E+04	9.43E-04	18	2.1	3.4	
LIBC523661	3.57E+04	2.75E-03	77	1.0	3.9	
LIBC524049	5.90E+04	9.04E-04	15	1.5	3.7	
LIBC527997	5.18E+04	1.36E-03	26	2.1	2.5	
LIBC527919	5.62E+04	3.39E-03	60	2.9	3.0	

*Calculated from fraction of free mAb vs LIBC523661 mAb
#a benchmark mAb in the clinic

Example 6. Improved Expression of the Affinity Matured Variants

[0234] The cognate and LC-swapped affinity matured variants were tested for transient and stable expression in HEK293 and CHO cells. Four out of the five most potent cognate 10D10 variants expressed transiently in HEK 293-6E cells at 100-200 mg/L (Table 17), representing >3× or >7× higher titer than 10D10 or a benchmark variant. Examination of the LC sequences suggests that reduction of hydrophobicity at key positions (especially L58Q) could be tied to improved expression titer (Table 17). The most potent LC-swapped 10D10 HC variants expressed transiently at 15-80 mg/L and expressed well at >670 mg/L in stable production runs from CHO-K1.

TABLE 17		
Improved expression of exemplary cognate affinity matured variants		
Name	Harvest (mg/L)	
10D10	34.6	
10D10 HC D83K/LC N35Y	14.4	
LIBC524049-1	26.2	
LIBC523797-1	91.7	
LIBC523661-1	177.1	
LIBC523844-1	100.9	
LIBC52369-1	121.9	

Example 7. Effects of Affinity Matured Antibodies in Animal Models

[0235] The LIBC528116-1 (37940-4) antibody in SEFL2 format (See Jacobsen F. W. et al., J Biol Chem. 2017 Feb. 3; 292(5): 1865-1875) was scaled up for testing in diet-induced obese (DIO) transgenic mice with huCB1 donor DNA knocked in using CRISPR-Cas9 technology to mediate the replacement of mouse Cnr1 coding sequence with the human CNR1 counterpart. Specifically, 6- to 10-week-old huCB1 knock in (KI) mouse was fed with high fat diet (HFD, Research Diets D12492 60% kcal from fat) for 13 weeks to induce obesity. The LIBC528116-1 antibody and a huIgG1 control antibody were administered to the mice at 30 mg/kg weekly (QW) for 60 days (n=7 mice/group), and the

body weight and food intake were measured every 3-4 days. Total fat mass was measured by MRI on day 58 of treatment, and triglyceride and insulin levels of the animals were analyzed and the results are shown in Table 18 and FIGS. 6A-6C. At day 60, the affinity-matured 10D10 variant reduced body weight by 15.9% vs huIgG1 control, and reduced fat mass by 22%, liver weight by 47%, liver triglycerides by 64%, and plasma insulin by 77% (Table 18, FIGS. 6A and 6C). By contrast, an earlier study with the D83K/N35Y variant of 10D10 (26970-2) resulted in no impact on weight gain compared to isotype control. These results showed that potent peripherally-acting anti-CB1 antibody described herein was an effective treatment for anti-obesity, hyperinsulinemia, and non-alcoholic steatohepatitis (NASH).

TABLE 18

Impact of affinity matured antibody on body weight, fat mass and liver weight in huCB1 KI mouse							
Name	IC50 (nM)	Mouse Model	BW % Change	BW % Change Day 45	BW % Change Day 60	Fat Mass % Change	Liver Weight % Change (Day)
Rimonabant 3 mg/kg (6.5 μmol/kg, QD) 26970-2 mAb 30 mg/kg (0.2 μmol/kg, QW)	2.3	JAX WT DIO (12 weeks HFD)	-12.9% (day 17)	N/A	N/A	-23% (Day 15)	-7% (Day 17)
37940-4 mAb 30 mg/kg (0.2 μmol/kg, QW)	9.5	huCNR1 TG (12 weeks HFD)	-1.9% (Day 31-32)	N/A	N/A	-2% (Day 30)	-19% (Day 32)
	2.7	huCNR1 TG (13 weeks HFD)	-12.1% (Day 31-32)	-15.2%	-15.9%	-22% (Day 58)	-47% (Day 60)

[0236] Fibrosis and inflammatory markers in kidney were also measured in the animals. Specifically, RNA was extracted and isolated from mouse kidney tissue via TRIzol extraction. 50-100 mg of tissue and a 5 mm stainless steel bead (Qiagen, #69989) was placed in a microcentrifuge tube with 1 mL of TRIzol (Ambion by life technologies, #15596018). The TissueLyser II machine was used at frequency (1/S) 30.0 for 3 minutes and repeated once. Lysates were then incubated for 5 minutes followed by a quick spin down. 200 μL of chloroform (Sigma-Aldrich, #372978-100 mL) was then added to each tube and vortexed for 30 seconds. The tubes were then incubated for 5 minutes before spinning down the tubes at 4° C. at 12,000 g for 15 minutes. The top aqueous layer was removed and placed into a new deep well plate and mixed with equal parts 70% EtOH. Upon mixing, the solution was transferred to the RNA Plate from the RNeasy Plus 96 Kit and RNA isolation was done per kit instructions (Qiagen, #74192) Qiagen DNase step was performed per kit instructions (Qiagen, #79254). RNA was diluted in RNase/DNase free water and the concentration of RNA was confirmed via NanoDrop 8000. RNA was then diluted to 50 ng/8.5 uL of water to be then run on Quantstudio 7 Flex Real-Time PCR system. TaqMan RNA-to Ct one-Step Kit (Applied Biosystems, #4392938) was used per instruction with TaqMan gene primer probe sets. CT values were used to calculate relative RNA expression levels. Treatment with the anti-CB1 antibody reduced the expression levels of the fibrosis and inflammatory markers in kidney compared to control IgG (FIG. 6D).

Example 8. New Xenomouse Antibody Campaign

[0237] Mouse immunization: Several xenomouse groups were immunized with different combinations of immunogens. Immunogens were: Plasmid DNA encoding full length huCB1 with E3k epitope for T-cell activation. DNA was coated on gold particles and delivered intradermally; HEK293T cells transiently transfected with full length huCB1 with E3k epitope. Cells were delivered subcutaneously; and Nanoparticles (nanodiscs) containing huCB1 protein engineered for enhanced stability. Nanoparticles were delivered subcutaneously.

[0238] Immunizations took several months; blood samples were collected from animals and CB1 antibody titers measured periodically. Because previous experience and pub-

lished CB1 structures suggested that large N-terminal domain of CB1 was likely to be irrelevant for drug function, only animals showing titers to both full length and truncated CB1 (i.e., devoid of the 1-98 aa N-term) were chosen for harvest to improve chances of finding functional antibodies during screening. Titer measurements were done by FACS on CHO—S cells transiently transfected with full length or truncated huCB1 or an empty vector.

[0239] Plasma B cell screening: Traditional hybridoma generation relies on immortalization of B cells by means of fusion with myeloma cells. Because of low fusion efficiency, valuable B cell clones can be difficult to recover through this process if they are rare. To complement hybridoma generation, direct screening of antibody secreting plasma B cells was implemented. The latter approach focuses on memory B cells and was used to increase the chance of finding valuable clones by sampling a different cell population.

[0240] B cells for screening were obtained from spleen, lymph nodes and bone marrow of sacrificed animals. Plasma B cells were separated from memory B cells according to CD138 cell surface marker expression using a commercial CD138 enrichment kit (Stemcell Technologies 17877). CD138+ cells were then combined with HEK293T cells transfected with truncated huCB1 and HEK293T cells transfected with an empty vector. The three types of cells were combined together in B cell media, and then mixed with a warm agarose solution (Sigma A5030, 1% final concentration), placed into fluorinated oil (RAN Biotechnologies 008-FluoroSurfactant-2 wtH-50G) and emulsified by vortexing. The result was a medium-in-oil emulsion that con-

tained microdroplets with cells trapped in them. After a 1 h incubation and a 10 min cooldown down, the agarose microgels were extracted from oil, washed with 1% FBS in PBS and screened using fluorescently labeled anti-human antibodies as a secondary detection reagent (Jackson 109-545-098). Microgels were examined under a fluorescence microscope and those containing fluorescent signal were manually picked using an Eppendorf micromanipulator. Isolated microgels were placed in individual PCR tubes with lysis buffer and subjected to RT-PCR reaction to rescue antibody sequences from B cells trapped in microgels for cloning and expression.

[0241] Molecular rescue and sequencing of CB1 antibodies from B cells: RNA (total or mRNA) was purified from samples containing CB1 antibody-producing single B cells using Agencourt RNA Clean XP magnetic beads. Purified RNA was used as template in a template-switching PCR to generate first strand cDNA via reverse transcription, followed by cDNA amplification. The cDNA was cleaned up using Agencourt AMPure XP PCR clean up kit and used as a template to amplify the antibody heavy and light chain variable region (V) genes. The fully human antibody gamma heavy chain was obtained using the cDNA as template and amplifying the variable region of the gamma heavy chain using multiplex PCR. The 5' gamma chain-specific primer annealed to the signal sequence of the antibody heavy chain, while the 3' primer annealed to a region of the gamma constant domain. The fully human antibody kappa light chain was obtained using the cDNA as template and amplifying the variable region of the kappa light chain using multiplex PCR. The 5' kappa light chain-specific primer annealed to the signal sequence of the antibody light chain while the 3' primer annealed to a region of the kappa constant domain. The fully human antibody lambda light chain was obtained using the cDNA as template and ampli-

fying the variable region of the lambda light chain using multiplex PCR. The 5' lambda light chain-specific primer annealed to the signal sequence of light chain while the 3' primer annealed to a region of the lambda constant domain. The primers used in the multiplex PCR reactions also have overhangs that facilitate Golden Gate cloning into an expression vector. The amplicons from the multiplex PCR reactions were purified using Agencourt AMPure XP PCR clean up kit. The clean PCR amplicons were then cloned into an expression vector using standard Golden Gate cloning methods followed by transformation into chemically competent *E. coli* bacterial cells. Bacterial colonies from the transformations were cultured and then sequenced directly. Amino acid sequences were deduced from the corresponding nucleic acid sequences bioinformatically. The derived amino acid sequences were then analyzed to determine the germline sequence origin of the antibodies and to identify deviations from the germline sequence.

[0242] Hybridoma generation and screening: hybridoma was generated by immortalization of B cells with mouse myeloma cells using standard technique. Hybridomas were first plated at a density of several clones per well and secreted antibodies were screened for binding to both full length and truncated CB1 transiently expressed on HEK293T or CHO—S cells. Polyclonal wells showing CB1 binding were single-cell sorted to obtain individual antibody-secreting clones.

[0243] A total number of 1325 Xenomouse clones were screened, of which 71 bound to both full length and truncated huCB1. Most of the 71 clones were non-functional and only 9 clones met potency standards set out for the mouse campaign. Three unique antibodies were isolated from the nine clones. The sequences of the antibodies (Tables 19 and 20) are similar to the sequence of 10D10. Notably, the Y57H mutation in HCDR2 that emerged after affinity maturation antibodies is also present in these antibodies (Table 19).

TABLE 19

CDR sequence of antibodies from the mouse campaign						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC529593-1 LV	471	RSSQSLHRSGYNYLD	472	LGSNRAS	473	MQSLQTPRT
LIBC529593-1 HV	474	RGGDYWS	475	HIYYSGSTNYNPSLRS	476	DYDILTGYSYYYYGLDV
LIBC560340-1 LV	477	RSSQSLLYSNGHNFLD	478	LGSNRAS	479	MQALQTPRT
LIBC560340-1 HV	480	RGGDYWN	481	HIYYSGSKNYNPSLKS	482	GYDSSGYSYYYYGMDV
LIBC560657-1 LV	483	RSSQSLLYSNGHNYLD	484	LGSNRAP	485	MQALQTPRT
LIBC560657-1 HV	486	RGGDYWN	487	HIYYTGTKYNYNPSLKS	488	GYDSSGYSYYYYGMDV
10D10 LV	91	RSSQSLLHSNGYNYLD	92	LGSNRAS	93	MQALQTPRT
10D10 HV	94	RGGDYWS	95	YIYYSGSTNYNPSLKS	96	DYDILTGYSYYYYGMDV

TABLE 20

Variable region sequence of antibodies from the mouse campaign					
Name	Seq No	HC HV	Seq No.	LC LV	
LIBC529593-1	489	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGDYG WSWIRQHPGKGLEWIGHIYYSGSTNYNPSLRSRVRI SVDTSKNQFSLKLSSVTAADTAVYYCARYDILTGYSS YYYYGLDVWGQGTITVTVSS	490	DIVMTQSPSLPVPVTPGEPASISCRSSQSLHRSYG NYLDWYLQKPGQSPQLLIYLGSNRAGVDPDRFS GSGSGTDFTLKISRVEAEDVGYYCMQSLQTPR TFGGGTKVEIKR	
LIBC560340-1	491	QVQLQESGPGLVKPSQTLSTLCIVSGGSIRRGDYG NWIRQHPGKGLEWIGHIYYSGSKNYNPSLKSRTVIS VDTSKNQFSLKLSSVTAADTAVYYCARGYDSSGYSY YYGMDVWGQGTITVTVSS	492	DIVMTQSPSLPVPVTPGEPASISCRSSQSLLYSNHG NFLDWYLQKPGQSPQLLIYLGSNRAGVDPDRFS GSGSGTDFTLKISRVEAEDVGYYCMQALQTPR TFGGGTKVEIKR	
LIBC560657-1	493	QVQLRESGPGLVKPSQTLSTLCIVSGGSIRRGDYG NWIRQHPGKGLEWIGHIYYTGKYYNPSLKSRTVIS VDTSKNQFSLKLSSVTAADTAVYYCARGYDSSGYSY YYGMDVWGQGTITVTVSS	494	DIVMTQSPSLPVPVTPGEPASISCRSSQSLLYSNHG NYLDWYLQKPGQSPQLLIYLGSNRAGVDPDRFS GSGSGTDFTLKISRVEAEDVGYYFCMQALQTPR TFGGGTKVEIKR	
10D10	127	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGDYG WSWIRQHPGKGLEWIGHIYYSGSTNYNPSLKSRTAT SVDTSKNQFSLKLSSVTAADTAVYYCARYDILTGYSS YYYYGMDVWGQGTITVTVSS	128	DIVMTQSPSLPVPVTPGEPASISCRSSQSLHRSNGY NYLDWYLQKPGQSPQLLIYLGSNRAGVDPDRFS GSGSGTDFTLKISRVEAEDVGYYCMQALQTPR TFGGGTKVEIKR	

[0244] Antibody characterization: Antibody clones identified above were tested for binding to full length human CB1 and N-term truncated human CB1 to confirm that antibodies bind to extracellular loops ECL1-3 and not to the CB1 N-terminus. Binding to human CB2 was also tested to make sure that there would be no side effects due to cross-reactivity to CB2. Finally, binding to mouse CB1 was tested. Following binding studies, functional screening using a TR-FRET cAMP assay was performed to measure antibody inhibitory effect on CB1.

[0245] To assess target specificity, binding was measured by FACS on HEK293T cells transiently transfected with the protein of interest and compared to nonspecific binding to cells transfected with an empty vector. Binding results are presented as ratios of Mean Fluorescence Intensities of target- and mock-transfected cells. The results showed that all three antibodies bound to full length and truncated human CB1, but not to human CB2 or murine CB1 (Table 21).

[0246] The inhibitory effect of the antibodies on CB1 function was assessed using a Time-Resolved Forster Resonance Energy Transfer (TR-FRET) cAMP assay. Antibodies that inhibit CB1 cause an increase of cAMP which, in turn, results in a decrease of FRET signal in the assay. Hence, lower assay signal means better inhibition of CB1 by the antibody. To improve assay sensitivity CP55,940 and Forskolin were used to activate cellular CB1 and adenylyl cyclase, respectively. Rimobant was used as a positive control.

[0247] Briefly, stably transfected CB1 CHO cells were cultured in DMEM (Sigma Cat #D5671) with 0.5% FBS, 25 mM HEPES overnight at 37° C./5% CO₂. On the day of assay, CB1 antibodies and Rimobant were titrated 1 in 4 at 4 times final concentration using DMEM (Sigma Cat #D5671) with 0.1% BSA (from PE kit), 25 mM HEPES, the titrated antibodies were transferred to the 384 well assay plate at a volume of 7.5 µL per well. The cells were harvested using Accutase (Sigma) and neutralized with DMEM (Sigma Cat #D5671) with 0.5% FBS, 25 mM HEPES. These cells were pelleted and resuspended to 1.33×10⁶ cells/mL using DMEM (Sigma Cat #D5671) with 0.1% BSA (from PE kit), 25 mM HEPES. The anti cAMP antibody from the Lance Ultra cAMP Detection kit (Perkin Elmer; TRF0263) was added to the cells at a 1 in 150 ratio

of antibody to cell volume, 3-Isobutyl-1-methylxanthine (IBMX; Sigma) was also added to the cells at a final concentration of 1 mM. The cells were transferred to the assay plate at a volume of 15 µL per well or 20,000 cells per well. Forskolin and CP55,940 cocktail was diluted to 4 times final concentration using DMEM (Sigma Cat #D5671) with 0.1% BSA (from PE kit), 25 mM HEPES and transferred to the assay plate at a volume of 7.5 µL per well. The assay plate was incubated at 37° C./5% CO₂ for ½ hour before EU-cAMP tracer (Perkin Elmer; TRF0263) was added to each well and further incubated at room temperature protected from light for 1 hour. After the final incubation, the assay plates were read using Envision plate reader with EU615 and APC 665 Emission filters. The data was analyzed using calculation of: (665 nm/615 nm)×10,000. Lower value (lower FRET) indicated stronger inhibition of CB1 and the data was used to calculated IC₅₀. The results showed that all three antibodies were much superior to 10D10 in terms of potency (Table 22 and FIG. 9).

TABLE 21

Binding data of the antibodies			
Target	Antibody binding, MFI ratio		
	LIBC529593	LIBC560340	LIBC560657
Full length hu CB1	422	578	440
Truncated hu CB1	495	705	444
Full length hu CB2	1.1	1.0	1.0
Full length mu CB1	1.0	1.0	1.0

TABLE 22

Potency data of the antibodies					
Mole- cule	LIBC529593	LIBC560340	LIBC560657	10D10	Rimona- bant
IC ₅₀ , nM	6.4	10.2	7.5	120	2.8

Example 9. Further Affinity Maturation of
Anti-CB1 Antibodies

[0248] Another round of affinity maturation was carried out to further improve functional potency. Analyses of the affinity matured antibodies from the previous round (Examples 1-3 above) showed that there was good correlation between GFP-tagged huCB1-SMALP binding and IC50 for the cognate 10D10 variants, and huCB1-GFP-SMALP binding predicted a majority of the most potent LC-swapped variants. Thus, improving binding to huCB1-GFP-SMALP was the focus during screening and sorting in this round of affinity maturation to obtain antibodies with subnanomolar potencies.

[0249] Two strategies were used in this affinity maturation. First, the VK1 LC shared by the top LC-swapped variants fortuitously belongs to a well-studied germline where there is high confidence in the LCDR positions predicted to be solvent-exposed and tolerant to mutation. Thus, to further optimize their LCs we sought to pair the HCs of LIBC528116-1 and other top LC-swapped variants with VK1/O2 LCDR libraries previously utilized in internal affinity maturation projects. While the co-crystal structure suggests that mutations at solvent-exposed LC positions are unlikely to create contacts with huCB1, it is hypothesized that LC mutations may provide more optimal structural support of HC loop conformations to enhance binding and potency. To increase selection stringency, monovalent yeast-display Fab libraries were constructed and sorted/screened for improved binding against wtCB1-SMALP while retaining huCB1-ND binding (vs LIBC58116-1 Fab).

[0250] Second, we leveraged the surprising discovery of more 10D10-like antibodies from the new mouse campaigns (see Example 8) to utilize better starting points for affinity maturation. The antibodies from the new mouse campaign indicated certain positions in HCDR2 and HC FR3 could be further explored to improve binding and potency. In addition, the new antibodies cluster into two highly-related HCDR3s that differ in length by one amino acid (e.g., LIBC560657-1 and LIBC529593-1 of Example 8) indicating that the length of HCDR3 could be varied. Yeast display libraries for further cognate affinity maturation merged HC and LC mutations from the most potent cognate 10D10 variants from Example 2 into the LIBC529593-1 sequence context and incorporated both HCDR3 lengths. HC library 2 incorporated the shorter HCDR3 sequence and additionally contained fixes at HCDR3 position 112 for the predicted DS isomerization site. The design of the further cognate affinity-maturation libraries is summarized in Tables 23-25.

TABLE 23

Design of 10D10-like HC library 1 for further cognate AM*		
10D10 residue	Location	Diversification strategy
Y57	HCDR2	H [#]
I58	HCDR2	I, V
Y60	HCDR2	Y, Q
S65	HCDR2	E, A, K, T
S67	HCDR2	S, Q
N69	HCDR2	E, A, K, T
S73	HCDR2	S, N
L74	HCDR2	L, F
K75	HCDR2	K, R [#]
S76	HCDR2	S, G, N, D
A78	HFR3	V [#]

TABLE 23-continued

Design of 10D10-like HC library 1 for further cognate AM*		
10D10 residue	Location	Diversification strategy
T79	HFR3	T, R [#]
D83	de loop (HFR3)	N
S85	de loop_(HFR3)	Y
M136	HCDR3	M, L [#]

[#]mutation found in LIBC529593-1

*Theoretical combinatorial diversity: 1.6E4

TABLE 24

Design of 10D10-like HC library 2 for further cognate AM*					
10D10 residue	location	Diversification strategy	10D10 residue	location	Diversification strategy
Y57	HCDR2	H [#]	A78	HFR3	V [#]
I58	HCDR2	I, V	T79	HFR3	T, R [#]
Y60	HCDR2	Y, Q	D83	de loop (HFR3)	N
S65	HCDR2	E, A, K, T	S85	de loop (HFR3)	Y
S67	HCDR2	S, Q	D109	HCDR3	G
N69	HCDR2	E, A, K, T	I112	HCDR3	I, A, T, V
S73	HCDR2	S, N	L113	HCDR3	deletion
L74	HCDR2	L, F	T114	HCDR3	S
K75	HCDR2	K, R [#]	M136	HCDR3	M, L [#]
S76	HCDR2	S, G, N, D			

[#]mutation found in LIBC529593-1

*Theoretical combinatorial diversity: 6.5E4

TABLE 25

Design of 10D10-like LC library for further cognate AM*		
10D10 residue	location	Diversification strategy
S34	LCDR1	R [#]
N35	LCDR1	S [#]
L58	LCDR2	L, Q
R94	LC FR3	S [#]
M107	LCDR3	R
A109	LCDR3	A, S [#]
L110	LCDR3	V, R
Q111	LCDR3	A, G
T135	LCDR3	I, L

[#]mutation found in LIBC529593-1

*Theoretical combinatorial diversity: 33 (32 + 1)

[0251] The 10D10 HC and LC library designs were combinatorially paired to construct Fab-yeast libraries to select for increased binding stringency. The cognate libraries for further affinity maturation were sorted/screened for improved binding against wtCB1-SMALP while retaining huCB1-ND binding (vs LIBC523797-1 Fab).

[0252] The libraries constructed for both further affinity maturation campaigns were sorted against successively lower concentrations of huCB1-GFP-SMALP, starting at 25 or 50 nM. We conducted clonal binding screens at huCB1-SMALP concentrations as low as 2-5 nM, representing a 20-50x lower concentration than what was employed in the previous affinity maturation campaign (Examples 2-4). The most improved binders exhibiting minimal non-specific binding and predicted sequence liabilities from the further VK1 LC optimization campaign and the further cognate

affinity maturation campaign were then selected based on similar principles as described before (Examples 2 and 3). Eight clones were obtained from the VK1 LC optimization, and 18 clones were obtained from the further cognate affinity maturation. These top clones showed improved huCB1-SMALP binding, similar huCB1-ND binding, and low levels of empty-ND binding versus the LIBC528116-1 and LIBC523797-1 Fab-yeast benchmarks from both LC-swapped and cognate campaigns, respectively.

[0253] CDR sequences, germline information and binding data of the top 8 further LC-swapped affinity maturation variants are summarized in Tables 26 and 27 and FIG. 7. All top clones showed improved binding against CB1-SMALP and CB1-ND (Table 28) while exhibiting minimal binding to irrelevant XCR1-SMALP and empty-ND. For LIBC681737-1, the LC P12S germline reversion mutation was included when making IgG1 SEFL2 mAb for functional testing.

TABLE 26

CDR sequence of top 8 further VK1 LC-optimized variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC680812-1_LC huCB1 LV	405	RASQSISSYLN	406	SARRLSS	407	QQYRSYPIT
LIBC680812-1_mol:fdChaincf huCB1 HV	408	RGGDYWG	409	HVYQTGSTKYNPSFKD	410	NYDTLTGYSYYYYGMDV
LIBC680562-1_LC huCB1 LV	411	RASQSISSYLN	412	SARRLSS	413	QQYAKSPIT
LIBC680562-1_mol:fdChaincf huCB1 HV	414	RGGDYWA	415	HIYQTGSTNYNPRFKG	416	NYDTLTGYSYYYYGMDV
LIBC680800-1_LC huCB1 LV	417	RASQSISSYLN	418	SARALSS	419	QQYRKPLT
LIBC680800-1_mol:fdChaincf huCB1 HV	420	RGGDYWA	421	HIYQTGSTNYNPRFKG	422	NYDTLTGYSYYYYGMDV
LIBC680574-1_LC huCB1 LV	423	RASQSISSYLN	424	GARRLGS	425	QQYSSLPVT
LIBC680574-1_mol:fdChaincf huCB1 HV	426	RGGDYWA	427	HVYYTGSTKYNPSFKD	428	NYDTLTGYSYYYYGMDV
LIBC680773-1_LC huCB1 LV	429	RASQSISSYLN	430	KARLLSS	431	QQYSRLPLT
LIBC680773-1_mol:fdChaincf huCB1 HV	432	RGGDYWA	433	HVYYTGSTNYNPRFKD	434	NYDVTGTGYSYYYYGMDV
LIBC681594-1_LC huCB1 LV	435	RASQSISSYLN	436	KARKLSS	437	QQFGSMPLT
LIBC681594-1_mol:fdChaincf huCB1 HV	438	RGGDYWA	439	HIYQTGSTNYNPRFKG	440	NYDTLTGYSYYYYGMDV
LIBC681737-1 (LC:P12S)_LC huCB1 LV	441	RASQSISSYLN	442	SARRLAS	443	QQYYHPFIT
LIBC681737-1 (LC:P12S)_mol:fdChaincf huCB1 HV	444	RGGDYWA	445	HVYYTGSTNYNPRFKD	446	NYDVTGTGYSYYYYGMDV
LIBC681593-1_LC huCB1 LV	447	RASQSISSYLN	448	NARRLGS	449	QQYRSSPVT

TABLE 26-continued

CDR sequence of top 8 further VK1 LC-optimized variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC681593-1_ mol:fdChaincf huCB1 HV	450	RGGDYWA	451	HIYQTGSTNYNPRFKG	452	NYDTLTGYSYYYYGMDV
LIBC681737-1_LC huCB1 LV	441	RASQSISSYLN	442	SARRLAS	443	QQYYHPPIT
LIBC681737-1_ mol:fdChaincf huCB1 HV	444	RGGDYWA	445	HVYYTGSTNYNPRFKD	446	NYDVTGTGYSYYYYGMDV

TABLE 27
Variable region sequence of top 8 further VK1 LC-optimized variants

Name	Seq No.	LC LV	Seq No.	mol:fdChaincf HV
LIBC680812-1	453	KPGKAPKLLIYSARLLSGVPSRFSGSGGDTFLTITSSL DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ QPEDFATYYCQQYRSYPITFGGGTKVEIKR	454	QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP KGLEWIGHYVYQTGSTKYNPFKDRATISVNTYKNQPSLKLSSVT AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS
LIBC680562-1	455	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ KPGKAPKLLIYSARLLSGVPSRFSGSGGDTFLTITSSL QPEDFATYYCQQYAKSPITFGGGTKVEIKR	456	QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP KGLEWIGHYVYQTGSTKYNPFKGRATISVNTYKNQPSLKLSSVT AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS
LIBC680800-1	457	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ KPGKAPKLLIYSARLLSGVPSRFSGSGGDTFLTITSSL QPEDFATYYCQQYRKPLTFGGGKVEIKR	458	QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP KGLEWIGHYVYQTGSTKYNPFKGRATISVNTYKNQPSLKLSSVT AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS
LIBC680574-1	459	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ KPGKAPKLLIYGARLLSGVPSRFSGSGGDTFLTITSSL QPEDFATYYCQQYSSLPVTFGGGKVEIKR	460	QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP KGLEWIGHYVYQTGSTKYNPFKDRATISVNTYKNQPSLKLSSVT AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS
LIBC680773-1	461	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ KPGKAPKLLIYKARLLSGVPSRFSGSGGDTFLTITSSL QPEDFATYYCQQYSRPLPTFGGGTKVEIKR	462	KGLEWIGHYVYQTGSTKYNPFKDRATISVNTYKNQPSLKLSSVT QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS
LIBC681594-1	463	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ KPGKAPKLLIYKARKLSSGVPSRFSGSGGDTFLTITSSL QPEDFATYYCQQFGSMPLTFGGGKVEIKR	464	QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP KGLEWIGHYVYQTGSTKYNPFKGRATISVNTYKNQPSLKLSSVT AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS
LIBC681737-1 (LC:P12S)	465	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ KPGKAPKLLIYSARLLSGVPSRFSGSGGDTFLTITSSL QPEDFATYYCQQYHHPITFGGGKVEIKR	466	QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP KGLEWIGHYVYQTGSTKYNPFKDRATISVNTYKNQPSLKLSSVT AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS
LIBC681593-1	467	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ KPGKAPKLLIYNARLLSGVPSRFSGSGGDTFLTITSSL QPEDFATYYCQQYRSPVTFGGGKVEIKR	468	QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP KGLEWIGHYVYQTGSTKYNPFKGRATISVNTYKNQPSLKLSSVT AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS
LIBC681737-1	469	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQ KPGKAPKLLIYSARLLSGVPSRFSGSGGDTFLTITSSL QPEDFATYYCQQYHHPITFGGGKVEIKR	470	QVQLQESGPGGLVKPSQTLSTCTVSGGSIRRGDYAWAIRQHP KGLEWIGHYVYQTGSTKYNPFKDRATISVNTYKNQPSLKLSSVT AADTAVYICARNYDITLTGYSYYYYGMDVWGQGTITVTVSS

TABLE 28

Germline information of top 8 further VK1 LC-optimized variants		
Name	VH Germline	VL Germline
LIBC680562-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC680574-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC680773-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC680800-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC680812-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC681593-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC681594-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4
LIBC681737-1	VH4 4-31/D3 3-9 RF2/JH6	VK1 O2/JK4

TABLE 29

Binding data of top 8 further VK1 LC-optimized variants				
Name	2.5 nM CB1- SMALP-GFP*	25 nM XCR1- SMALP-GFP*	0.2 nM CB1-ND*	50 nM Empty ND*
LIBC680562-1	1.94	0.64	3.34	0.68
LIBC680574-1	1.72	0.56	NA	0.72
LIBC680773-1	1.9	0.84	2.4	0.75
LIBC680800-1	1.76	0.47	2.52	0.78
LIBC680812-1	1.71	0.59	2.23	0.74
LIBC681593-1	2.37	0.72	NA	0.91
LIBC681594-1	2.57	1.04	3.56	0.83

TABLE 29-continued

Binding data of top 8 further VK1 LC-optimized variants				
Name	2.5 nM CB1- SMALP-GFP*	25 nM XCR1- SMALP-GFP*	0.2 nM CB1-ND*	50 nM Empty ND*
LIBC681737-1	1.94	0.62	1.89	0.73
LIBC528116 (Fab)	1	1	1	1

*Relative binding fold change compared to LIBC528116 (Fab), calculated after background subtraction

[0254] CDR sequences, germline information and binding data of the top 18 further cognate affinity maturation variants are summarized in Tables 30 and 31 and FIGS. 8A and 8B. Most of the 18 binders utilize the shorter HCDR3 and possess the HC T79R FR3 mutation in the library design for HC library 2, highlighting the benefit of rationally merging features from all available sequence-function relationships. All eighteen cognate variants met the stringent binding improvement threshold against huCB11-GFP-SMALP and had minimal binding to irrelevant XCR1-SMALP and empty-ND (Table 32). These variants also exhibited better binding to CB 1 SMALP relative to LIBC523797-1. For LIBC674043-1 and LIBC674200-1, VH R149S and VL G76D germline reversion mutations were also included in the panel of IgG1 SEFL2 mAbs produced for functional testing.

TABLE 30

CDR sequence of top 18 further cognate variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC673972-1_LC huCB1 LV	257	RSSQSLHRSGYNYLD	258	LGSNRAS	259	MQSLQTPRT
LIBC673972-1_mol: fdChaincf huCB1 HV	260	RGGDYWS	261	HIYYTGQTAYNPNLRN	262	GYDASGYSYYYYGLDV
LIBC673952-1_LC huCB1 LV	263	RSSQSLHRSGYNYLD	264	QGSNRAS	265	RQARALPRT
LIBC673952-1_mol: fdChaincf huCB1 HV	266	RGGDYWS	267	HVYYTGQTTYNPSPFRG	268	DYDILTGYSYYYYGLDV
LIBC673982-1_LC huCB1 LV	269	RSSQSLHRSGYNYLD	270	LGSNRAS	271	RQSVALPRT
LIBC673982-1_mol: fdChaincf huCB1 HV	272	RGGDYWS	273	HIYYTGSTTYNPSPFRG	274	DYDISGYSYYYYGMDV
LIBC674043-1_LC huCB1 LV	275	RSSQSLHRSGYNYLD	276	LGSNRAS	277	MQSLQTPRT
LIBC674043-1_mol: fdChaincf huCB1 HV	278	RGGDYWS	279	HIYYTGQTTYNPNFKG	280	GYDASGYSYYYYGMDV
LIBC673948-1_LC huCB1 LV	281	RSSQSLHRSGYNYLD	282	LGSNRAS	283	MQSLQTPRT
LIBC673948-1_mol: fdChaincf huCB1 HV	284	RGGDYWS	285	HIYYTGSTKYNPSLKN	286	GYDVSGYSYYYYGLDV
LIBC674090-1_LC huCB1 LV	287	RSSQSLHRSGYNYLD	288	QGSNRAS	289	MQSLQTPRT
LIBC674090-1_mol: fdChaincf huCB1 HV	290	RGGDYWS	291	HIYYAGSTAYNPNFRG	292	GYDTSGYSYYYYGLDV
LIBC674200-1(LC:G76D)_ LC huCB1 LV	293	RSSQSLHRSGYNYLD	294	LGSNRAS	295	MQSLQTPRT
LIBC674200- 1(LC:G76D)_mol: fdChaincf huCB1 HV	296	RGGDYWS	297	HVYYTGQTYNPNFRN	298	GYDVSGYSYYYYGMDV

TABLE 30-continued

CDR sequence of top 18 further cognate variants						
Name	Seq No.	CDR1	Seq No.	CDR2	Seq No.	CDR3
LIBC673965-1_LC huCB1 LV	299	RSSQSLHRSGYNYLD	300	LGSNRAS	301	MQSLQTPRT
LIBC673965-1_mol: fdChaincf huCB1 HV	302	RGGDYWS	303	HVYYTGQTAYNP SLRG	304	GYDTSGYSYYYYGLDV
LIBC674216-1_LC huCB1 LV	305	RSSQSLHRSGYNYLD	306	LGSNRAS	307	MQSLQTPRT
LIBC674216-1_mol: fdChaincf huCB1 HV	308	RGGDYWS	309	HIYYTGQTAYNP SLRN	310	GYDISGYSYYYYGLDV
LIBC674043-1 (mol:fdChaincf:R149S)_ LC huCB1 LV	275	RSSQSLHRSGYNYLD	276	LGSNRAS	277	MQSLQTPRT
LIBC674043-1 (mol:fdChaincf:R149S)_ mol:fdChaincf huCB1 HV	278	RGGDYWS	279	HIYYTGQTTYNP NFKG	280	GYDASGYSYYYYGMDV
LIBC674214-1_LC huCB1 LV	311	RSSQSLHRSGYNYLD	312	LGSNRAS	313	MQSLQTPRT
LIBC674214-1_mol: fdChaincf huCB1 HV	314	RGGDYWS	315	HVYYEGSTKYNP SFRG	316	GYDVSGYSYYYYGLDV
LIBC674235-1_LC huCB1 LV	317	RSSQSLHRSGYNYLD	318	LGSNRAS	319	MQSLQTPRT
LIBC674235-1_mol: fdChaincf huCB1 HV	320	RGGDYWS	321	HVYYAGSTTYNP SLKS	322	GYDVSGYSYYYYGLDV
LIBC674257-1_LC huCB1 LV	323	RSSQSLHRSGYNYLD	324	LGSNRAS	325	MQSLQTPRT
LIBC674257-1_mol: fdChaincf huCB1 HV	326	RGGDYWS	327	HIYYTGQTKYNP SLRG	328	GYDISGYSYYYYGLDV
LIBC674200-1_LC huCB1 LV	293	RSSQSLHRSGYNYLD	294	LGSNRAS	295	MQSLQTPRT
LIBC674200-1_mol: fdChaincf huCB1 HV	296	RGGDYWS	297	HVYYTGQTAYNP NFRN	298	GYDVSGYSYYYYGMDV
LIBC674153-1_LC huCB1 LV	329	RSSQSLHRSGYNYLD	330	LGSNRAS	331	MQSLQTPRT
LIBC674153-1_mol: fdChaincf huCB1 HV	332	RGGDYWS	333	HVYYTGSTAYNP SFRG	334	GYDTSGYSYYYYGLDV
LIBC674002-1_LC huCB1 LV	335	RSSQSLHRSGYNYLD	336	LGSNRAS	337	MQSLQTPRT
LIBC674002-1_mol: fdChaincf huCB1 HV	338	RGGDYWS	339	HIYYKGSTEYNP SLRG	340	GYDVSGYSYYYYGLDV
LIBC674276-1_LC huCB1 LV	341	RSSQSLHRSGYNYLD	342	LGSNRAS	343	MQSLQTPRT
LIBC674276-1_mol: fdChaincf huCB1 HV	344	RGGDYWS	345	HVYYTGSTTYNP NFKG	346	GYDASGYSYYYYGLDV
LIBC674024-1_LC huCB1 LV	347	RSSQSLHRSGYNYLD	348	LGSNRAS	349	MQSLQTPRT
LIBC674024-1_mol: fdChaincf huCB1 HV	350	RGGDYWS	351	HVYQTGSTKYNP SFKN	352	GYDVSGYSYYYYGLDV
LIBC674035-1_LC huCB1 LV	353	RSSQSLHRSGYNYLD	354	LGSNRAS	355	MQSLQTPRT
LIBC674035-1_mol: fdChaincf huCB1 HV	356	RGGDYWS	357	HIYYAGQTKYNP SFKN	358	GYDISGYSYYYYGLDV
LIBC674229-1_LC huCB1 LV	359	RSSQSLHRSGYNYLD	360	QGSNRAS	361	MQSLQTPRT
LIBC674229-1_mol: fdChaincf huCB1 HV	362	RGGDYWS	363	HIYQTGQTKYNP SFKG	364	GYDVSGYSYYYYGMDV

TABLE 31

Variable regions sequence of top 18 further cognate variants				
Name	Seq No.	LC LV	Seq No.	mol:fdChaincf HV
LIBC67 3972-1	365	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	366	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTAYNPRLNRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDASGYSYYYGLDVGW QGTTTVTVSS
LIBC67 3952-1	367	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYQGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCRQARALPRTF QGQTKVEIKR	368	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTTYNPSFRGRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARDYDILTSYYYGLDVGW QGTTTVTVSS
LIBC67 3982-1	369	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCRQSVLPRTF QGQTKVEIKR	370	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGSTTYPNPSFRGRVRI SVNTYKN FSLKLSSVTAADTAVYYCARDYDISGYSYYYGMDVWG GTTTVTVSS
LIBC67 4043-1	371	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	372	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTTYNPNFRGRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDASGYSYYYGMDVWG QGTTTVTVSR
LIBC67 3948-1	373	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	374	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGSTKYNPSLKNRVRI SVNTYKN FSLKLSSVTAADTAVYYCARGYDVSIGSYYYGLDVGW GTTTVTVSS
LIBC67 4090-1	375	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYQGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRT FGQTKVEIKR	376	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYAGSTAYNPFRGRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDTSIGSYYYGLDVGW QGTTTVTVSS
LIBC67 4200-1 (LC: G76D)	377	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	378	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTAYNPFRNRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDVSIGSYYYGMDVWG QGTTTVTVSS
LIBC67 3965-1	379	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	380	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTAYNPRLGRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDTSIGSYYYGLDVGW QGTTTVTVSS
LIBC67 4216-1	381	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	382	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTAYNPRLNRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDISIGSYYYGLDVGW GTTTVTVSS
LIBC67 4043-1 (mol: fdChain cf: R149S)	383	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	384	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTTYNPNFRGRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDASGYSYYYGMDVWG QGTTTVTVSS
LIBC67 4214-1	385	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	386	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYEGSTKYNPSFRGRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDVSIGSYYYGLDVGW QGTTTVTVSS
LIBC67 4235-1	387	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	388	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYAGSTTYPNPSLKRVRRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDVSIGSYYYGLDVGW QGTTTVTVSS
LIBC67 4257-1	389	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	390	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTKYNPSLRGRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDISGYSYYYGLDVGW GTTTVTVSS
LIBC67 4200-1	391	DIVMTQSPPLSLPVTGPGEPAISCRSSQSLLRSGY NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLTKISRVEAEDVGVIYCMQSLQTPRTF QGQTKVEIKR	392	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDYWSWI RQHPGKGLEWIGHIYYTGQTAYNPFRNRVRI SVNTYKN QFSLKLSSVTAADTAVYYCARGYDVSIGSYYYGMDVWG QGTTTVTVSS

TABLE 31-continued

Variable regions sequence of top 18 further cognate variants					
Name	Seq No.	LC LV	Seq No.	mol:fdChaincf HV	
LIBC67 4153-1	393	DIVMTQSPPLSLPVTPGEPASISCRSSQSLHRS NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLKISRVEAEDVGVYYCMQSLQTPRTF QGQTKVEIKR	394	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDY RQHPGKGLEWIGHVYVTGSTAYNPSRGRVRI QFSLKLSVTAADTAVYYCARGYDTSYSGYGLDVWG QGTTVTVSS	
LIBC67 4002-1	395	DIVMTQSPPLSLPVTPGEPASISCRSSQSLHRS NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLKISRVEAEDVGVYYCMQSLQTPRTF QGQTKVEIKR	396	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDY RQHPGKGLEWIGHIYKGSSTEYNPSLRGRVRI FSLKLSVTAADTAVYYCARGYDVSYSGYGLDVWG GTTVTVSS	
LIBC67 4276-1	397	DIVMTQSPPLSLPVTPGEPASISCRSSQSLHRS NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLKISRVEAEDVGVYYCMQSLQTPRTF QGQTKVEIKR	398	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDY RQHPGKGLEWIGHVYVTGSTYNNPSRGRVRI QFSLKLSVTAADTAVYYCARGYDASGYGLDVWG QGTTVTVSS	
LIBC67 4024-1	399	DIVMTQSPPLSLPVTPGEPASISCRSSQSLHRS NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLKISRVEAEDVGVYYCMQSLQTPRTF QGQTKVEIKR	400	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDY RQHPGKGLEWIGHVYQTGSKYNPSFKNRVRI QFSLKLSVTAADTAVYYCARGYDVSYSGYGLDVWG QGTTVTVSS	
LIBC67 4035-1	401	DIVMTQSPPLSLPVTPGEPASISCRSSQSLHRS NYLDWYLQKPGQSPQLLIYLGSNRASGVDRFSG SGSGTDFTLKISRVEAEDVGVYYCMQSLQTPRTF QGQTKVEIKR	402	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDY RQHPGKGLEWIGHIYAGQTKYNPSFKNRVRI QFSLKLSVTAADTAVYYCARGYDISGYGLDVWG GTTVTVSS	
LIBC67 4229-1	403	DIVMTQSPPLSLPVTPGEPASISCRSSQSLHRS NYLDWYLQKPGQSPQLLIYQGSNRASGVDRFS SGSGTDFTLKISRVEAEDVGVYYCMQSLQTPRT FGQTKVEIKR	404	QVQLQESGPGLVKPSQTLSTCTVSGGSIRRGGDY RQHPGKGLEWIGHIYQTGSKYNPSFKNRVRI QFSLKLSVTAADTAVYYCARGYDVSYSGYGLDVWG QGTTVTVSS	

TABLE 32

Binding data of top 18 further cognate variants					
HTB Barcode	10 nM CB1 SMALP*	5 nM CB1 SMALP*	50 nM XCR1 SMALP*	0.5 nM CB1 ND*	50 nM Empty ND*
LIBC529593-1 (Fab)	1.06	1.42	1.50	0.38	1.14
LIBC523797-1 (Fab)	1.00	1.00	1.00	1.00	1.00
LIBC673948-1	2.46	1.37	0.88	0.64	0.48
LIBC673952-1	1.91	1.23	0.59	0.77	0.62
LIBC673965-1	2.50	1.30	0.74	0.65	0.43
LIBC673982-1	2.16	1.24	0.79	0.78	0.60
LIBC673972-1	2.46	1.46	0.80	0.63	0.40
LIBC674024-1	2.39	1.24	0.86	0.49	0.55
LIBC674035-1	2.74	1.29	0.91	0.52	0.56
LIBC674043-1	3.23	1.20	0.62	1.85	0.75
LIBC674090-1	2.85	1.20	0.74	1.35	0.99
LIBC674002-1	2.59	1.49	0.91	0.46	0.03
LIBC674153-1	2.62	1.35	0.68	1.83	0.33
LIBC674200-1	3.31	1.33	0.79	2.08	0.74
LIBC674214-1	2.10	1.17	0.68	1.76	1.09
LIBC674216-1	1.96	1.10	0.52	1.66	0.65
LIBC674229-1	3.17	1.11	0.80	2.22	0.75
LIBC674235-1	2.05	1.30	0.68	1.28	0.72
LIBC674257-1	3.34	1.19	0.77	1.86	0.73
LIBC674276-1	2.06	1.17	0.54	2.04	0.51

*Relative binding fold change compared to LIBC523797-1(Fab), calculated after background subtraction

Example 10. Functional Analysis of Further Affinity Matured Variants

[0255] Clones isolated from Example 9 above were cloned and expressed as IgG1 SEFL2 mAbs for functional analysis.

Antagonist activity of the antibodies was analyzed using the TR-FRET cAMP assay as described above (e.g., Example 8) and potency of the antibodies are summarized in Table 33. The potency of all the antibodies was in single digit nM, with several having potency of less than 1 nM in all four experiments.

TABLE 33

IC50 values of exemplary antibodies from further affinity maturation			
Molecule	IC50 (nM, n = 4)	Molecule	IC50 (nM, n = 4)
LIBC673952-1	2.48	LIBC680562-1	0.71
LIBC673965-1	2.21	LIBC680574-1	0.75
LIBC673972-1	2.43	LIBC680773-1	0.74
LIBC673982-1	2.11	LIBC680800-1	1.57
LIBC674090-1	1.76	LIBC680812-1	0.68
LIBC674200-1	2.39	LIBC681593-1	0.61
LIBC674200-1 (LC: G76D)	2.23	LIBC681594-1	1.24
LIBC674229-1	4.16	LIBC681737-1	0.55
26970-1	14.41		
10D10	346.54		
Rimonabant	3.40		
LIBC528116-1	2.02		

Example 11. Effects of Further Affinity Matured Variants in Animal Model

[0256] The further affinity matured antibodies LIBC673952 (52777) and LIBC680574 (52778) in YTE format (Dall'Acqua et al., The Journal of Immunology, 169: 5171-5180 (2002)) was scaled up for testing in diet-induced

obese (DIO) transgenic mice with huCB1 knocked in. 6- to 10-week-old huCB1 knock in (KI) mice were fed with high fat diet (HFD) for 15 weeks to induce obesity. The antibodies (52777, 52778, and 37940) and a control antibody (huIgG control) were administered to the mice at 30 mg/kg QW for 35 days, and the body weight and food intake of the animals were measured every 3-4 days and the results are shown in FIGS. 10A and 10B. The administration of the further affinity matured antibodies reduced body weight of the animals (FIG. 10A), while the average daily food intake did not change significantly (FIG. 10B). Fat mass and liver weight of the animals were also measured, and the results are shown in FIGS. 10C and 10D. These results showed that potent peripherally-acting anti-CB1 antibody described herein was an effective treatment for anti-obesity, hyperinsulinemia, and NASH.

Example 12. Effects of Anti-CB1 mAb in Energy Expenditure and Insulin Tolerance Test

[0257] Materials and Methods: All studies using mice complied with all relevant ethical regulations and approval for the studies performed were obtained from Amgen's IACUC institutional review board (Thousand Oaks, CA). Lighting in animal holding rooms was maintained on 12:12 hr light:dark cycle, and the ambient temperature and humidity range was at 68-79° F. and 30-70%, respectively. Animals had ad libitum access to irradiated pelleted feed and reverse-osmosis chlorinated (0.3 to 0.5 ppm) water via an automatic watering system or by water bottles as indicated. Cages were changed weekly.

[0258] Male huCNR1 TG mice (4-7 weeks old) were shipped from Charles River Laboratories (Hollister, CA) to Amgen (Thousand Oaks, CA), singly housed upon arrival, and started high-fat diet (HFD) feeding (60% kcal fat, Research Diets D12492i) for 12 weeks. After 12 weeks HFD feeding, mice were acclimated to water bottles for one week and then n=24 mice were transferred to the Comprehensive Laboratory Animal Monitoring System (CLAMS, Columbus Instruments, Columbus, Ohio—Oxymax Model 2018, 0233-004M, Oxymax for Windows v5.53 software, Hardware configuration 190395) for an additional two weeks of acclimation. Volume of oxygen consumption (V02), carbon dioxide production (VCO2), respiratory exchange ratio (RER), and energy expenditure (HEAT) were measured during that period. At the start of treatment, mice had been fed HFD for 15 weeks total and n=23 mice were randomized into two groups based on body weight, age, V02, VCO2, RER, and Heat during baseline measurements from days -3 to -1. Mice were dosed with huIgG1 control PL-42466 (n=11 mice) or CB1 mAb 37940-17 (n=12 mice) at 30 mg/kg (2 mL/kg) IP weekly for a total of 5 injections. Mice were housed in the CLAMS system from days -14 to day 7, were returned to home caging on days 7-14, returned to the CLAMS system from days 14 to 28, and were returned to home caging on day 28 throughout the duration of the study. Body weight and food intake were measured weekly. For simplicity, data is displayed as lights on and lights off cycles (12 hours on: 12 hours off).

[0259] For the insulin tolerance test (ITT), on day 32 of treatment (4 days post 5th IP injection), mice were fasted at 6 am for 6 hours. At 12 pm, blood glucose was measured by retroorbital bleed using a glucometer (AlphaTrak) and mice were immediately IP dosed with insulin (0.75 U/kg). Blood glucose was measured by retroorbital bleed at 30-, 60-, 90-,

and 120-minutes post insulin administration. Area under the curve (AUC) was calculated for each mouse using GraphPad Prism (version 9.5.1).

[0260] Results: Energy expenditure measured as heat (kcal/hr/kg) and respiratory exchange ratio (RER) were averaged per week (acclimation, week 1, week 3, and week 4) for the light cycle (6 am-6 pm) and the dark cycle (6 pm-6 am). In the light cycle when mice are resting, energy expenditure during week 4 of treatment was significantly higher in the CB1 mAb treated mice (+12% compared to baseline for CB1 vs. +4% compared to baseline for control, p=0.0325; two-way repeated measures ANOVA with Sidak's test for multiple comparisons, FIG. 11A). Additionally, RER was significantly lower in the CB1 mAb treatment group during week 3 of treatment (treatment p=0.0125; two-way repeated measures ANOVA with Sidak's test for multiple comparisons), indicating increased lipid oxidation (FIG. 11A).

[0261] To assess insulin sensitivity, an insulin tolerance test (ITT) was conducted. After a 6-hour fast, mice did not differ in baseline blood glucose measurements; however, after insulin administration, CB1 mAb treated mice had statistically lower blood glucose at 30-, 60-, and 90-mins post-insulin injection (p<0.01, two-way repeated measures ANOVA with Sidak's test for multiple comparisons) and 25% lower AUC (p=0.0006; Student's unpaired t-test), indicating significantly improved insulin sensitivity after CB1 mAb treatment (FIG. 11B).

Example 13. Brain Uptake Study of Anti-CB1 Mab in Tg32 Mice

[0262] This study was carried out to characterize the single-dose plasma pharmacokinetics and brain uptake of an exemplary anti-CB1 antagonist mAb following intravenous bolus administration in male huFcRn Tg 32 homozygous mice.

[0263] Materials and Methods: Food and water were provided ad libitum throughout the study duration. On the day before dosing, the stock test articles were removed from -70° C. ($\pm 10^{\circ}$ C.) storage and placed at 2-8° C. to thaw overnight. On the morning of dose administration, the stock test article was removed from 2-8° C. storage, mixed by gentle inversion, and placed on wet ice for the dilution procedures. The vehicle was removed from 2-8° C. storage and mixed by gentle inversion and placed on wet ice for the dilution procedures. For all groups, the stock test articles were diluted in the appropriate formulation buffer to obtain the final dose solution concentrations listed in the experimental design. Following dose solution preparation, all remaining vehicles (if applicable) were returned to 2-8° C. storage. Following dose administration, all remaining stock test articles (if applicable) and remaining dosing solutions (if applicable) were placed at -70° C. ($\pm 10^{\circ}$ C.) and transferred to the Bioanalytical Principal Investigator for subsequent analysis and storage. Test animals received an intravenous bolus dose of the appropriate dosing solution via the lateral tail vein.

[0264] SARSTEDT Microvette® K3 EDTA plasma separation tubes were used to collect approximately 0.05 mL of whole blood per test system at each serial time point via submandibular venipuncture. After putting the whole blood into the SARSTEDT Microvette® plasma tubes, the tubes were mixed by 8-10 gentle inversions and placed on wet ice. The specimens were centrifuged at 2-8° C. at approximately

14,000 ref for 5 minutes using a calibrated Eppendorf 5417R Centrifuge System (Brinkmann Instruments, Inc., Westbury, NY 11590). All plasma specimens were stored at approximately -70° C. (±10° C.) until analysis.

[0265] Transcardial perfusion was performed at the end of the study. Each test system was deeply anesthetized with isoflurane (5% in O2(g) at a rate of 1.5 L/min) and remained under a surgical plane of anesthesia throughout the perfusion procedure. A cut was made in the skin right below the sternum and the tip of the sternum was located. Grasping the sternum, a cut was made through the diaphragm and along both sides of the rib cage to expose the heart, giving the animal a pneumothorax. The heart was steadied, and a 20 gauge, ½" needle was placed in the left ventricle. The needle was connected via tubing to a perfusion/syringe pump, and 0.9% saline was allowed to run at a rate of approximately 4 mL/minute for approximately 5 minutes. Once flow was established, the right atrium was incised to allow fluid to flow out of the animal, preventing over-pressurizing of the vasculature. After 5 minutes of perfusion with saline, 10% neutral buffered formalin was perfused for an additional 5 minutes. Once perfusion was complete, the needle was removed from the left ventricle and the appropriate tissue(s) was harvested for subsequent analysis.

[0266] Tissues were harvested from each test system as outlined below immediately following exsanguination at the terminal time point. Tissues were rinsed with 0.9% sterile saline upon extraction, blotted with dry gauze, immediately placed into cryovials (Eppendorf Protein LoBind, cat. #0030108116), weighed, and submerged in liquid nitrogen to snap freeze the tissues. Once the tissues were snapped frozen, they were stored on dry ice during necropsy procedures and at -70° C. (±10° C.) for subsequent analysis.

[0267] Plasma and tissue specimens for PK test article concentration determination were analyzed using an Enzyme-Linked ImmunoSorbant Assay (ELISA). Specimens were maintained at -70° C. (±10° C.) prior to analysis. Noncompartmental analysis was performed on the mean plasma test article concentration vs. time data from all mice at each sampling time per dose group. Parameters of interest include tissue-to-plasma percentage ratio on day 10, and half-life (t_{1/2}).

[0268] Results: the PK and brain exposure of the antibody are shown in Table 34 below. The results show that brain exposure of the anti-CB1 mAb is 0.2% of plasma concen-

tration of the antibody. Transgenic Tg32 mice expressing human neonatal Fc receptor (hFcRn) are known to be predictive of human PK for mAb therapeutics (Avery et. al. mAbs 20161). These results show that a 0.002 fraction of mAb plasma concentration can distribute to the brain. These results are also consistent with the previously published literature on Immunoglobulin G1 (IgG1) mAbs in other mouse models (e.g., Avery L. B. et al., MAb. 2016 August-September; 8(6):1064-78. doi: 10.1080/19420862.2016.1193660. Epub 2016 May 27. PMID: 27232760; PMCID: PMC4968115; Shah D. K. et al., MAb. 2013 March-April; 5(2):297-305. doi: 10.4161/mabs.23684. Epub 2013 Feb. 13. PMID: 23406896; PMCID: PMC3893240; and Garg A. and Balthasar J. P., AAPS J. 2009 September; 11(3):553-7. doi: 10.1208/s-12248-009-9129-9. Epub 2009 Jul. 28. PMID: 19636712; PMCID: PMC2758122).

TABLE 34

PK and brain exposure of the anti-CB1 mAb		
	Animal system	Anti-CB1 mAb
PK (T _{1/2})	Tg32 mice (huFcRn)	10.37 days
Brain exposure (% of plasma concentration on day 10)	Tg32 mice (huFcRn)	0.21 ± 0.07%

Example 14. Cryo-Em Structures of Cb1-Fab Complexes

[0269] Cryo-EM structures of huCB1 in complex with Fab of 10D10 and four affinity matured (AM) antibodies were resolved in a resolution of from ~3.1 to 3.7 Å, which is sufficient to determine epitope/paratope interactions. The potency of the antibodies ranges from 1.77 to 47 nM. The cryo-EM structures confirmed that only heavy chain of the antibodies is involved in binding and ECL2 of CB1 is the major epitope. In addition, HCDR1 plays a major role in binding of CB1 ECL2, this is consistent with the fact that HCDR1 sequences of AM antibodies are highly conserved. R32 (Aho numbering) that is adjacent to HCDR1 is also involved in epitope interaction. HCDR2 are diverse in sequence and certain residues (e.g., Y59, Aho numbering) in HCDR2 appear to contribute to binding affinity. Y131 and Y132 (Aho numbering) of HCDR3 are conserved among the tested antibodies and bind to CB1.

TABLE 35

Consensus sequences of affinity matured variants		
Seq. No.	Name	Sequence
530	Consensus sequence affinity matured variants HV CDR1	RGGDYWX1 X1 is A, S, or G
531	Consensus sequence affinity matured variants HV CDR2	HX2YX3X4GX5TX6YNPX7X8X9X10 X2 is I or V; X3 is H, Y, or Q; X4 is E, T, or S; X5 is K, S, or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N
532	Consensus sequence affinity matured variants HV CDR3	X11YDX12X13X14GX15SYYYGMDV X11 is D, E, or N; X12 is A, I, P, T, or V; X13 is F, L, or V; X14 is S or T; and X15 is H, N, or Y

TABLE 35-continued

Consensus sequences of affinity matured variants		
Seq. No.	Name	Sequence
533	Consensus sequence	RSSQSLLX16SX17GX18NYX19D
	affinity matured variants LV CDR1	X16 is H, S, or T; X17 is S, T, or Y; X18 is A, N, I or Y; and X19 is L or V
534	Consensus sequence	X20GSNRA
	affinity matured variants LV CDR2	X20 is L or Q
535	Consensus sequence	X21QAX22X23X24PRT
	affinity matured variants LV CDR3	X21 is M or R; X22 is L, I, R, or V; X23 is Q, E, T, A, or G; and X24 is T, I, L, or Q
564	Consensus sequence_further	RGGDYWX1
	affinity matured LC swapped variants HV CDR1	X1 is A or G
565	Consensus sequence_further	HX2YX3X4GX5TX6YNPX7X8X9X10
	affinity matured LC swapped variants HV CDR2	X2 is I or V; X3 is Y, or Q; X4 is T; X5 is S; X6 is K or N; X7 is S, or R; X8 is F; X9 is K; and X10 is G or D
566	Consensus sequence_further	X11YDX12X13X14GX15SYYYGMDV
	affinity matured LC swapped variants HV CDR3	X11 is N; X12 is T; X13 is L, or V; X14 is T; and X15 is Y
405	Consensus sequence_further	RASQSISSYLN
	affinity matured LC swapped variants LV CDR1	
536	Consensus sequence_further	X39ARX40LX41S
	affinity matured LC swapped variants LV CDR2	X39 is N, S, K or G; X40 is R, K, L or A; and X41 is A, G or S
537	Consensus sequence_further	QQX42X43X44X45PX46T
	affinity matured LC swapped variants LV CDR3	X42 is Y or F; X43 is R, A, S, G, or Y; X44 is S, K, R or H; X45 is S, L, F, Y, P, or M; and X46 is L, I or V
1	Consensus sequence_further	RGGDYWS
	affinity matured cognate variants HV CDR1	
538	Consensus sequence_further	HX25YX26X27GX28TX29YNPX30X31X32X33
	affinity matured cognate variants HV CDR2	X25 is I or V; X26 is Y or Q; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L; X32 is K or R; and X33 is G, S, or
539	Consensus sequence_further	X34YDX35X36X37GYSYYYGX38DV
	affinity matured cognate variants HV CDR3	X34 is D or G; X35 is A, I, T, or V; X36 is L or absent; X37 is S or T; and X38 is M or L
257	Consensus sequence_further	RSSQSLLHRSRGYNYLD
	affinity matured cognate variants LV CDR1	
534	Consensus sequence_further	X20GSNRA
	affinity matured cognate variants LV CDR2	X20 is L or Q
540	Consensus sequence_further	X47QX48X49X50X51PRT
	affinity matured cognate variants LV CDR3	X47 is M or R; X48 is A or S; X49 is L, R, or V; X50 is Q or A; and X51 is T or L

[0270] While this invention has been described with an emphasis upon preferred aspects, it will be obvious to those of ordinary skill in the art that variations of the preferred compounds and methods may be used and that it is intended

that the invention may be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications encompassed within the spirit and scope of the invention as defined by the following claims.

SEQUENCE LISTING

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SEQUENCE: 11 QGSNRAS		7
SEQ ID NO: 12 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
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FEATURE	Location/Qualifiers	
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	mol_type = protein	

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SEQ ID NO: 49 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 49 RGGDYWS		7
SEQ ID NO: 50 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 50 HIYQTGSTAY NPSLKG		16
SEQ ID NO: 51 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
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SEQUENCE: 53 LGSNRAS		7
SEQ ID NO: 54 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
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SEQ ID NO: 55 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
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	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 59		
LGSNRAS		7
SEQ ID NO: 60	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 60		
RQARTLPRT		9
SEQ ID NO: 61	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 61		
RGGDYWS		7
SEQ ID NO: 62	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 62		
HVYYTGQTAY NPKFKD		16
SEQ ID NO: 63	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 63		
DYDILTGYSY YYYGMDV		17
SEQ ID NO: 64	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 64		
RSSQSLSSS GANYVD		16
SEQ ID NO: 65	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 65		
LGSNRAS		7
SEQ ID NO: 66	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	

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SEQUENCE: 66 MQARGIPRT	organism = synthetic construct	9
SEQ ID NO: 67 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 67 RGGDYWS		7
SEQ ID NO: 68 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 68 HIYQSGQTKY NPSLKD		16
SEQ ID NO: 69 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 69 DYDILTGYSY YYYGMDV		17
SEQ ID NO: 70 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 70 RSSQSLTSS GANYVD		16
SEQ ID NO: 71 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 71 LGSNRAS		7
SEQ ID NO: 72 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 72 RQAREQPRT		9
SEQ ID NO: 73 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 73 RGGDYWA		7
SEQ ID NO: 74 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 74 HVIYQSGQTAY NPNFEN		16
SEQ ID NO: 75 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 75 DYDAVSGYSY YYYGMDV		17

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SEQ ID NO: 76	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 76		
RSSQSLSSY GNNYVD		16
SEQ ID NO: 77	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 77		
QGSNRAS		7
SEQ ID NO: 78	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 78		
RQAVGLPRT		9
SEQ ID NO: 79	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 79		
RGDDYWS		7
SEQ ID NO: 80	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 80		
HVYQSGQTAY NPNFED		16
SEQ ID NO: 81	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 81		
EYDVVSGYSY YYYGMDV		17
SEQ ID NO: 82	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 82		
RSSQSLTSS GYNYVD		16
SEQ ID NO: 83	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 83		
QGSNRAS		7
SEQ ID NO: 84	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 84		
MQAVKIPRT		9
SEQ ID NO: 85	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	

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SEQUENCE: 85 RGGDYWA	organism = synthetic construct	7
SEQ ID NO: 86 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 86 HVVQTGQTAY NPNLES		16
SEQ ID NO: 87 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 87 NYDVLSGYSY YYYGMDV		17
SEQ ID NO: 88 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 88 RSSQSLTYSY GANYLD		16
SEQ ID NO: 89 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 89 QGSNRAS		7
SEQ ID NO: 90 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 90 MQALQIPRT		9
SEQ ID NO: 91 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 91 RGGDYWS		7
SEQ ID NO: 92 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 92 YIYSGSTNY NPSLKS		16
SEQ ID NO: 93 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 93 DYDILTGYSY YYYGMDV		17
SEQ ID NO: 94 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 94 RSSQSLHSN GYNYLD		16

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SEQ ID NO: 95	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 95		
LGSNRAS		7
SEQ ID NO: 96	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 96		
MQALQTPRT		9
SEQ ID NO: 97	moltype = AA length = 113	
FEATURE	Location/Qualifiers	
source	1..113	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 97		
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL TSSGYNYVDW YLQKPGQSPQ LLIYQGSNRA		60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQARAIP RTFGQGKVE IKR		113
SEQ ID NO: 98	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 98		
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYEGSTK		60
YNPSPFKGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDPVTGHSYY YYGMDVWGQG		120
TTVTVSS		127
SEQ ID NO: 99	moltype = AA length = 113	
FEATURE	Location/Qualifiers	
source	1..113	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 99		
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL TSYGINYVDW YLQKPGQSPQ LLIYQGSNRA		60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQAVGLP RTFGQGKVE IKR		113
SEQ ID NO: 100	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 100		
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYYTGSTK		60
YNPNFKGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSYY YYGMDVWGQG		120
TTVTVSS		127
SEQ ID NO: 101	moltype = AA length = 113	
FEATURE	Location/Qualifiers	
source	1..113	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 101		
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL SSTGANYLDW YLQKPGQSPQ LLIYQGSNRA		60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQAVGLP RTFGQGKVE IKR		113
SEQ ID NO: 102	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 102		
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQTGQTK		60
YNPNLKGRAIT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSYY YYGMDVWGQG		120
TTVTVSS		127
SEQ ID NO: 103	moltype = AA length = 113	
FEATURE	Location/Qualifiers	
source	1..113	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 103
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HSYGANYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQARALP RTFGQGKVE IKR 113

SEQ ID NO: 104      moltype = AA length = 127
FEATURE            Location/Qualifiers
source             1..127
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 104
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYTGSTK 60
YNPSFEGRAT ISVKTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 105      moltype = AA length = 113
FEATURE            Location/Qualifiers
source             1..113
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 105
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL SSTGANYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQAVELP RTFGQGKVE IKR 113

SEQ ID NO: 106      moltype = AA length = 127
FEATURE            Location/Qualifiers
source             1..127
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 106
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQTGQTA 60
YNPRLEGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 107      moltype = AA length = 113
FEATURE            Location/Qualifiers
source             1..113
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 107
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL TSYGINYVDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCMQARQIP RTFGQGKVE IKR 113

SEQ ID NO: 108      moltype = AA length = 127
FEATURE            Location/Qualifiers
source             1..127
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 108
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYEGSTA 60
YNPSFEDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 109      moltype = AA length = 113
FEATURE            Location/Qualifiers
source             1..113
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 109
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL SSSGANYVDW YLQKPGQSPQ LLIYQGSNRA 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQARALP RTFGQGKVE IKR 113

SEQ ID NO: 110      moltype = AA length = 127
FEATURE            Location/Qualifiers
source             1..127
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 110
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYTGSTK 60
YNPSFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 111      moltype = AA length = 113
FEATURE            Location/Qualifiers
source             1..113

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mol_type = protein
organism = synthetic construct

SEQUENCE: 111
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL TSSGANYVDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQAVGTP RTFGQGKVE IKR 113

SEQ ID NO: 112      moltype = AA length = 127
FEATURE            Location/Qualifiers
source              1..127
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 112
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYQTGSTK 60
YNPKFKGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 113      moltype = AA length = 113
FEATURE            Location/Qualifiers
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 113
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HSTGANYVDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQAIALP RTFGQGKVE IKR 113

SEQ ID NO: 114      moltype = AA length = 127
FEATURE            Location/Qualifiers
source              1..127
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 114
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQTGSTA 60
YNPSLKGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 115      moltype = AA length = 113
FEATURE            Location/Qualifiers
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 115
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HSYGANYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQARTLP RTFGQGKVE IKR 113

SEQ ID NO: 116      moltype = AA length = 127
FEATURE            Location/Qualifiers
source              1..127
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 116
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYTGSTA 60
YNPSLKSRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 117      moltype = AA length = 113
FEATURE            Location/Qualifiers
source              1..113
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 117
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL SSSGANYVDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPRDRFSGS GSGTDFTLKI SRVEAEDVGV YYCMQARGIP RTFGQGKVE IKR 113

SEQ ID NO: 118      moltype = AA length = 127
FEATURE            Location/Qualifiers
source              1..127
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 118
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYYTGQTA 60
YNPKFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 119      moltype = AA length = 113
FEATURE            Location/Qualifiers
source              1..113

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mol_type = protein
organism = synthetic construct

SEQUENCE: 119
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL TSSGANYVDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQAREQP RTFGQGKVE IKR 113

SEQ ID NO: 120 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
mol_type = protein
organism = synthetic construct

SEQUENCE: 120
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHYQSGQTK 60
YNPNSLKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSYY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 121 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
mol_type = protein
organism = synthetic construct

SEQUENCE: 121
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL SSGYNNYVDW YLQKPGQSPQ LLIYQGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCRQAVGLP RTFGQGKVE IKR 113

SEQ ID NO: 122 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
mol_type = protein
organism = synthetic construct

SEQUENCE: 122
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHVYQSGQTA 60
YNPNFENRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDAVSGYSYY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 123 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
mol_type = protein
organism = synthetic construct

SEQUENCE: 123
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL TSSGYNYVDW YLQKPGQSPQ LLIYQGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCMQAVKIP RTFGQGKVE IKR 113

SEQ ID NO: 124 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
mol_type = protein
organism = synthetic construct

SEQUENCE: 124
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYQSGQTA 60
YNPNFEDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARE YDVVSGYSYY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 125 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
mol_type = protein
organism = synthetic construct

SEQUENCE: 125
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL TSYGANYLDW YLQKPGQSPQ LLIYQGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCMQALQIP RTFGQGKVE IKR 113

SEQ ID NO: 126 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
mol_type = protein
organism = synthetic construct

SEQUENCE: 126
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHVYQTGQTA 60
YNPNLES RAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARN YDVLSGYSYY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 127 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113

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	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 127		
DIVMTQSPPLS LPVTPGEPAS	ISCRSSQSLL HSENGYNYLDW YLQKPGQSPQ LLIYLGSNRA	60
SGVPDRFSGS GSGTDFTLKI	RRVEAEDVGV YYCMQALQTP RTFGQGTVKE IKR	113
SEQ ID NO: 128	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 128		
QVQLQESGPG LVKPSQTLST	TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GYIYSGSTN	60
YNPSLKSRAT ISVDTSKNQF	SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGMDVWQGQ	120
TTVTVSS		127
SEQ ID NO: 129	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 129		
RGGDYWA		7
SEQ ID NO: 130	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 130		
HIYQTGSTNY NPRFKG		16
SEQ ID NO: 131	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 131		
NYDTLTGYSY YYGMDV		17
SEQ ID NO: 132	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 132		
RASQSSISNYL N		11
SEQ ID NO: 133	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 133		
AASSLHS		7
SEQ ID NO: 134	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 134		
QQYQSYPLT		9
SEQ ID NO: 135	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 135		
RGGDYWA		7
SEQ ID NO: 136	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	

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SEQUENCE: 136 HVVYTGSTNY NPRFKD	organism = synthetic construct	16
SEQ ID NO: 137 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 137 NYDTVTGYSY YYYGMDV		17
SEQ ID NO: 138 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 138 RASQISISNYL N		11
SEQ ID NO: 139 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 139 AASSLHS		7
SEQ ID NO: 140 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 140 QQYQSYPLT		9
SEQ ID NO: 141 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 141 RGGDYWS		7
SEQ ID NO: 142 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 142 HIYQSGSTKY NPSFED		16
SEQ ID NO: 143 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 143 DYDILTGYSY YYYGMDV		17
SEQ ID NO: 144 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 144 RASQISISNYL N		11
SEQ ID NO: 145 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 145 AASSLHS		7

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SEQ ID NO: 146	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 146		
QQYQSYPLT		9
SEQ ID NO: 147	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 147		
RGGDYWG		7
SEQ ID NO: 148	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 148		
HIYQSGSTNY NPKLKS		16
SEQ ID NO: 149	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 149		
DYDIFTGYSY YYYGMDV		17
SEQ ID NO: 150	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 150		
RASQSIISYL N		11
SEQ ID NO: 151	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 151		
AASSLRS		7
SEQ ID NO: 152	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 152		
QQYSNYPLT		9
SEQ ID NO: 153	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 153		
RGGDYWS		7
SEQ ID NO: 154	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 154		
HVYQTGSTKY NPRFKG		16
SEQ ID NO: 155	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	

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SEQUENCE: 155 DYDILTGYSY YYYGMDV	organism = synthetic construct	17
SEQ ID NO: 156 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 156 RASQISISNYL N		11
SEQ ID NO: 157 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 157 AASSLHS		7
SEQ ID NO: 158 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 158 QQYQSYPLT		9
SEQ ID NO: 159 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 159 RGGDYWG		7
SEQ ID NO: 160 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 160 HVVQTGSTKY NPSFKD		16
SEQ ID NO: 161 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 161 NYDTLTGYSY YYYGMDV		17
SEQ ID NO: 162 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 162 RASQISISNYL N		11
SEQ ID NO: 163 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 163 AASSLHS		7
SEQ ID NO: 164 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 164 QQYQSYPLT		9

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SEQ ID NO: 165	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 165		
RGGDYWS		7
SEQ ID NO: 166	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 166		
HIYQTGQTAY NPNLEG		16
SEQ ID NO: 167	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 167		
DYDILTGYSY YYYGMDV		17
SEQ ID NO: 168	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 168		
RASQSVSSYL G		11
SEQ ID NO: 169	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 169		
GASSRAT		7
SEQ ID NO: 170	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 170		
QQYGSSPRT		9
SEQ ID NO: 171	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 171		
RGGDYWS		7
SEQ ID NO: 172	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 172		
HIYQTGKTNY NPNFKS		16
SEQ ID NO: 173	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 173		
DYDILTGYSY YYYGMDV		17
SEQ ID NO: 174	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	

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SEQUENCE: 174 RASQISISNYL N	organism = synthetic construct	11
SEQ ID NO: 175 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 175 AASSLHS		7
SEQ ID NO: 176 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 176 QQYQSYPLT		9
SEQ ID NO: 177 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 177 RGGDYWG		7
SEQ ID NO: 178 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 178 HIYQTGSTNY NPNFKD		16
SEQ ID NO: 179 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 179 DYDILTGYSY YYYGMDV		17
SEQ ID NO: 180 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 180 RASQISISNYL N		11
SEQ ID NO: 181 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 181 AASSLHS		7
SEQ ID NO: 182 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 182 QQYQSYPLT		9
SEQ ID NO: 183 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 183 RGGDYWS		7

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SEQ ID NO: 184	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 184		
YIYQTGQTKY NPNFEG		16
SEQ ID NO: 185	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 185		
DYDILTGYSY YYYGMDV		17
SEQ ID NO: 186	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 186		
RASQSISSYL N		11
SEQ ID NO: 187	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 187		
AASSLRS		7
SEQ ID NO: 188	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 188		
QQYSNYPLT		9
SEQ ID NO: 189	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 189		
RGGDYWS		7
SEQ ID NO: 190	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 190		
HIYQSGQTKY NPKLKG		16
SEQ ID NO: 191	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 191		
DYDILTGYSY YYYGMDV		17
SEQ ID NO: 192	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 192		
RASQSISSYL N		11
SEQ ID NO: 193	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	

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SEQUENCE: 193 AASSLHS	organism = synthetic construct	7
SEQ ID NO: 194 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 194 QQYQSYPLT		9
SEQ ID NO: 195 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 195 RGGDYWS		7
SEQ ID NO: 196 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 196 HIYQSGSTKY NPSFKD		16
SEQ ID NO: 197 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 197 DYDILTGYSY YYYGMDV		17
SEQ ID NO: 198 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 198 RASQSIISNYL N		11
SEQ ID NO: 199 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 199 AASSLHS		7
SEQ ID NO: 200 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 200 QQYQSYPLT		9
SEQ ID NO: 201 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 201 RGGDYWA		7
SEQ ID NO: 202 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 202 HIYHTGKTNY NPSFKG		16

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SEQ ID NO: 203	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 203		
NYDTLSGNSY YYYGMDV		17
SEQ ID NO: 204	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 204		
RASQSIISYL N		11
SEQ ID NO: 205	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 205		
AASSLHS		7
SEQ ID NO: 206	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 206		
QQYQSYPLT		9
SEQ ID NO: 207	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 207		
RGDYWA		7
SEQ ID NO: 208	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 208		
HVYQTGSTAY NPNFKD		16
SEQ ID NO: 209	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 209		
NYDALSGYSY YYYGMDV		17
SEQ ID NO: 210	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 210		
RASQSIISYL N		11
SEQ ID NO: 211	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 211		
AASSLRS		7
SEQ ID NO: 212	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	

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SEQUENCE: 212 QQYSNYPLT	organism = synthetic construct	9
SEQ ID NO: 213 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 213 RGGDYWG		7
SEQ ID NO: 214 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 214 HVIQSGQTAY NPKFEN		16
SEQ ID NO: 215 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 215 NYDTLTGHSY YYYGMDV		17
SEQ ID NO: 216 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 216 RASQSSISNYL N		11
SEQ ID NO: 217 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 217 AASSLHS		7
SEQ ID NO: 218 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 218 QQYQSYPLT		9
SEQ ID NO: 219 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 219 RGGDYWS		7
SEQ ID NO: 220 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 220 YIIYSGSTNY NPSLKS		16
SEQ ID NO: 221 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 221 DYDILTGYSY YYYGMDV		17

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SEQ ID NO: 222	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 222		
RSSQSLLHSY GYNYLD		16
SEQ ID NO: 223	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 223		
LGSNRAS		7
SEQ ID NO: 224	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 224		
MQALQTPRT		9
SEQ ID NO: 225	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 225		
DIQMTQSPSS LSASVGDVRT ITCRASQSSIS NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS		60
RFSGSGSGTD FTLTISSLQP EDFASYICQQ YQSYPLTFGQ GTKVEIKR		108
SEQ ID NO: 226	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 226		
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYAWIR QHPGKGLEWI GHYQTGSTN		60
YNPRFKGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARN YDTLTGYSYY YGMDVWGQG		120
TTVTVSS		127
SEQ ID NO: 227	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 227		
DIQLTQSPST LSASVGDVRT ITCRASQSSIS NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS		60
RFSGSGSGTD FTLTISSLQP EDFASYICQQ YQSYPLTFGQ GTKVEIKR		108
SEQ ID NO: 228	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 228		
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYAWIR QHPGKGLEWI GHVYYTGSTN		60
YNPRFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARN YDTVTGYSYY YGMDVWGQG		120
TTVTVSS		127
SEQ ID NO: 229	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 229		
DIQLTQSPST LSASVGDVRT ITCRASQSSIS NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS		60
RFSGSGSGTD FTLTISSLQP EDFASYICQQ YQSYPLTFGQ GTKVEIKR		108
SEQ ID NO: 230	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 230
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHYQSGSTK 60
YNPSFEDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 231 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 231
DIQLTQSPSS LSASVGDVRT ITCRASQSII SYLNWYQQKP GKAPKLLIYA ASSLRSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YSNYPLTFGG GTRVEIKR 108

SEQ ID NO: 232 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 232
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWGWR QHPGKGLEWI GHYQSGSTN 60
YNPKLKSRAIT ISVDYKNQF SLKLSSVTAA DTAVYYCARD YDIFTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 233 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 233
DIQLTQSPST LSASVGDVRT ITCRASQSI SYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFASYCQQ YQSYPLTFGG GTKVEIKR 108

SEQ ID NO: 234 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 234
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYQTGSTK 60
YNPRFKGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 235 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 235
DIQMTQSPST LSASVGDVRT ITCRASQSI SYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFASYCQQ YQSYPLTFGG GTKVEIKR 108

SEQ ID NO: 236 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 236
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWGWR QHPGKGLEWI GHVYQTGSTK 60
YNPSFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARN YDTLTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 237 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 237
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLGWYQQKP GQAPRLLIYG ASSRATGIPD 60
RFSGSGSGTD FTLTISRLEP EDFAVYYCQQ YGSSPRTFGG GTKVEIKR 108

SEQ ID NO: 238 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

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SEQUENCE: 238
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQTGQTA 60
YNPNLEGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 239 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 239
DIQMTQSPST LSASVGDRTV ITCRASQSSIS NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFASYCQQ YQSYPLTFGQ GTKVEIKR 108

SEQ ID NO: 240 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 240
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQTGKTN 60
YNPNFKSRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 241 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 241
DIQMTQSPST LSASVGDRTV ITCRASQSSIS NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFASYCQQ YQSYPLTFGQ GTKVEIKR 108

SEQ ID NO: 242 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 242
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQTGSTN 60
YNPNFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 243 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 243
DIQMTQSPST LSASVGDRTV ITCRASQSSIS NYLNWYQQKP GKAPKLLIYA ASSLRSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YSNYPLTFGG GTRVEIKR 108

SEQ ID NO: 244 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 244
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GYIYQTGQTK 60
YNPNFEGRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 245 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 245
DIQMTQSPST LSASVGDRTV ITCRASQSSIS NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFASYCQQ YQSYPLTFGQ GTKVEIKR 108

SEQ ID NO: 246 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

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SEQUENCE: 246
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQSGQTK 60
YNPKLKGRAT ISVDTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 247 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 247
DIQLTQSPST LSASVGDRVT ITCRASQSI NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
RFGSGSGSTD FTLTISSLQP EDFASYCQQ YQSYPLTFGQ GTKVEIKR 108

SEQ ID NO: 248 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 248
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQSGSTK 60
YNPSFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 249 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 249
DIQLTQSPST LSASVGDRVT ITCRASQSI NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
RFGSGSGSTD FTLTISSLQP EDFASYCQQ YQSYPLTFGQ GTKVEIKR 108

SEQ ID NO: 250 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 250
QVQLQESGPG LVKPSQTLST TCTVSGGNIR RGGDYWAWIR QHPGKGLEWI GHIYHTGKTN 60
YNPSFKGRAT ISVKTYKNQF SLKLSSVTAA DTAVYYCARN YDTLSGNSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 251 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 251
DIQLTQSPSS LSASVGDRVT ITCRASQSI SYLNWYQQKP GKAPKLLIYA ASSLRSGVPS 60
RFGSGSGSTD FTLTISSLQP EDFATYYCQQ YSNYPLTFGG GTRVEIKR 108

SEQ ID NO: 252 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 252
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHVYQTGSTA 60
YNPNFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARN YDALSGYSY YGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 253 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 253
DIQMTQSPST LSASVGDRVT ITCRASQSI NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
RFGSGSGSTD FTLTISSLQP EDFASYCQQ YQSYPLTFGQ GTKVEIKR 108

SEQ ID NO: 254 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

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SEQUENCE: 254
QVQLQESGPG LVKPSQTLNL TCTVSGGSIR RGGDYWGWR QHPGKGLEWI GHVYQSGQTA 60
YNPKFENRAT ISVKTYKNQF SLKLSSVTAA DTAVYYCARN YDILTGHSSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 255 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 255
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HSYGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI RRVEAEVGV YYCMQALQTP RTFGQGTKVE IKR 113

SEQ ID NO: 256 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 256
QVQLQESGPG LVKPSQTLNL TCTVSGGSIR RGGDYWSWR QHPGKGLEWI GYIYSGSTN 60
YNPSLKSRAI ISVDTSKNQF SLKLSSVTAA DTAVYYCARD YDILTGSY YYGMDVWGQG 120
TTVTVSS 127

SEQ ID NO: 257 moltype = AA length = 16
FEATURE Location/Qualifiers
source 1..16
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 257
RSSQSLLHRS GYNYLD 16

SEQ ID NO: 258 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 258
LGSNRAS 7

SEQ ID NO: 259 moltype = AA length = 9
FEATURE Location/Qualifiers
source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 259
MQSLQTPRT 9

SEQ ID NO: 260 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 260
RGGDYWS 7

SEQ ID NO: 261 moltype = AA length = 16
FEATURE Location/Qualifiers
source 1..16
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 261
HIYYTGQTAY NPNLRN 16

SEQ ID NO: 262 moltype = AA length = 16
FEATURE Location/Qualifiers
source 1..16
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 262
GYDASGYSY YYGLDV 16

SEQ ID NO: 263 moltype = AA length = 16
FEATURE Location/Qualifiers
source 1..16
 mol_type = protein

-continued

SEQUENCE: 263 RSSQSLHRS GYNYLD	organism = synthetic construct	16
SEQ ID NO: 264 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 264 QGSNRAS		7
SEQ ID NO: 265 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 265 RQARALPRT		9
SEQ ID NO: 266 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 266 RGGDYWS		7
SEQ ID NO: 267 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 267 HVYYTGQTTY NPSFRG		16
SEQ ID NO: 268 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 268 DYDILTGYSY YYYGLDV		17
SEQ ID NO: 269 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 269 RSSQSLHRS GYNYLD		16
SEQ ID NO: 270 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 270 LGSNRAS		7
SEQ ID NO: 271 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 271 RQSVALPRT		9
SEQ ID NO: 272 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 272 RGGDYWS		7

-continued

SEQ ID NO: 273	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 273		
HIYYTGSTTY NPSFRG		16
SEQ ID NO: 274	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 274		
DYDISGYSYY YGMDV		16
SEQ ID NO: 275	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 275		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 276	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 276		
LGSNRAS		7
SEQ ID NO: 277	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 277		
MQSLQTPRT		9
SEQ ID NO: 278	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 278		
RGGDYWS		7
SEQ ID NO: 279	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 279		
HIYYTGQTTY NPNFKG		16
SEQ ID NO: 280	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 280		
GYDASGYSYY YGMDV		16
SEQ ID NO: 281	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 281		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 282	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	

-continued

SEQUENCE: 282 LGSNRAS	organism = synthetic construct	7
SEQ ID NO: 283 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 283 MQSLQTPRT		9
SEQ ID NO: 284 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 284 RGGDYWS		7
SEQ ID NO: 285 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 285 HIYYTGSTKY NPSLKN		16
SEQ ID NO: 286 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 286 GYDVSGYSYY YYGLDV		16
SEQ ID NO: 287 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 287 RSSQSLHRS GYNYLD		16
SEQ ID NO: 288 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 288 QGSNRAS		7
SEQ ID NO: 289 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 289 MQSLQTPRT		9
SEQ ID NO: 290 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 290 RGGDYWS		7
SEQ ID NO: 291 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 291 HIYYAGSTAY NPNFRG		16

-continued

SEQ ID NO: 292	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 292		
GYDTSGYSYY YYGLDV		16
SEQ ID NO: 293	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 293		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 294	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 294		
LGSNRAS		7
SEQ ID NO: 295	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 295		
MQSLQTPRT		9
SEQ ID NO: 296	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 296		
RGGDYWS		7
SEQ ID NO: 297	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 297		
HVYYTGQTAY NPNFRN		16
SEQ ID NO: 298	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 298		
GYDVSGYSYY YYGMDV		16
SEQ ID NO: 299	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 299		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 300	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 300		
LGSNRAS		7
SEQ ID NO: 301	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	

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SEQUENCE: 301 MQSLQTPRT	organism = synthetic construct	9
SEQ ID NO: 302 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 302 RGGDYWS		7
SEQ ID NO: 303 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 303 HVYYTGQTAY NPSLRG		16
SEQ ID NO: 304 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 304 GYDTSGYSYY YYGLDV		16
SEQ ID NO: 305 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 305 RSSQSLLHRS GYNYLD		16
SEQ ID NO: 306 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 306 LGSNRAS		7
SEQ ID NO: 307 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 307 MQSLQTPRT		9
SEQ ID NO: 308 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 308 RGGDYWS		7
SEQ ID NO: 309 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 309 HIYYTGQTAY NPSLRN		16
SEQ ID NO: 310 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 310 GYDISGYSYY YYGLDV		16

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SEQ ID NO: 311	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 311		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 312	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 312		
LGSNRAS		7
SEQ ID NO: 313	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 313		
MQSLQTPRT		9
SEQ ID NO: 314	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 314		
RGGDYWS		7
SEQ ID NO: 315	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 315		
HVYEGSTKY NPSFRG		16
SEQ ID NO: 316	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 316		
GYDVSGYSYY YYGLDV		16
SEQ ID NO: 317	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 317		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 318	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 318		
LGSNRAS		7
SEQ ID NO: 319	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 319		
MQSLQTPRT		9
SEQ ID NO: 320	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	

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SEQUENCE: 320 RGGDYWS	organism = synthetic construct	7
SEQ ID NO: 321 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 321 HVIYAGSTTY NPSLKS		16
SEQ ID NO: 322 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 322 GYDVSGYSYY YYGLDV		16
SEQ ID NO: 323 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 323 RSSQSLHRS GYNYLD		16
SEQ ID NO: 324 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 324 LGSNRAS		7
SEQ ID NO: 325 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 325 MQSLQTPRT		9
SEQ ID NO: 326 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 326 RGGDYWS		7
SEQ ID NO: 327 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 327 HIYYTGQTKY NPSLRG		16
SEQ ID NO: 328 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 328 GYDISGYSYY YYGLDV		16
SEQ ID NO: 329 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 329 RSSQSLHRS GYNYLD		16

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SEQ ID NO: 330	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 330		
LGSNRAS		7
SEQ ID NO: 331	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 331		
MQSLQTPRT		9
SEQ ID NO: 332	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 332		
RGGDYWS		7
SEQ ID NO: 333	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 333		
HVYYTGSTAY NPSFRG		16
SEQ ID NO: 334	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 334		
GYDTSGYSYY YYGLDV		16
SEQ ID NO: 335	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 335		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 336	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 336		
LGSNRAS		7
SEQ ID NO: 337	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 337		
MQSLQTPRT		9
SEQ ID NO: 338	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 338		
RGGDYWS		7
SEQ ID NO: 339	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	

-continued

SEQUENCE: 339 HIYYKGSTGY NPSLRG	organism = synthetic construct	16
SEQ ID NO: 340 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 340 GYDVSGYSYY YYGLDV		16
SEQ ID NO: 341 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 341 RSSQSLHRS GYNYLD		16
SEQ ID NO: 342 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 342 LGSNRAS		7
SEQ ID NO: 343 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 343 MQSLQTPRT		9
SEQ ID NO: 344 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 344 RGGDYWS		7
SEQ ID NO: 345 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 345 HVYTGSTTY NPNFKG		16
SEQ ID NO: 346 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 346 GYDASGYSYY YYGLDV		16
SEQ ID NO: 347 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 347 RSSQSLHRS GYNYLD		16
SEQ ID NO: 348 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 348 LGSNRAS		7

-continued

SEQ ID NO: 349	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 349		
MQSLQTPRT		9
SEQ ID NO: 350	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 350		
RGGDYWS		7
SEQ ID NO: 351	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 351		
HVYQTGSTKY NPSFKN		16
SEQ ID NO: 352	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 352		
GYDVSGYSYY YGLDV		16
SEQ ID NO: 353	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 353		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 354	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 354		
LGSNRAS		7
SEQ ID NO: 355	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 355		
MQSLQTPRT		9
SEQ ID NO: 356	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 356		
RGGDYWS		7
SEQ ID NO: 357	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 357		
HIYYAGQTKY NPSFKN		16
SEQ ID NO: 358	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	

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SEQUENCE: 358 GYDISGYSYY YYGLDV	organism = synthetic construct	16
SEQ ID NO: 359 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 359 RSSQSLHRS GYNYLD		16
SEQ ID NO: 360 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 360 QGSNRAS		7
SEQ ID NO: 361 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 361 MQSLQTPRT		9
SEQ ID NO: 362 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 362 RGGDYWS		7
SEQ ID NO: 363 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 363 HIYQTGQTKY NPSFKG		16
SEQ ID NO: 364 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 364 GYDVSIGYSYY YYGMDV		16
SEQ ID NO: 365 FEATURE source	moltype = AA length = 113 Location/Qualifiers 1..113 mol_type = protein organism = synthetic construct	
SEQUENCE: 365 DIVMTQSPLS LPVTPGEPAS ISCRSSQSL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA SGVPDRFSGS GSGTDFTLKI SRVEAEDVG YVCMQSLQTP RTFGQGKVE IKR		60 113
SEQ ID NO: 366 FEATURE source	moltype = AA length = 126 Location/Qualifiers 1..126 mol_type = protein organism = synthetic construct	
SEQUENCE: 366 QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYTGQTA YNPNLRNRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDASGYSYY YGLDVWGQGT TVTVSS		60 120 126
SEQ ID NO: 367 FEATURE source	moltype = AA length = 113 Location/Qualifiers 1..113 mol_type = protein organism = synthetic construct	

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SEQUENCE: 367
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYQGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCRQARALP RTFGQGKVE IKR 113

SEQ ID NO: 368 moltype = AA length = 127
FEATURE Location/Qualifiers
source 1..127
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 368
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYYTGQTT 60
YNPSFRGRVT ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSTY YYGLDVWGQG 120
TTVTVSS 127

SEQ ID NO: 369 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 369
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCRQSVALP RTFGQGKVE IKR 113

SEQ ID NO: 370 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 370
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYTGSTT 60
YNPSFRGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARD YDISGYSYYY YGMDVWGQGT 120
TVTVSS 126

SEQ ID NO: 371 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 371
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 372 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 372
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYTGQTT 60
YNPNFKGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDASGYSYYY YGMDVWGQGT 120
TVTVSR 126

SEQ ID NO: 373 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 373
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 374 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 374
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYTGSTK 60
YNPSLKNRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDVSGYSYYY YGLDVWGQGT 120
TVTVSS 126

SEQ ID NO: 375 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 375			
DIVMTQSPLS	LPVTPGEPAS	ISCRSSQSL	HRSGYNYLDW YLQKPGQSPQ LLIIYQGSNRA 60
SGVPDRFSGS	SGGTDFTLKI	SRVEAEDVG	YVCMQSLQTP RTFGQGTKEV IKR 113
SEQ ID NO: 376	moltype = AA length = 126		
FEATURE	Location/Qualifiers		
source	1..126		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 376			
QVQLQESGPG	LVKPSQTL	TCTVSGGSIR	RGGDYWSWIR QHPGKGLEWI GHIYYAGSTA 60
YNPNFRGRVR	ISVNTYKNQF	SLKLSSVTAA	DTAVIYCARG YDTSGYSYYY YGLDVWGQGT 120
TVTVSS			126
SEQ ID NO: 377	moltype = AA length = 113		
FEATURE	Location/Qualifiers		
source	1..113		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 377			
DIVMTQSPLS	LPVTPGEPAS	ISCRSSQSL	HRSGYNYLDW YLQKPGQSPQ LLIIYLSNRA 60
SGVPDRFSGS	SGGTDFTLKI	SRVEAEDVG	YVCMQSLQTP RTFGQGTKEV IKR 113
SEQ ID NO: 378	moltype = AA length = 126		
FEATURE	Location/Qualifiers		
source	1..126		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 378			
QVQLQESGPG	LVKPSQTL	TCTVSGGSIR	RGGDYWSWIR QHPGKGLEWI GHVYYTGQTA 60
YNPNFRNRVR	ISVNTYKNQF	SLKLSSVTAA	DTAVIYCARG YDVSYSYYY YGMDVWGQGT 120
TVTVSS			126
SEQ ID NO: 379	moltype = AA length = 113		
FEATURE	Location/Qualifiers		
source	1..113		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 379			
DIVMTQSPLS	LPVTPGEPAS	ISCRSSQSL	HRSGYNYLDW YLQKPGQSPQ LLIIYLSNRA 60
SGVPDRFSGS	SGGTDFTLKI	SRVEAEDVG	YVCMQSLQTP RTFGQGTKEV IKR 113
SEQ ID NO: 380	moltype = AA length = 126		
FEATURE	Location/Qualifiers		
source	1..126		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 380			
QVQLQESGPG	LVKPSQTL	TCTVSGGSIR	RGGDYWSWIR QHPGKGLEWI GHVYYTGQTA 60
YNPSLRGRVR	ISVNTYKNQF	SLKLSSVTAA	DTAVIYCARG YDTSGYSYYY YGLDVWGQGT 120
TVTVSS			126
SEQ ID NO: 381	moltype = AA length = 113		
FEATURE	Location/Qualifiers		
source	1..113		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 381			
DIVMTQSPLS	LPVTPGEPAS	ISCRSSQSL	HRSGYNYLDW YLQKPGQSPQ LLIIYLSNRA 60
SGVPDRFSGS	SGGTDFTLKI	SRVEAEDVG	YVCMQSLQTP RTFGQGTKEV IKR 113
SEQ ID NO: 382	moltype = AA length = 126		
FEATURE	Location/Qualifiers		
source	1..126		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 382			
QVQLQESGPG	LVKPSQTL	TCTVSGGSIR	RGGDYWSWIR QHPGKGLEWI GHIYYTGQTA 60
YNPNFRNRVR	ISVNTYKNQF	SLKLSSVTAA	DTAVIYCARG YDISGYSYYY YGLDVWGQGT 120
TVTVSS			126
SEQ ID NO: 383	moltype = AA length = 113		
FEATURE	Location/Qualifiers		
source	1..113		
	mol_type = protein		
	organism = synthetic construct		

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SEQUENCE: 383
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 384 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 384
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYTGQTT 60
YNPNPKGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDASGYSYYY YGMDVWGQGT 120
TVTVSS 126

SEQ ID NO: 385 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 385
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 386 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 386
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYEGSTK 60
YNPSFRGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDVS GYSYYY YGLDVWGQGT 120
TVTVSS 126

SEQ ID NO: 387 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 387
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 388 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 388
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYAGSTT 60
YNPSLKSRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDVS GYSYYY YGLDVWGQGT 120
TVTVSS 126

SEQ ID NO: 389 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 389
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 390 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 390
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYTGQTK 60
YNPSLRGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDISGYSYYY YGLDVWGQGT 120
TVTVSS 126

SEQ ID NO: 391 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

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SEQUENCE: 391
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPGRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 392 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 392
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYYTGQTA 60
YNPNFRNRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDVS GYSYYY YGLDVWGQGT 120
TVTVSS 126

SEQ ID NO: 393 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 393
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 394 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 394
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYYTGSTA 60
YNPSFRGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDTS GYSYYY YGLDVWGQGT 120
TVTVSS 126

SEQ ID NO: 395 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 395
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 396 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 396
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYKGSTE 60
YNPSLRGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDVS GYSYYY YGLDVWGQGT 120
TVTVSS 126

SEQ ID NO: 397 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 397
DIVMTQSPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 398 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 398
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYYTGSTT 60
YNPNFKGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDAS GYSYYY YGLDVWGQGT 120
TVTVSS 126

SEQ ID NO: 399 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

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SEQUENCE: 399
 DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
 SGVPDRFSGS GSGTDFTLKI SRVEAEDVG YVCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 400 moltype = AA length = 126
 FEATURE Location/Qualifiers
 source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 400
 QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHVYQTGSTK 60
 YNPSPKNNRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDVGSGSYYY YGLDVWGQGT 120
 TVTVSS 126

SEQ ID NO: 401 moltype = AA length = 113
 FEATURE Location/Qualifiers
 source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 401
 DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
 SGVPDRFSGS GSGTDFTLKI SRVEAEDVG YVCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 402 moltype = AA length = 126
 FEATURE Location/Qualifiers
 source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 402
 QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYAGQTK 60
 YNPSPKNNRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDISGYSYY YGLDVWGQGT 120
 TVTVSS 126

SEQ ID NO: 403 moltype = AA length = 113
 FEATURE Location/Qualifiers
 source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 403
 DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYQGSNRA 60
 SGVPDRFSGS GSGTDFTLKI SRVEAEDVG YVCMQSLQTP RTFGQGKVE IKR 113

SEQ ID NO: 404 moltype = AA length = 126
 FEATURE Location/Qualifiers
 source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 404
 QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYQTGQTK 60
 YNPSPKGRVR ISVNTYKNQF SLKLSSVTAA DTAVYYCARG YDVGSGSYYY YGMDVWGQGT 120
 TVTVSS 126

SEQ ID NO: 405 moltype = AA length = 11
 FEATURE Location/Qualifiers
 source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 405
 RASQSISSYL N 11

SEQ ID NO: 406 moltype = AA length = 7
 FEATURE Location/Qualifiers
 source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 406
 SARRLSS 7

SEQ ID NO: 407 moltype = AA length = 9
 FEATURE Location/Qualifiers
 source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 407
 QQYRSYPIT 9

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SEQ ID NO: 408	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 408		
RGGDYWG		7
SEQ ID NO: 409	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 409		
HVYQTGSTKY NPSFKD		16
SEQ ID NO: 410	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 410		
NYDTLTGYSY YYYGMDV		17
SEQ ID NO: 411	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 411		
RASQSISSYL N		11
SEQ ID NO: 412	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 412		
SARRLSS		7
SEQ ID NO: 413	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 413		
QQYAKSPIT		9
SEQ ID NO: 414	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 414		
RGGDYWA		7
SEQ ID NO: 415	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 415		
HIYQTGSTNY NPRFKG		16
SEQ ID NO: 416	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 416		
NYDTLTGYSY YYYGMDV		17
SEQ ID NO: 417	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	

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SEQUENCE: 417 RASQSISSYL N	organism = synthetic construct	11
SEQ ID NO: 418 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 418 SARALSS		7
SEQ ID NO: 419 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 419 QQYRKPLT		9
SEQ ID NO: 420 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 420 RGGDYWA		7
SEQ ID NO: 421 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 421 HIYQTGSTNY NPRPKG		16
SEQ ID NO: 422 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 422 NYDTLTGYSY YYYGMDV		17
SEQ ID NO: 423 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 423 RASQSISSYL N		11
SEQ ID NO: 424 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 424 GARRLGS		7
SEQ ID NO: 425 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 425 QQYSSLPVT		9
SEQ ID NO: 426 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 426 RGGDYWA		7

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SEQ ID NO: 427	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 427		
HVYYTGSTKY NPSFKD		16
SEQ ID NO: 428	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 428		
NYDTLTGYSY YYYGMDV		17
SEQ ID NO: 429	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 429		
RASQSISSYL N		11
SEQ ID NO: 430	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 430		
KARLLSS		7
SEQ ID NO: 431	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 431		
QQYSRLPLT		9
SEQ ID NO: 432	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 432		
RGGDYWA		7
SEQ ID NO: 433	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 433		
HVYYTGSTNY NPRFKD		16
SEQ ID NO: 434	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 434		
NYDTVTGYSY YYYGMDV		17
SEQ ID NO: 435	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 435		
RASQSISSYL N		11
SEQ ID NO: 436	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	

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SEQUENCE: 436 KARKLSS	organism = synthetic construct	7
SEQ ID NO: 437 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 437 QQFGSMPLT		9
SEQ ID NO: 438 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 438 RGGDYWA		7
SEQ ID NO: 439 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 439 HIYQTGSTNY NPRFKG		16
SEQ ID NO: 440 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 440 NYDTLTGYSY YYYGMDV		17
SEQ ID NO: 441 FEATURE source	moltype = AA length = 11 Location/Qualifiers 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 441 RASQSISSYL N		11
SEQ ID NO: 442 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 442 SARRLAS		7
SEQ ID NO: 443 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 443 QQYYHPPIT		9
SEQ ID NO: 444 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 444 RGGDYWA		7
SEQ ID NO: 445 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 445 HVYTGSTNY NPRFKD		16

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SEQ ID NO: 446	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 446		
NYDTVTGYSY YYYGMDV		17
SEQ ID NO: 447	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 447		
RASQSISSYL N		11
SEQ ID NO: 448	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 448		
NARRLGS		7
SEQ ID NO: 449	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 449		
QQYRSSPVT		9
SEQ ID NO: 450	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 450		
RGGDYWA		7
SEQ ID NO: 451	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 451		
HIYQTGSTNY NPRFKG		16
SEQ ID NO: 452	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 452		
NYDTLTGYSY YYYGMDV		17
SEQ ID NO: 453	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 453		
DIQMTQSPSS LSASVGRVT ITCRASQSI SYLNWYQQK GKAPKLLIYS ARRLSSGVPS		60
RFSGSGSGTD PTLTISSLQP EDFATYYCQQ YRSYPITFGG GTKVEIKR		108
SEQ ID NO: 454	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 454		
QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWGWR QHPGKGLEWI GHVYQTGSTK		60
YNPSFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARN YDTLTGYSY YGMDVWGQG		120
TTVTVSS		127
SEQ ID NO: 455	moltype = AA length = 108	

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FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 455		
DIQMTQSPSS LSASVGDVRT	ITCRASQSSIS SYLNWYQQKP GKAPKLLIYS ARRLSSGVPS	60
RFSGSGSGTD FTLTISSLQP	EDFATYYCQQ YAKSPITFGG GTKVEIKR	108
SEQ ID NO: 456	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 456		
QVQLQESGPG LVKPSQTLST	TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHYYQTGSTN	60
YNPRFKGRAT ISVNTYKNQF	SLKLSSVTAA DTAVYYCARN YDTLTGYSYY YGMDVWGQG	120
TTVTVSS		127
SEQ ID NO: 457	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 457		
DIQMTQSPSS LSASVGDVRT	ITCRASQSSIS SYLNWYQQKP GKAPKLLIYS ARALSSGVPS	60
RFSGSGSGTD FTLTISSLQP	EDFATYYCQQ YRKPLTFGG GTKVEIKR	108
SEQ ID NO: 458	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 458		
QVQLQESGPG LVKPSQTLST	TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHYYQTGSTN	60
YNPRFKGRAT ISVNTYKNQF	SLKLSSVTAA DTAVYYCARN YDTLTGYSYY YGMDVWGQG	120
TTVTVSS		127
SEQ ID NO: 459	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 459		
DIQMTQSPSS LSASVGDVRT	ITCRASQSSIS SYLNWYQQKP GKAPKLLIYG ARRLSGGVPS	60
RFSGSGSGTD FTLTISSLQP	EDFATYYCQQ YSSLPVTFGG GTKVEIKR	108
SEQ ID NO: 460	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 460		
QVQLQESGPG LVKPSQTLST	TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHVYYTGSTK	60
YNPSFKDRAT ISVNTYKNQF	SLKLSSVTAA DTAVYYCARN YDTLTGYSYY YGMDVWGQG	120
TTVTVSS		127
SEQ ID NO: 461	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 461		
DIQMTQSPSS LSASVGDVRT	ITCRASQSSIS SYLNWYQQKP GKAPKLLIYK ARRLSSGVPS	60
RFSGSGSGTD FTLTISSLQP	EDFATYYCQQ YSRLPLTFGG GTKVEIKR	108
SEQ ID NO: 462	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 462		
QVQLQESGPG LVKPSQTLST	TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHVYYTGSTN	60
YNPRFKDRAT ISVNTYKNQF	SLKLSSVTAA DTAVYYCARN YDTVTGYSYY YGMDVWGQG	120
TTVTVSS		127
SEQ ID NO: 463	moltype = AA length = 108	

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FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 463		
DIQMTQSPSS	LSASVGDVRT	ITCRASQSSIS SYLNWYQQKP GKAPKLLIYK ARKLSSGVPS 60
RFSGSGSGTD	FTLTISSLQP	EDFATYYCQQ FGSMPITFGG GTKVEIKR 108
SEQ ID NO: 464	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 464		
QVQLQESGPG	LVKPSQTLST	TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHYQTGSTN 60
YNPRFKGRAT	ISVNTYKNQF	SLKLSSVTAA DTAVYYCARN YDTLTGYSYY YGMDVWGQG 120
TTVTVSS		127
SEQ ID NO: 465	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 465		
DIQMTQSPSS	LSASVGDVRT	ITCRASQSSIS SYLNWYQQKP GKAPKLLIYS ARRLASGVPS 60
RFSGSGSGTD	FTLTISSLQP	EDFATYYCQQ YHPPITFGG GTKVEIKR 108
SEQ ID NO: 466	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 466		
QVQLQESGPG	LVKPSQTLST	TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHVYYTGSTN 60
YNPRFKDRAT	ISVNTYKNQF	SLKLSSVTAA DTAVYYCARN YDVTGTGYSYY YGMDVWGQG 120
TTVTVSS		127
SEQ ID NO: 467	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 467		
DIQMTQSPSS	LSASVGDVRT	ITCRASQSSIS SYLNWYQQKP GKAPKLLIYN ARRLSGGVPS 60
RFSGSGSGTD	FTLTISSLQP	EDFATYYCQQ YRSPVTFGG GTKVEIKR 108
SEQ ID NO: 468	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 468		
QVQLQESGPG	LVKPSQTLST	TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHYQTGSTN 60
YNPRFKGRAT	ISVNTYKNQF	SLKLSSVTAA DTAVYYCARN YDTLTGYSYY YGMDVWGQG 120
TTVTVSS		127
SEQ ID NO: 469	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 469		
DIQMTQSPSS	LPASVGDVRT	ITCRASQSSIS SYLNWYQQKP GKAPKLLIYS ARRLASGVPS 60
RFSGSGSGTD	FTLTISSLQP	EDFATYYCQQ YHPPITFGG GTKVEIKR 108
SEQ ID NO: 470	moltype = AA length = 127	
FEATURE	Location/Qualifiers	
source	1..127	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 470		
QVQLQESGPG	LVKPSQTLST	TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHVYYTGSTN 60
YNPRFKDRAT	ISVNTYKNQF	SLKLSSVTAA DTAVYYCARN YDVTGTGYSYY YGMDVWGQG 120
TTVTVSS		127
SEQ ID NO: 471	moltype = AA length = 16	

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FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 471		
RSSQSLHRS GYNYLD		16
SEQ ID NO: 472	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 472		
LGSNRAS		7
SEQ ID NO: 473	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 473		
MQSLQTPRT		9
SEQ ID NO: 474	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 474		
RGGDYWS		7
SEQ ID NO: 475	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 475		
HIYYSGSTNY NPSLRS		16
SEQ ID NO: 476	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 476		
DYDILTGYSY YYYGLDV		17
SEQ ID NO: 477	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 477		
RSSQSLLYSN GHNFLD		16
SEQ ID NO: 478	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 478		
LGSNRAS		7
SEQ ID NO: 479	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 479		
MQALQTPRT		9
SEQ ID NO: 480	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 480 RGGDYWN		7
SEQ ID NO: 481 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 481 HIYYSGSKNY NPSLKS		16
SEQ ID NO: 482 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 482 GYDSSGYSYY YGMDV		16
SEQ ID NO: 483 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 483 RSSQSLLYSN GHNYLD		16
SEQ ID NO: 484 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 484 LGSNRAP		7
SEQ ID NO: 485 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 485 MQALQTPRT		9
SEQ ID NO: 486 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 486 RGGDYWN		7
SEQ ID NO: 487 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 487 HIYYTGTKYY NPSLKS		16
SEQ ID NO: 488 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	
SEQUENCE: 488 GYDSSGYSYY YGMDV		16
SEQ ID NO: 489 FEATURE source	moltype = AA length = 127 Location/Qualifiers 1..127 mol_type = protein organism = synthetic construct	
SEQUENCE: 489 QVQLQESGPG LVKPSQTLST TCTVSGGSIR RGGDYWSWIR QHPGKGLEWI GHIYYSGSTN YNPSLRSRVR ISVDTSKNQF SLKLSSVTAA DTAVYYCARD YDILTGYSSY YYGLDVWGQG TTVTVSS		60 120 127

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SEQ ID NO: 490 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 490
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL HRSGYNYLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQSLQTP RTFGGQTKVE IKR 113

SEQ ID NO: 491 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 491
QVQLQESGPG LVKPSQTLISL ICIVSGGSIR RGGDYNNWIR QHPGKGLEWI GHIYYSKSN 60
YNPSLKSRTV ISVDTSKNQF SLKLSSVTAA DTAVYYCARG YDSSGYSYYY YGMDVWGQGT 120
TVTVSS 126

SEQ ID NO: 492 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 492
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL YSNGHNFLDW YLQKPGQSPQ LLIYLGSNRA 60
SGVPDRFSGS GSGTDFTLKI SRVEAEDVG VYCMQALQTP RTFGGQTKVE IKR 113

SEQ ID NO: 493 moltype = AA length = 126
FEATURE Location/Qualifiers
source 1..126
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 493
QVQLRESGPG LVKPSQTLISL ICIVSGGSIR RGGDYNNWIR QHPGKGLEWI GHIYYTGTKY 60
YNPSLKSRTV ISVDTSKNQF SLKLSSVTAA DTAVYYCARG YDSSGYSYYY YGMDVWGQGT 120
TVTVSS 126

SEQ ID NO: 494 moltype = AA length = 113
FEATURE Location/Qualifiers
source 1..113
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 494
DIVMTQSPPLS LPVTPGEPAS ISCRSSQSLL YSNGHNYLDW YLQKPGQSPQ LLIYLGSNRA 60
PGVPDRFSGS GSGTDFTLKI SRVEAEDVG VFCMQALQTP RTFGGQTKVE IKR 113

SEQ ID NO: 495 moltype = AA length = 12
FEATURE Location/Qualifiers
source 1..12
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 495
RASQSVSSSY LA 12

SEQ ID NO: 496 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 496
GASSRAT 7

SEQ ID NO: 497 moltype = AA length = 9
FEATURE Location/Qualifiers
source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 497
QQGYSNPRT 9

SEQ ID NO: 498 moltype = AA length = 11
FEATURE Location/Qualifiers
source 1..11
 mol_type = protein

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SEQUENCE: 498	organism = synthetic construct	
RASQSIISYL N		11
SEQ ID NO: 499	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 499		
AASSLRS		7
SEQ ID NO: 500	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 500		
QQYSNYPLT		9
SEQ ID NO: 501	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 501		
RASQSVSNYL A		11
SEQ ID NO: 502	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 502		
DASNRAT		7
SEQ ID NO: 503	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 503		
QQYGSLPRT		9
SEQ ID NO: 504	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 504		
RASQSVSSYL G		11
SEQ ID NO: 505	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 505		
GASSRAT		7
SEQ ID NO: 506	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 506		
QQYGSSPRT		9
SEQ ID NO: 507	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 507		
RASQSVSSYL A		11

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SEQ ID NO: 508	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 508		
DASNRAT		7
SEQ ID NO: 509	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 509		
QQYGSLPRT		9
SEQ ID NO: 510	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 510		
RASQSSISNYL N		11
SEQ ID NO: 511	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 511		
AASSLHS		7
SEQ ID NO: 512	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 512		
QQYQSYPLT		9
SEQ ID NO: 513	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 513		
GQPKANPTVT LFPPSSEELQ ANKATLVCLI SDFYPGAVTV AWKADGSPVK AGVETTKPSK		60
QSNNKYAASS YLSLTPEQWK SHRSYSCQVT HEGSTVEKTV APTECS		106
SEQ ID NO: 514	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 514		
GQPKAAPSVT LFPPSSEELQ ANKATLVCLI SDFYPGAVTV AWKADSSPVK AGVETTTPSK		60
QSNNKYAASS YLSLTPEQWK SHRSYSCQVT HEGSTVEKTV APTECS		106
SEQ ID NO: 515	moltype = AA length = 105	
FEATURE	Location/Qualifiers	
source	1..105	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 515		
QPKAAPSVTL FPPSSEELQA NKATLVCLIS DFYPGAVTVA WKADSSPVKA GVETTTPSKQ		60
SNNKYAASSY LSLTPEQWKS HRSYSCQVTH EGSTVEKTV PTECS		105
SEQ ID NO: 516	moltype = AA length = 106	
FEATURE	Location/Qualifiers	
source	1..106	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 516		
GQPKAAPSVT LFPPSSEELQ ANKATLVCLI SDFYPGAVTV AWKADSSPVK AGVETTTPSK		60
QSNNKYAASS YLSLTPEQWK SHRSYSCQVT HEGSTVEKTV APTECS		106

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SEQ ID NO: 517 moltype = AA length = 106
FEATURE Location/Qualifiers
source 1..106
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 517
GQPKAAPSVT LPPSSSEELQ ANKATLVCLV SDFYPGAVTV AWKADGSPVK VGVETTKPSK 60
QSNNKYAASS YLSLTPEQWK SHRSYSCRVT HEGSTVEKTV APAECS 106

SEQ ID NO: 518 moltype = AA length = 106
FEATURE Location/Qualifiers
source 1..106
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 518
TVAAPSVFI PPSDEQLKSG TASVVCLLNN FYPREAKVQW KVDNALQSGN SQESVTEQDS 60
KDSTYLSST LTLSKADYEK HKVYACEVTH QGLSSPVTKS FNRGEC 106

SEQ ID NO: 519 moltype = AA length = 107
FEATURE Location/Qualifiers
source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 519
RTVAAPSVFI PPSDEQLKS GTASVVCLLN NFYPREAKVQ WKVDNALQSG NSQESVTEQD 60
SKDSTYLSST TLTLKADYE KHKVYACEVT HQGLSSPVTK SFNRGEC 107

SEQ ID NO: 520 moltype = AA length = 330
FEATURE Location/Qualifiers
source 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 520
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 180
STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPVV LDSDGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 521 moltype = AA length = 330
FEATURE Location/Qualifiers
source 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 521
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 180
STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSRDE 240
LTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPVV LDSDGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 522 moltype = AA length = 330
FEATURE Location/Qualifiers
source 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 522
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 180
STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 240
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPVV LDSDGSFFLY SKLTVDKSRW 300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 330

SEQ ID NO: 523 moltype = AA length = 330
FEATURE Location/Qualifiers
source 1..330
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 523
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVEP KSCDKTHTCP PCPAPELLGG 120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 180
STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSRDE 240

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LTKNQVSLTC	LVKGFYPSDI	AVEWESNGQP	ENNYKTTTPV	LDSDGSFFLY	SKLTVDKSRW	300
QQGNVFSCSV	MHEALHNHYT	QKSLSLSPGK				330

SEQ ID NO: 524	moltype = AA length = 330
FEATURE	Location/Qualifiers
source	1..330
	mol_type = protein
	organism = synthetic construct

SEQUENCE: 524	
ASTKGPSVFP	LAPSSKSTSG
GLYSLSSVVT	VPSSSLGTQT
PSVFLFPPKP	KDTLMISRTT
STYRVSVLT	VLHQDWLNGK
MTKNQVSLTC	LVKGFYPSDI
QQGNVFSCSV	MHEALHNHYT
	QKSLSLSPGK

SEQ ID NO: 525	moltype = AA length = 330
FEATURE	Location/Qualifiers
source	1..330
	mol_type = protein
	organism = synthetic construct

SEQUENCE: 525	
ASTKGPSVFP	LAPSSKSTSG
GLYSLSSVVT	VPSSSLGTQT
PSVFLFPPKP	KDTLMISRTT
STYRCVSVLT	VLHQDWLNGK
MTKNQVSLTC	LVKGFYPSDI
QQGNVFSCSV	MHEALHNHYT
	QKSLSLSPGK

SEQ ID NO: 526	moltype = AA length = 330
FEATURE	Location/Qualifiers
source	1..330
	mol_type = protein
	organism = synthetic construct

SEQUENCE: 526	
ASTKGPSVFP	LAPSSKSTSG
GLYSLSSVVT	VPSSSLGTQT
PSVFLFPPKP	KDTLYITREP
STYRCVSVLT	VLHQDWLNGK
MTKNQVSLTC	LVKGFYPSDI
QQGNVFSCSV	MHEALHNHYT
	QKSLSLSPGK

SEQ ID NO: 527	moltype = AA length = 330
FEATURE	Location/Qualifiers
source	1..330
	mol_type = protein
	organism = synthetic construct

SEQUENCE: 527	
ASTKGPSVFP	LAPSSKSTSG
GLYSLSSVVT	VPSSSLGTQT
PSVFLFPPKP	KDTLYITREP
STYRVVSVLT	VLHQDWLNGK
MTKNQVSLTC	LVKGFYPSDI
QQGNVFSCSV	MHEALHNHYT
	QKSLSLSPGK

SEQ ID NO: 528	moltype = AA length = 326
FEATURE	Location/Qualifiers
source	1..326
	mol_type = protein
	organism = synthetic construct

SEQUENCE: 528	
ASTKGPSVFP	LAPCSRSTSE
GLYSLSSVVT	VPSSNFGTQT
LFPPKPKDTL	MISRTPEVTC
VVSVLTVVHQ	DWLNGKEYKC
QVSLTCLVKG	FYPSTIAVEW
VFSCSVMHEA	LHNHYTQKSL
	SLSPGK

SEQ ID NO: 529	moltype = AA length = 327
FEATURE	Location/Qualifiers
source	1..327
	mol_type = protein
	organism = synthetic construct

SEQUENCE: 529	
ASTKGPSVFP	LAPCSRSTSE
GLYSLSSVVT	VPSSSLGTGT
	YTCNVDHKPS

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FLFPPKPKDT	LMISRTPEVT	CVVVDVSQED	PEVQFNWYVD	GVEVHNAKTK	PREEQFNSTY	180
RVVSVLTVLH	QDWLNGKEYK	CKVSNKGLPS	SIEKTISKAK	GQPREPQVYT	LPPSQEEMTK	240
NQVSLTCLVK	GFYPSDIAVE	WESNGQPENN	YKTPPVLDL	DGSFFLYSRL	TVDKSRWQEG	300
NVFSCSVME	ALHNHYTQKS	LSLSLGK				327
SEQ ID NO: 530	moltype = AA length = 7					
FEATURE	Location/Qualifiers					
source	1..7					
	mol_type = protein					
	organism = synthetic construct					
VARIANT	7					
	note = Xaa is A, S or G					
SEQUENCE: 530						
RGGDYWX						7
SEQ ID NO: 531	moltype = AA length = 16					
FEATURE	Location/Qualifiers					
source	1..16					
	mol_type = protein					
	organism = synthetic construct					
VARIANT	2					
	note = Xaa is I or V					
VARIANT	4					
	note = Xaa is H, Y or Q					
VARIANT	5					
	note = Xaa is E, T, or S					
VARIANT	7					
	note = Xaa is K, S, or Q					
VARIANT	9					
	note = Xaa is A, K or N					
VARIANT	13					
	note = Xaa is N, S, K, or R					
VARIANT	14					
	note = Xaa is F or L					
VARIANT	15					
	note = Xaa is E or K					
VARIANT	16					
	note = Xaa is G, D, S, or N					
SEQUENCE: 531						
HXYXXGXTXY	NPXXXX					16
SEQ ID NO: 532	moltype = AA length = 17					
FEATURE	Location/Qualifiers					
source	1..17					
	mol_type = protein					
	organism = synthetic construct					
VARIANT	1					
	note = Xaa is N					
VARIANT	4					
	note = Xaa is T					
VARIANT	5					
	note = Xaa is L, or V					
VARIANT	6					
	note = Xaa is T					
VARIANT	8					
	note = Xaa is Y					
SEQUENCE: 532						
XYDXXGXSY	YYYGMDV					17
SEQ ID NO: 533	moltype = AA length = 16					
FEATURE	Location/Qualifiers					
source	1..16					
	mol_type = protein					
	organism = synthetic construct					
VARIANT	8					
	note = Xaa is H, S, or T					
VARIANT	10					
	note = Xaa is S, T, or Y;					
VARIANT	12					
	note = Xaa is A, N, I or Y					
VARIANT	15					
	note = Xaa is L or V					
SEQUENCE: 533						
RSSQSLXSY	GXNYXD					16
SEQ ID NO: 534	moltype = AA length = 6					

-continued

FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
NON_STD	1	
	note = Xaa is L or V	
SEQUENCE: 534		
XGSNRA		6
SEQ ID NO: 535	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
VARIANT	1	
	note = Xaa is M or R	
VARIANT	4	
	note = Xaa is L, I, R, or V	
VARIANT	5	
	note = X is Q, E, T, A, or G	
VARIANT	6	
	note = Xaa is T, I, L, or Q	
SEQUENCE: 535		
XQAXXPRT		9
SEQ ID NO: 536	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
VARIANT	1	
	note = X is N, S, K or G	
VARIANT	4	
	note = Xaa is R, K, L or A	
VARIANT	6	
	note = Xaa is A, G or S	
SEQUENCE: 536		
XARXLXS		7
SEQ ID NO: 537	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
VARIANT	3	
	note = Xaa is Y or F	
VARIANT	4	
	note = Xaa is R, A, S, G, or Y	
VARIANT	5	
	note = Xaa is S, K, R or H	
VARIANT	6	
	note = Xaa is S, L, F, Y, P, or M	
VARIANT	8	
	note = Xaa is L, I or V	
SEQUENCE: 537		
QQXXXXPXT		9
SEQ ID NO: 538	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
VARIANT	2	
	note = Xaa is I or V	
VARIANT	4	
	note = Xaa is Y or Q	
VARIANT	5	
	note = Xaa is A, E, K, or T	
VARIANT	7	
	note = Xaa is S or Q	
VARIANT	9	
	note = Xaa is A, E, K, or T	
VARIANT	13	
	note = Xaa is N or S	
VARIANT	14	
	note = X is F or L	

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VARIANT	15	
	note = Xaa is K or R	
VARIANT	16	
	note = Xaa is G, S, or N	
SEQUENCE: 538		
HXYXXGXTXY NPXXXX		16
SEQ ID NO: 539	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
VARIANT	1	
	note = Xaa is D or G	
VARIANT	4	
	note = Xaa is A, I, T, or V	
VARIANT	5	
	note = Xaa is L or absent	
VARIANT	6	
	note = Xaa is S or T	
VARIANT	15	
	note = Xaa is M or L	
SEQUENCE: 539		
XYDXXGYSY YYYGXDV		17
SEQ ID NO: 540	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
VARIANT	1	
	note = Xaa is M or R	
VARIANT	3	
	note = Xaa is A or S	
VARIANT	4	
	note = Xaa is L, R, or V	
VARIANT	5	
	note = Xaa is Q or A	
VARIANT	6	
	note = Xaa is T or L	
SEQUENCE: 540		
XQXXXXPRT		9
SEQ ID NO: 541	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 541		
MDMRVPAQLL GLLLLWLRGA RC		22
SEQ ID NO: 542	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 542		
MAWALLLT LTQGTGSWA		19
SEQ ID NO: 543	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 543		
MTCSPLLLT LIHCTGSWA		19
SEQ ID NO: 544	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 544		
MEAPAQLLFL LLLWLPTTG		20
SEQ ID NO: 545	moltype = AA length = 19	

-continued

FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 545		
MEWTWRVLFL VAAATGAHS		19
SEQ ID NO: 546	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 546		
METPAQLLFL LLLWLPDTTG		20
SEQ ID NO: 547	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 547		
METPAQLLFL LLLWLPDTTG		20
SEQ ID NO: 548	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 548		
MKHLWFFLLL VAAPRWVLS		19
SEQ ID NO: 549	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 549		
MEWSWVFLPF LSVTTGVHS		19
SEQ ID NO: 550	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 550		
MDIRAPTQLL GLLLLWLPGA KC		22
SEQ ID NO: 551	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 551		
MDIRAPTQLL GLLLLWLPGA RC		22
SEQ ID NO: 552	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 552		
MDTRAPTQLL GLLLLWLPGA TF		22
SEQ ID NO: 553	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 553		
MDTRAPTQLL GLLLLWLPGA RC		22
SEQ ID NO: 554	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 554
METGLRWLLL VAVLKGVQC 19

SEQ ID NO: 555 moltype = AA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 555
METGLRWLLL VAVLKGVQCQ E 21

SEQ ID NO: 556 moltype = AA length = 22
FEATURE Location/Qualifiers
source 1..22
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 556
MDMRAPTQLL GLLLLWLPGA RC 22

SEQ ID NO: 557 moltype = AA length = 19
FEATURE Location/Qualifiers
source 1..19
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 557
MKILILGIFL FLCSTPAWA 19

SEQ ID NO: 558 moltype = AA length = 19
FEATURE Location/Qualifiers
source 1..19
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 558
MRTLAILAAI LLVALQAQA 19

SEQ ID NO: 559 moltype = AA length = 472
FEATURE Location/Qualifiers
source 1..472
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 559
MKSILDGLAD TTFRTITDLD LYVGSNDIQY EDIKGDMASK LGYFPQKFPL TSFRGSPFQE 60
KMTAGDNPQL VPADQVNITE FYNKSLSSF ENEENIQCGE NFMDIECFMV LNPSQQLAIA 120
VLSLTLTGTFT VLENLLVLCV ILHSRSLRCR PSYHFIGSLA VADLLGSVIF VYSFIDFHVF 180
HRKDSRNVFL FKLGGVTASF TASVGSFLFL AIDRYISIRH PLAYKRIVTR PKAVVAFCLM 240
WTIAIVIAVL PLLGWNCEKL QSVCSDFPH IDETYLMEFI GVTSVLLLFV VYAYMYILWK 300
AHSHAVRMIQ RGTQKSIH TSEDGKVQVT RPDQARMDIR LAKTLVLILV VLIICWGPLL 360
AIMVYDVFGK MNKLIKTVFA FCSMLCLLNS TVNPIIYALR SKDLRHAFRS MFPSCEGTAQ 420
PLDNSMGDSL CLHKHANNA SVHRAESCI KSTVKIAKVT MSVSTDTSAE AL 472

SEQ ID NO: 560 moltype = AA length = 359
FEATURE Location/Qualifiers
source 1..359
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 560
MEECVVTEIA NGSKDGLDSN PMKDYMILSG POKTAVAVLC TLLGLLSALE NVAVLYLILS 60
SHQLRRKPSTY LFIGSLAGAD FLASVVFACS FVNPHVPHGV DSKAVFLKLI GSVTMTFTAS 120
VGSLLLTAID RYLCLRYPPS YKALLTRGRA LVTLGIMWVL SALVSYLPLM GWTCPPRPS 180
ELFPLIPNDY LLSWLLFIAF LFSGIIYTYG HVLWKAHQHV ASLSGHQDRQ VGMARMRLDV 240
RLAKTLGLVL AVLLICWFPV LALMAHSLAT TLSQVKKAF AFCSMLCLIN SMVNPVIYAL 300
RSGEIRSSAH HCLAHWKKCV RGLGSEAKEE APRSSVTETE ADGKITPWPD SRDLDSLDC 359

SEQ ID NO: 561 moltype = AA length = 473
FEATURE Location/Qualifiers
source 1..473
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 561
MKSILDGLAD TTFRTITDLD LYVGSNDIQY EDIKGDMASK LGYFPQKFPL TSFRGSPFQE 60
KMTAGDNPQL VPADQVNITE FYNKSLSSF KENEDNIQCG ENFMDMECFM ILNPSQQLAI 120
AVLSLTLTGTFT VLENLLVLCV VILHSRSLRC RPSYHFIGSL AVADLLGSVI FVYSFVDFHV 180
FHRKDSNPVFL LFKLGGVTASF FTASVGSFLFL TAIDRYISIRH RPLAYKRIVT RPKAVVAFCL 240
MWITIAIVIAV LPLLGNCKK LQSVCSDFPH LIDETYLMEFI IGVTSVLLLFV IVYAYMYILW 300
KAHSHAVRMI RGTQKSIH HTSEDGKVQV TRPDQARMDI RLAKTLVLIL VLIICWGPLL 360
LAIMVYDVFGK MNKLIKTVFA FCSMLCLLNS STVNPIIYAL RSKDLRHAFR SMFPSCEGTA 420

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QPLDMSMGDS DCLHKHANNT ASMHRAAESC IKSTVKIAKV TMSVSTDTSA EAL 473

SEQ ID NO: 562 moltype = AA length = 457
 FEATURE Location/Qualifiers
 source 1..457
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 562
 QVQLQESGPG LVKPSQTLSL TCTVSGGSIR RGGDYWAWIR QHPGKGLEWI GHVYTGSTN 60
 YNPRFKDRAT ISVNTYKNQF SLKLSSVTAA DTAVYYCARN YDVTGYSY YGMDVMGQG 120
 TTVTSSAST KGPVSFPLAP SSKSTSGGTA ALGCLVKDYF PEPVTVSWNS GALTSGVHTF 180
 PAVLQSSGLY SLSSVVTVP SSSLGTQTYIC NVNHKPSNTK VDKKVEPKSC DKTHTCPPCP 240
 APELLGGPSV FLFPPPKPKDT LMISRTPEVT CVVVDVSHED PEVKFNWYVD GVEVHNAKTK 300
 PCEEQYGSTY RCVSVLTVLH QDWLNGKEYK CKVSNKALPA PIEKTISKAK GQPREPQVYT 360
 LPPSREEMTK NQVSLTCLVK GFYPSDIAVE WESNGQPENN YKTTTPVLDS DGSFPLYSKL 420
 TVDKSRWQGG NVFSCSVMEH ALHNHYTQKS LSLSPGK 457

SEQ ID NO: 563 moltype = AA length = 214
 FEATURE Location/Qualifiers
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 563
 DIQLTQSPST LSASVGRVIT ITCRASQSSIS NYLNWYQQKP GKAPKLLIYA ASSLHSGVPS 60
 RPSGSGSGTD FTLTISSLPQ EDFASYCQQ YQSYPLTFGQ GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 564 moltype = AA length = 7
 FEATURE Location/Qualifiers
 source 1..7
 mol_type = protein
 organism = synthetic construct
 VARIANT 7
 note = Xaa is A or G

SEQUENCE: 564
 RGGDYWX 7

SEQ ID NO: 565 moltype = AA length = 16
 FEATURE Location/Qualifiers
 source 1..16
 mol_type = protein
 organism = synthetic construct
 VARIANT 2
 note = Xaa is I or V
 VARIANT 4
 note = Xaa is H, Y, or Q
 VARIANT 5
 note = Xaa is E, T, or S
 VARIANT 7
 note = Xaa is K, S, or Q
 VARIANT 9
 note = Xaa is A, K or N
 VARIANT 13
 note = Xaa is N, S, K, or R
 VARIANT 14
 note = Xaa is F or L
 VARIANT 15
 note = Xaa is E or K
 VARIANT 16
 note = Xaa is G, D, S, or N

SEQUENCE: 565
 HXYXXGXTXY NPXXXX 16

SEQ ID NO: 566 moltype = AA length = 17
 FEATURE Location/Qualifiers
 source 1..17
 mol_type = protein
 organism = synthetic construct
 VARIANT 1
 note = Xaa is N
 VARIANT 4
 note = Xaa is T
 VARIANT 5
 note = Xaa is L, or V
 VARIANT 6

-continued

VARIANT note = Xaa is T
 8
 note = Xaa is Y
 SEQUENCE: 566
 XYDXXGXS YYYGMDV

17

1. An isolated antibody that binds to human cannabinoid receptor 1 (huCB1), wherein the antibody comprises a heavy chain variable region (HV) comprising heavy chain complementary regions (CDR) HCDR1, HCDR2, and HCDR3, and a light chain variable region (LV) comprising light chain complementary regions (CDR) LCDR1, LCDR2, and LCDR3, wherein:

- (a) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 135-137, respectively, and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 138-140, respectively;
- (b) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 426-428, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 423-425, respectively;
- (c) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 266-268, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 263-235, respectively;
- (d) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 129-131, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 132-134, respectively;
- (e) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 159-161, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 162-164, respectively;
- (f) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 207-209, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 210-212, respectively;
- (g) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 1-3, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 4-6, respectively; or
- (h) HCDR1, HCDR2 and HCDR3 comprise the amino acid sequence of SEQ ID NOS: 55-57, respectively and LCDR1, LCDR2 and LCDR3 comprise the amino acid sequence of SEQ ID NOS: 58-60, respectively.

2. The antibody of claim 1, wherein HV comprises an amino acid sequence selected from SEQ ID NO: 228, SEQ 1-N-DO ID NO: 460, SEQ ID NO: 368, SEQ ID NO: 100, SEQ ID NO: 236, SEQ ID NO: 226, SEQ ID NO: 252, SEQ ID NO: 98, and SEQ ID NO: 116.

3. The antibody of claim 1, wherein LV comprises an amino acid sequence selected from SEQ ID NO: 227, SEQ ID NO: 459, SEQ ID NO: 367, SEQ ID NO: 99, SEQ ID NO: 235, SEQ ID NO: 225, SEQ ID NO: 251, SEQ ID NO: 97, and SEQ ID NO: 115.

4. The antibody of claim 1, wherein

- i) HV comprises the amino acid sequence of SEQ ID NO: 228, and LV comprises the amino acid sequence of SEQ ID NO: 227;

- ii) HV comprises the amino acid sequence of SEQ ID NO: 460, and LV comprises the amino acid sequence of SEQ ID NO: 459;
- iii) HV comprises the amino acid sequence of SEQ ID NO: 368, and LV comprises the amino acid sequence of SEQ ID NO: 367;
- iv) HV comprises the amino acid sequence of SEQ ID NO: 100, and LV comprises the amino acid sequence of SEQ ID NO: 99;
- v) HV comprises the amino acid sequence of SEQ ID NO: 236, and LV comprises the amino acid sequence of SEQ ID NO: 235;
- vi) HV comprises the amino acid sequence of SEQ ID NO: 226, and LV comprises the amino acid sequence of SEQ ID NO: 225;
- vii) HV comprises the amino acid sequence of SEQ ID NO: 252, and LV comprises the amino acid sequence of SEQ ID NO: 251;
- viii) HV comprises the amino acid sequence of SEQ ID NO: 98, and LV comprises the amino acid sequence of SEQ ID NO: 97; or
- ix) HV comprises the amino acid sequence of SEQ ID NO: 116, and LV comprises the amino acid sequence of SEQ ID NO: 115.

5. An isolated antibody that binds to huCB1, wherein the antibody comprises a HV and a LV, wherein

- a) the HV comprises HCDR1, HCDR2 and HCDR3, wherein

HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A, S, or G;

HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is H, Y, or Q; X4 is E, T, or S; X5 is K, S, or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and

HCDR3 comprises the amino acid sequence of X11YDX12X13X14GX15SYYYYGMDV (SEQ ID NO: 532), wherein X11 is D, E, or N; X12 is A, I, P, T, or V; X13 is F, L, or V; X14 is S or T; and X15 is H, N, or Y;

and

- b) the LV is of a VK1/O2/JK4, a VK2/A19/JK1, or a VK3/A27/JK1 germline.

6. The antibody of claim 5, wherein

- a) HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A or S;

HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is Y or Q; X4 is E, T, or S; X5 is S or Q; X6 is A or K; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and

- HCDR3 comprises the amino acid sequence of X11YDX12X13X14GX15SYYYYGMDV (SEQ ID NO: 532), wherein X11 is D, E, or N; X12 is A, I, P, or V; X13 is L or V; X14 is S or T; and X15 is H or Y; and
- b) the LV is of a VK2/A19/JK1 germline.
7. The antibody of claim 6, wherein X4 is T, X15 is Y.
8. The antibody of claim 5, wherein
- b) the LV comprises LCDR1, LCDR2 and LCDR3, and wherein
- LCDR1 comprises the amino acid sequence RSSQSLLX16SX17GX18NYX19D (SEQ ID NO: 533), wherein X16 is H, S, or T; X17 is S, T, or Y; X18 is A, N, I or Y; and X19 is L or V;
- LCDR2 comprises the amino acid sequence of X20GSNRA (SEQ ID NO: 534), wherein X20 is L or Q; and
- LCDR3 comprises the amino acid sequence of X21QAX22X23X24PRT (SEQ ID NO: 535), wherein X21 is M or R; X22 is L, I, R, or V; X23 is Q, E, T, A, or G; and X24 is T, I, L, or Q.
9. The antibody of claim 8, wherein X24 is L.
10. The antibody of claim 5, wherein
- a) HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A, G or S;
- HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is H, Y, or Q; X4 is T or S; X5 is K, S or Q; X6 is A, K or N; X7 is N, S, K, or R; X8 is F or L; X9 is E or K; and X10 is G, D, S, or N; and
- HCDR3 comprises the amino acid sequence of X11YDX12X13X14GX15SYYYYGMDV (SEQ ID NO: 532), wherein X11 is D or N; X12 is A, I, or T; X13 is F, L, or V; X14 is S or T; and X15 is H, N or Y; and
- b) the LV is of a VK1/O2/JK4 or a VK3/A27/JK1 germline.
11. The antibody of claim 10, wherein a) X1 is A or G; X3 is Q or Y, X4 is T, X8 is F, X14 is T; and X15 is Y.
12. The antibody of claim 10 wherein b) the LV is of a VK1/O2/JK4 germline.
13. The antibody of claim 11, wherein
- b) the LV is of a VK1/O2/JK4 germline and comprises light chain LCDR1, LCDR2 and LCDR3, and wherein
- LCDR1 comprises the amino acid sequence of RASQ-SISNYLN (SEQ ID NO: 132), RASQSIISYLN (SEQ ID NO: 150), or RASQSISSYLN (SEQ ID NO: 186);
- LCDR2 comprises the amino acid sequence of AASSLHS (SEQ ID NO: 133) or AASSLRS (SEQ ID NO: 151); and
- LCDR3 comprises the amino acid sequence of QQYQSY-PLT (SEQ ID NO: 134) or QQYSNYPLT (SEQ ID NO: 152);
- or
- b) the LV is of a VK3/A27/JK1 germline and comprises light chain LCDR1, LCDR2 and LCDR3, and wherein
- LCDR1 comprises the amino acid sequence of RASQSVSSYL (SEQ ID NO: 168);
- LCDR2 comprises the amino acid sequence of GASSRAT (SEQ ID NO: 169); and
- LCDR3 comprises the amino acid sequence of QQYGSSPRT (SEQ ID NO: 170).
14. The antibody of claim 13, wherein LCDR1 comprises the amino acid sequence of RASQSISSNYLN (SEQ ID NO: 132), LCDR2 comprises the amino acid sequence of AASSLHS (SEQ ID NO: 133), and LCDR3 comprises the amino acid sequence of QQYQSYPLT (SEQ ID NO: 134).
15. The antibody of claim 10, wherein
- a) HCDR1 comprises the amino acid sequence of RGGDYWX1 (SEQ ID NO: 530), wherein X1 is A or G,
- HCDR2 comprises the amino acid sequence of HX2YX3X4GX5TX6YNPX7X8X9X10 (SEQ ID NO: 531), wherein X2 is I or V; X3 is Y or Q; X4 is T; X5 is S; X6 is K or N; X7 is S or R; X8 is F; X9 is K; and X10 is G or D; and
- HCDR3 comprises the amino acid sequence of X11YDX12X13X14GX15SYYYYGMDV (SEQ ID NO: 532), wherein X11 is N; X12 is T; X13 is L or V; X14 is T; and X15 is Y, and
- b) the LV is of a VK1/O2/JK4 germline.
16. The antibody of claim 15, wherein
- a) HCDR1 comprises the amino acid sequence of RGGDYWA (SEQ ID NO: 414) or RGGDYWG (SEQ ID NO: 408);
- HCDR2 comprises the amino acid sequence of HVYYTGSTKYNPSFKD (SEQ ID NO: 427), HVYYTGSTNYNPRFKD (SEQ ID NO: 445), HIYQTGSTNYNPRFKG (SEQ ID NO: 415), or HVYQTGSTKYNPSFKD (SEQ ID NO: 409); and
- HCDR3 comprises the amino acid sequence of NYDTLTGYSYYYYGMDV (SEQ ID NO: 410) or NYDTVTGYSYYYYGMDV (SEQ ID NO: 446).
17. The antibody of claim 15, wherein b) the LV comprises LCDR1, LCDR2 and LCDR3, and wherein LCDR1 comprises the sequence of RASQSISSYLN (SEQ ID NO: 405); LCDR2 comprises the sequence of X39ARX40LX41S (SEQ ID NO: 536), wherein X39 is N, S, K or G; X40 is R, K, L or A; and X41 is A, G or S; and LCDR3 comprises the sequence of QQX42X43X44X45PX46T (SEQ ID NO: 537), wherein X42 is Y or F, X43 is R, A, S, G, or Y; X44 is S, K, R or H; X45 is S, L, F, Y, P, or M; and X46 is L, I or V.
18. The antibody of claim 17, wherein X42 is Y.
19. The antibody of claim 5, wherein the amino acid at position 83 of the HV is N and the amino acid at position 85 of the HV is Y.
20. The antibody of claim 19, wherein the amino acid at position 94 of the LV is S.
21. An isolated antibody that binds to huCB1, the antibody comprises a HV and a LV, wherein
- a) the HV comprises CDRH1, CDRH2 and CDRH3, wherein
- CDRH1 comprises the amino acid sequence of RGGDYWS (SEQ ID NO: 1);
- CDRH2 comprises the amino acid sequence of HX25YX26X27GX28TX29YNPX30X31X32X33 (SEQ ID NO: 538), wherein X25 is I or V; X26 is Y or Q; X27 is A, E, K, or T; X28 is S or Q; X29 is A, E, K, or T; X30 is N or S; X31 is F or L, X32 is K or R; and X33 is G, S, or N; and
- CDRH3 comprises the amino acid sequence of X34YDX35X36X37GYSYYYYGX38DV (SEQ ID NO: 539), wherein X34 is D or G; X35 is A, I, T, or V; X36 is L or absent; X37 is S or T; and X38 is M or L; and
- b) the LV is of a VK2/A19/JK1 germline.
22. The antibody of claim 21, wherein a) X26 is Y, X34 is G, X36 is absent, and X37 is S.

23. The antibody of claim 21, wherein b) the LV comprises LCDR1, LCDR2 and LCDR3, wherein LCDR1 comprises the amino acid sequence of RSSQSLHRSYNYLD (SEQ ID NO: 257); LCDR2 comprises the amino acid sequence of LGSNRAS (SEQ ID NO: 258) or QGSNRAS (SEQ ID NO: 264); and LCDR3 comprises the amino acid sequence of MQLQTPRT (SEQ ID NO: 259), RQSVALPRT (SEQ ID NO: 271), or RQARALPRT (SEQ ID NO: 265).

24. The antibody of claim 21, wherein the amino acid at position 83 of the HV is N and the amino acid at position 85 of the HV is Y.

25. The antibody of claim 24, wherein the amino acid at position 79 of the HV is R.

26. The antibody of claim 21, wherein the amino acid at position 94 of the LV is S, or the amino acid at position 76 is D, or both.

27. The antibody of claim 21, wherein HCDR2 of the antibody comprises 5 or fewer mutations compared to HVYYTGSTNYPNPFKD (SEQ ID NO: 136).

28. The antibody of 27, wherein HCDR3 of the antibody comprises 3 or fewer mutations compared to NYDITVTGYSYYYYGMDV (SEQ ID NO: 137).

29. The antibody of claim 21, wherein HCDR2 of the antibody comprises 5 or fewer mutations compared to HVYYTGSTKYNPNFKG (SEQ ID NO: 8).

30. The antibody of claim 29, wherein HCDR3 of the antibody comprises 4 or fewer mutations compared to DYDILTGYSYYYYGMDV (SEQ ID NO: 9).

31. An isolated antibody or antigen binding fragment thereof that binds to huCB1, wherein the antigen binding molecule comprises a HV comprising HCDR1, HCDR2 and HCDR3 and a LV comprising LCDR1, LCDR2 and LCDR3, wherein HCDR1 comprises the amino acid sequence of RGGDYWS (SEQ ID NO: 474) or RGGDYWN (SEQ ID NO: 480), HCDR2 comprises the amino acid sequence of HIYYSGSTNYPNPSLRS (SEQ ID NO: 475), HIYYSGSKNYPNPSLKS (SEQ ID NO: 481), or HIYYTGTKYYNPSLKS (SEQ ID NO: 487), and HCDR3 comprises the amino acid sequence of DYDILYGYSYYYYGLDV (SEQ ID NO: 476), or GYDSSGYSYYYYGMDV (SEQ ID NO: 475), and wherein LCDR1 comprises the amino acid sequence of RSSQSLHRSYNYLD (SEQ ID NO: 471), RSSQSLLYSNGHNFLD (SEQ ID NO: 477), or RSSQSLLYSNGHNYLD (SEQ ID NO: 483), LCDR2 comprises the amino acid sequence of LGSNRAS (SEQ ID NO: 472) or LGSNRAP (SEQ ID NO: 484), and LCDR3 comprises the amino acid sequence of MQLQTPRT (SEQ ID NO: 473) or MQALQTPRT (SEQ ID NO: 479).

32. The antibody of claim 31, wherein the HV comprises the amino acid sequence of SEQ ID NO: 489, SEQ ID NO: 491 or SEQ ID NO: 493, and the LV comprises the amino acid sequence of SEQ ID NO: 490, SEQ ID NO: 492 or SEQ ID NO 494.

33. The antibody of claim 21, wherein the amino acid at position 32 according to AHo numbering in the heavy chain is an R.

34. The antibody of claim 1, wherein the antibody is a monoclonal antibody.

35. (canceled)

36. The antibody of claim 34, wherein the monoclonal antibody is an antagonist antibody of huCB1.

37. The antibody of claim 34, wherein the monoclonal antibody is an inverse agonist antibody of huCB1.

38. The antibody of claim 36, wherein the antibody has a binding affinity for huCB1 that is at least 3 times higher compared to an antibody comprising a heavy chain variable region having the amino acid sequence of SEQ ID NO: 256 and a light chain variable region having the amino acid sequence of SEQ ID NO: 255 (10D10 LC N35Y).

39. The antibody of claim 36, wherein the antibody has an IC50 of 10 nM or less as determined using a cell-based cAMP assay.

40. The antibody of claim 39, wherein the antibody has an IC50 of 5 nM or less, or of 1 nM or less as determined using a cell-based cAMP assay.

41. (canceled)

42. The antibody of claim 41, wherein the monoclonal antibody is a human IgG1 antibody.

43. The antibody of claim 42, wherein the antibody comprises a mutation at amino acid position N297 according to EU numbering in its heavy chain.

44. The antibody of claim 43, wherein the mutation is N297G.

45. The antibody of claim 43, wherein the antibody further comprises R292C and V302C mutations according to EU numbering in its heavy chain.

46. The antibody of claim 42, wherein the antibody comprises mutations at amino acid positions M252, S254, and T256 according to EU numbering in its heavy chain.

47. The antibody of claim 46, wherein the mutations are M252Y, S254T, and T256E.

48. The antibody of claim 40, wherein the antibody comprises a heavy chain constant region amino acid sequence selected from SEQ ID NOs: 520-529 and a light chain constant region amino acid sequence selected from SEQ ID NOs: 513-519.

49. The antibody of claim 40, wherein the antibody comprises:

- i) a heavy chain constant region amino acid sequence of SEQ ID NOs: 525, 526 or 527 and a light chain constant region amino acid sequence of SEQ ID NOs: 518 or 519; or
- ii) a heavy chain constant region amino acid sequence of SEQ ID NOs: 525, 526 or 527 and a light chain constant region amino acid sequence of SEQ ID NOs: 518.

50. A pharmaceutical composition comprising the antibody of claim 1 and a pharmaceutically acceptable excipient.

51. An isolated polynucleotide that encodes the antibody of claim 1.

52. An expression vector comprising the polynucleotide of claim 51.

53. A host cell comprising the expression vector of claim 52.

54. A method of producing an antibody that binds to huCB1 comprising culturing the host cell of claim 53 under conditions that allow expression of the antibody and recovering the antibody from the culture medium or host cell.

55. A method for treating a subject in need of antagonizing or inverse agonizing the CB1 receptor, the method comprises administering an effective amount of the pharmaceutical composition of claim 50 to the subject.

56. A method for treating a disease or disorder in a subject responsive to antagonizing or inverse agonizing the CB1 receptor, the method comprises administering to the subject the antibody of claim 1.

57. The method of claim **56**, wherein the CB1 receptor is a peripheral CB1 receptor.

58. The method of claim **57**, wherein the administration results in one or more of reduced body weight, reduced appetite, improved metabolic parameters, reduced blood glucose levels, reduced insulin levels, reduced triglyceride levels, reduced kidney injury, reduced kidney fibrosis, reduced kidney inflammation, and improved kidney function.

59. The method of claim **57**, wherein the disease or disorder is selected from obesity, diabetes, dyslipidemia, metabolic diseases, liver disease, fibrosis, non-alcoholic steatohepatitis (NASH), primary biliary cirrhosis, renal disease, kidney fibrosis, chronic kidney disease, IgA nephropathy, osteoporosis, atherosclerosis, cardiovascular disease, cancer, inflammatory disease, and a combination thereof.

60. A method for treating a disease or disorder in a subject in need thereof, the method comprises administering to the subject the antibody of claim **1**, wherein the disease or disorder is obesity, diabetes, dyslipidemia, metabolic diseases, liver disease, fibrosis, NASH, primary biliary cirrhosis, renal disease, kidney fibrosis, chronic kidney disease,

IgA nephropathy, osteoporosis, atherosclerosis, cardiovascular disease, cancer, inflammatory disease or a combination thereof.

61. (canceled)

62. The method of claim **60**, wherein the administration results in one or more of reduced body weight, reduced appetite, improved metabolic parameters, reduced blood glucose levels, reduced insulin levels, reduced triglyceride levels, reduced kidney injury, reduced kidney fibrosis, reduced kidney inflammation, and improved kidney function.

63. The method of claim **56**, wherein the subject is a human.

64.-66. (canceled)

67. The method of claim **59**, wherein the disease or disorder is obesity.

68. The method of claim **56**, wherein the subject has a BMI of at least 27 kg/m².

69. The method or antibody of claim **68**, wherein the subject has a BMI of at least 30 kg/m².

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